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w/CHANGE 1
14 February 2024

SUPERSEDING
MIL-STD-3040A
24 March 2022

DEPARTMENT OF DEFENSE MANUFACTURING PROCESS STANDARD

QUALIFICATION, FABRICATION, WELDING, AND INSPECTION OF ARMOR AND NON-ARMOR STEEL WELDMENTS

AMSC N/A AREA THJM

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FOREWORD

1. This standard is approved for use by all Departments and Agencies of the Department of Defense (DoD).

2. This standard does not specify armor capability in terms of protection. The user is responsible for defining the threat that the armor system must withstand.

3. This revision has resulted in many changes from the base document, but the most significant changes are:

a. Gramarrical and editorial changes throughout the document to clarify or correct issues from the previous version.

b. Base materials groupings clarified.

c. Hold times have been updated.

4. Comments, suggestions, or questions on this document should be addressed to U.S. Army Ground Vehicle Systems Center, ATTN: FCDD-GVS-SAT MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000, or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

5. Questions or information with regards to this document related to ground vehicle systems should be addressed to gvscjoining@army.mil.

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CONTENTS

PARAGRAPH PAGE

FOREWORD II

1. SCOPE 1
1.1 Scope 1

2. APPLICABLE DOCUMENTS 1
2.1 General 1
2.2 Government documents 1
2.2.1 Specifications, standards, and handbooks 1
2.2.2 Other Government documents, drawings, and publications 3
2.3 Non-Government publications 4
2.4 Order of precedence 8

3. DEFINITIONS 8
3.1 Active flux 8
3.2 Amplitude Rejection Level (ARL) 8
3.3 Active supervision 8
3.4 Contractor 9
3.5 Deposition thickness 9
3.6 Design engineering activity 9
3.7 Digital radiography (DR) 9
3.8 Entity 9
3.9 Entity responsible representative 9
3.10 Essential variables 9
3.11 Field repair 9
3.12 Forced cooling 9
3.13 Inadvertent inspection 9
3.14 Manufacturer 9
3.15 Material Review Board (MRB) 10
3.16 Neutral flux 10
3.17 Nominal thickness 10
3.18 Procuring contracting officer (PCO) 10
3.19 Proofing Projectile Impact Testing (PPIT) 10
3.20 Reference standard 10
3.21 Repair 10
3.22 Steel 10
3.22.1 Armor grade steel 10
3.22.2 Non-armor grade steel 11
3.22.3 Sheet steel 11
3.23 Standard welding terms and definitions 11
3.24 Technical authority (TA) 11

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Check the source to verify that this is the current version before use.

PARAGRAPH PAGE

3.25 Validation test plate 11
3.26 Weld repair procedure 11
3.27 Welder qualification continuity record 11
3.28 Welding engineer 11
3.29 Welding equipment 12
3.30 Welding symbols 12

4. GENERAL REQUIREMENTS 12
4.1 Qualification requirements 12
4.1.1 Submission format 12
4.2 Qualification applicability 13
4.3 Responsibility 13
4.4 Previous qualification to MIL-STD-3040 revisions 13
4.4.1 Previous qualification to other standards 13
4.5 Safety precautions 14
4.6 Joint design 14
4.6.1 Weld joint classification requirements 14
4.6.2 Minimum fillet weld size 16
4.6.3 Intermittent fillet welds 16
4.7 Welding processes 16
4.7.1 Acceptable welding processes 16
4.7.2 Restricted welding processes 17
4.7.3 Additional welding processes 17
4.8 Weld positions 17
4.9 Quality control 17
4.9.1 Quality assurance system 18
4.9.2 Welding equipment calibration and verification 18
4.10 Welding prohibitions 18
4.11 Company name change 18

5. DETAILED REQUIREMENTS 18
5.1 Base material 18
5.1.1 Base material certification 19
5.1.2 Armor base material 19
5.1.3 Base material grouping 19
5.1.4 Backing material 19
5.1.5 Run off tab material 20
5.1.6 Material record retention 20
5.2 Welding materials 20
5.2.1 Identification 20
5.2.2 Filler materials 20
5.2.3 Shielding 23
5.3 Procedure Qualification Records (PQRs) 24
5.3.1 Requirements 24

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Check the source to verify that this is the current version before use.

PARAGRAPH PAGE

5.3.2 PQR cross qualification 27
5.3.3 PQR revision 28
5.3.4 Supplemental PQRs 28
5.3.5 Previously qualified welding procedures 28
5.3.6 Bead width limitation for weaving 28
5.3.7 Base material repair qualification 28
5.3.8 PQR record retention 29
5.4 Essential variables 29
5.4.1 Essential variables 29
5.4.2 Base material essential variables 29
5.4.3 Filler material essential variables 30
5.4.4 Shielding 31
5.4.5 Joint design 31
5.4.6 Electrical parameters 33
5.4.7 General procedure parameters 34
5.5 Welding Procedure Specification (WPS) 35
5.5.1 Requirements 36
5.5.2 WPS essential variables 36
5.5.3 Minimum travel speed tables 36
5.5.4 WPS transfer requirements 36
5.5.5 Contracted sub-entity qualification 37
5.5.6 WPS revision 37
5.5.7 WPS record retention 38
5.6 Welder performance qualification 38
5.6.1 Requirements 38
5.6.2 Welder Qualification Records (WQR) 38
5.6.3 Welder and welding operator performance qualification 38
5.6.4 Welder and welding operator essential variables 40
5.6.5 Qualifying welder or welding operator through PQR 41
5.6.6 Previous welder or welding operator qualification 41
5.6.7 Transfer of qualification 41
5.6.8 Renewal of qualification 41
5.6.9 Maintenance of qualification 41
5.6.10 Retest requirements 42
5.7 Destructive testing of welds 42
5.7.1 Requirements 42
5.7.2 Tension/reduced-section tension testing 42
5.7.3 Bend testing 42
5.7.4 Macroetch testing 43
5.7.5 Hardness testing 43
5.7.6 Fillet weld break test 44
5.7.7 Proofing projectile impact testing 44
5.7.8 Charpy V Notch (CVN) testing 44
5.7.9 Destructive test failures 45

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Check the source to verify that this is the current version before use.

PARAGRAPH PAGE

5.7.10 Destructive testing record retention 46
5.8 Nondestructive Testing (NDT) personnel qualification 46
5.8.1 Requirements 46
5.8.2 Visual inspector qualification 46
5.8.3 Other NDT methods 47
5.8.4 NDT re-examinations 47
5.8.5 Alternative NDT specifications and standards 48
5.8.6 Personnel qualification documentation 48
5.9 Nondestructive testing (NDT) 48
5.9.1 Nondestructive Testing (NDT) requirements 48
5.9.2 Nondestructive Testing (NDT) methods 50
5.9.3 Nondestructive testing record retention 54
5.10 Welding requirements 54
5.10.1 Welding environment 54
5.10.2 Welding Procedure Specification (WPS) accessibility 55
5.10.3 Position 55
5.10.4 Preheat and interpass temperatures 55
5.10.5 Post Weld Heat Treatment (PWHT) 57
5.10.6 Heat input calculations 57
5.10.7 Tack welding 58
5.10.8 Temporary welds for fabrication purposes 58
5.10.9 Weld profiles 59
5.10.10 Peening 59
5.10.11 Arc strikes 59
5.10.12 Weave and stringer passes 59
5.10.13 Backgouged roots 59
5.11 Workmanship requirements 59
5.11.1 Weld joint surface preparation 59
5.11.2 Weld cleaning 60
5.11.3 Welder and welding operator workmanship check 60
5.11.4 In-process hardness checks 60
5.11.5 Joint fit-up inspection 60
5.11.6 Distortion control 61
5.12 Repairs 61
5.12.1 Requirement 62
5.12.2 Repair procedures 62
5.12.3 Repair cycles 62
5.12.4 Mill and foundry defects 63
5.12.5 Repair of armor base material 63
5.12.6 Repairs of non-armor base material 63
5.12.7 Weld metal removal 64
5.12.8 Crack repair 65
5.12.9 Field repairs 66
5.12.10 Repair record retention 67

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Check the source to verify that this is the current version before use.

PARAGRAPH PAGE

6. NOTES 67
6.1 Intended use 67
6.2 Acquisitions requirements 67
6.3 Standard units of measure 68
6.4 Subject term (keyword) listing 68
6.5 Change notations 68

WELDING PROCEDURE QUALIFICATION OF CLASS I WELDS 111

A.1 SCOPE 111
A.1.1 Scope 111
A.1.2 Application 111

A.2 APPLICABLE DOCUMENTS 111
A.2.1 General 111

A.3 REQUIRMENTS 111
A.3.1 Requirements 111

A.4 PROCEDURE RECORDS 111
A.4.1 Procedure Qualification Records (PQRs) 111
A.4.1.1 Previous qualifications 111
A.4.1.2 PQR base material 111
A.4.1.3 Welding materials 111
A.4.1.4 Process qualification essential variables 111
A.4.1.5 Nondestructive Testing (NDT) 111
A.4.1.6 Destructive testing 112
A.4.2 Welding Procedure Specification (WPS) 112

A.5 TEST PLATE 112
A.5.1 Dimensions of test plate 112
A.5.2 Plate flatness 112
A.5.3 Test plate thickness 112
A.5.3.1 Dissimilar thickness welds 112
A.5.4 Test plate armor types 112
A.5.5 Test plate joint type 112
A.5.6 Preparation of test plate 113
A.5.6.1 Preheat and interpass temperature 113
A.5.6.2 Order of welding 113
A.5.6.3 Backgouging of welds 113
A.5.6.4 Tack welds 113
A.5.7 Identification marking of test plates 113
A.5.8 Marking of retest plates 113

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Check the source to verify that this is the current version before use.

PARAGRAPH PAGE

A.6 PROOFING PROJECTILE IMPACT TESTING 113
A.6.1 Proofing Projectile Impact Testing (PPIT) requirements 114
A.6.2 Proofing projectile impact testing address 114
A.6.3 NDT examination of test plate 114
A.6.3.1 Visual examination of test plates 114
A.6.3.2 Volumetric NDT examination 114

A.7 TEST RESULTS 114
A.7.1 Evaluation of test 114
A.7.2 Nondestructive testing after test 114
A.7.3 PPIT Reports 114
A.7.3.1 Photographic record 114
A.7.4 Repair of test plate 115
A.7.5 Rejection of test plate 115
A.7.6 Retest 115

REFERENCE MATERIALS 119

B.1 SCOPE 119
B.1.1 Scope 119
B.1.2 Application 119

TABLE PAGE

I. Base material groupings. 69
II. Welding specification and weld classification cross reference 73
III. Weld classification decision matrix. 74
IV. Filler material groupings. 75
V. Essential variables. 80
VI. Procedure qualification testing requirements 81
VII. Procedure qualification positions for plate. 82
VIII. Procedure qualification positions for pipe and tube. 82
IX. Procedure qualification test criteria for fillet welds 83
X. Procedure qualification thickness limitations. 83
XI. Procedure cross qualification. 84
XII. Heat input guidance for 120ksi wires. 87
XIII. Preheat and interpass temperatures. 88
XIV. Minimum travel speed example (IPM). 90
XV. Welder and welding operator qualification criteria. 90
XVI. Welder and welding operator qualification positions for plate. 91
XVII. Welder and welding operator qualification positions for pipe and tube. 91
XVIII. Welder and welding operator essential variables 92
XIX. Welder and welding operator cross qualification categories. 93
XX. Visual testing acceptance criteria 94

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Check the source to verify that this is the current version before use.

TABLE PAGE

XXI. NDT per weld classification. 96
A-I. H-test plate thickness and qualification range. 115

FIGURE PAGE

1. Overlap visual. 97
2. Recommended engineering drawing weld note. 97
3. Welding symbol classifications. 98
4. Groove procedure qualification mechanical test plate. 98
5. Fillet procedure qualification test plate. 99
6. Class I welder and welding operator qualification plate thickness less than 0.5 inches inclusive. 100
7. Class I welder and welding operator qualification plate thickness greater than 1.0 inches. 101
8. Class II, III, and IV welder qualification groove plate. 102
9. Class II, III, and IV welder qualification groove plate horizontal. 103
10. Class II, III, and IV welding operator qualification groove plate. 104
11. Class II, III, and IV welder and welding operator qualification plug weld assembly. 105
12. Class II, III, and IV welder qualification fillet weld assembly. 106
13. Class II, III, and IV PQR, welder, and welding operator qualification pipe or tube assembly. 106
14. Excavated weld metal flow chart. 107
15. Visual inspection. 107
16. Temperature measurement technique. 108
17. Hole repair. 109
18. Excavated weld metal. 110
A-1. Proofing projectile impact testing test plate (H plate). 116
A-2. Type I-A, single V-groove butt joint, without backing. 117
A-3. Type I-B, single V-groove butt joint with backing material. 117
A-4. Type II, double V-groove butt joint. 118
B-1. Procedure Qualification Record (PQR) example. 120
B-2. Welding Procedure Specification (WPS) example. 123
B-3. Welder/Welding Operator Qualification Record (WQR) example. 124

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1. SCOPE

1.1 Scope. This standard contains the minimum requirements for qualification of welding procedures, qualification of personnel, testing, fabrication, inspection, and repair of armor and non-armor steel weldments. This standard is intended to be used to facilitate the fabrications of new weldments, repair of existing weldments, and govern the requirements associated with any weldment made in accordance with this document. The qualification tests required by this document are devised to demonstrate the adequacy of the welding procedures and to demonstrate the ability of welders or welding operators to produce sound welds. This standard should not be applied in the cases of: structural tubular T-Y-K- connections; pressure vessels, or media moving or process piping under high or low pressure.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-11356	-	Armor, Steel, Cast, Homogeneous, Combat-Vehicle Type (1/4 to 8 Inches, Inclusive)
MIL-DTL-12560	-	Armor Plate, Steel, Wrought, Homogeneous (For Use in Combat-Vehicles and For Ammunition Testing)
MIL-DTL-22200/1	-	Electrodes, Welding, Mineral Covered, Iron-Powder, Low-Hydrogen Medium and High Tensile Steel, As-Welded or Stress-Relieved Weld Application
MIL-DTL-24403/1	-	Electrodes - Welding, Flux Cored, Ordinary Strength and Low Alloy Steel
MIL-DTL-32332	-	Armor Plate, Steel, Wrought, Ultra-High-Hardness

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MIL-DTL-46100	-	Armor Plate, Steel, Wrought, High-Hardness
MIL-DTL-46186	-	Armor Plate, Steel, Wrought, Very High Hardness
MIL-E-19933	-	Electrodes and Rods - Welding, Bare, Chromium and Chromium-Nickel Steels
MIL-E-21562	-	Electrodes and Rods-Welding, Bare, Nickel Alloy
MIL-E-22200/2	-	Electrode, Welding Covered (Austenitic Chromium-Nickel Steel)
MIL-E-22200/3	-	Electrodes, Welding, Covered, Nickel Base Alloy, and Cobalt Base Alloy
MIL-E-22200/8	-	Electrodes, Welding, Covered, Low - Hydrogen, and Iron - Powder Low - Hydrogen, Chromium - Molybdenum Alloy Steel and Corrosion Resisting Steel
MIL-E-23765/1	-	Electrodes and Rods – Welding, Bare, Solid and Alloyed Cored, Ordinary Strength and Low Alloy Steel
MIL-E-23765/2	-	Electrodes and Rods - Welding, Bare, Solid, or Alloy Cored; and Fluxes, Low Alloy Steel
MIL-E-23765/4	-	Electrodes – Welding, Bare, Solid; and Fluxes, Submerged Arc Welding, Carbon and Low Alloy Steels
MIL-E-24403/2	-	Electrodes - Welding, Flux Cored, Low-Alloy Steel

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-22	-	Welded Joint Design
MIL-STD-1916	-	DoD Preferred Methods for Acceptance of Product
MIL-STD-2035	-	Nondestructive Testing Acceptance Criteria

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-1222	-	Guide to the General Style and Format of U.S. Army Work Package Technical Manuals

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(Copies of these documents are available from http://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ARMY TECHNICAL MANUALS

TM 1-1500-335-23	-	Technical Manual - Nondestructive Inspection Methods, Basic Theory
TM 1-1500-366-23	-	Technical Manual – Nondestructive Inspection General Procedures and Process Controls

(Copies of these documents are available from https://armypubs.army.mil/.)

ARMY TEST AND EVALUATION COMMAND

TOP 2-2-711	-	US Army and Evaluation Command Test Operations Procedure “Ballistic testing of armor weldments”

(Copies of this document are available from https://apps.dtic.mil.)

NAVY TECHNICAL PUBLICATIONS

NAVSEA S9074-AQ-GIB-010/248	-	Requirements for Welding and Brazing Procedure and Performance Qualification
NAVSEA T9074-AS-GIB-010/271	-	Requirements for Nondestructive Testing Methods
NAVSEA S9074-A1-GIB-010/1628	-	Fillet weld and partial penetration groove tee weld size, strength and efficiency determination
NAVSEA T9074-BC-GIB-010/0200	-	Filler Materials for Critical Applications Requirements for Flux Cored Welding Electrodes, Bare Welding Electrodes, and Fluxes, and Covered Welding Electrodes for Low-Alloy Steel Applications

(Copies of this document are available from https://www.navsup.navy.mil/public/navsup/nll/.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME BPVC Section IX	-	ASME Boiler and Pressure Vessel Code Section IX Welding, Brazing, and Fusing Qualifications

(Copies of these documents are available online at http://www.asme.org.)

AMERICAN SOCIETY FOR NONDESTRUCTIVE TESTING (ASNT)

ANSI/ASNT CP-189	-	Qualification and Certification of Nondestructive Testing Personnel
ASNT SNT-TC-1A	-	Personnel Qualification and Certification in Nondestructive Testing

(Copies of this document are available online at https://www.asnt.org.)

ASTM INTERNATIONAL

ASTM E10	-	Standard Test Method for Brinell Hardness of Metallic Materials.
ASTM E18	-	Standard Test Methods for Rockwell Hardness of Metallic Materials
ASTM E23	-	Standard Test Methods for Notched Bar Impact Testing of Metallic Materials
ASTM E92	-	Standard Test Methods for Vickers Hardness of Metallic Materials
ASTM E94/E94M	-	Standard Guide for Radiographic Examination Using Industrial Radiographic Film
ASTM E140	-	Standard Hardness Conversion Tables for Metals Relationship Among Brinell Hardness, Vickers Hardness, Rockwell Hardness, Superficial Hardness, Knoop Hardness, Scleroscope Hardness, and Leeb Hardness
ASTM E165/E165M	-	Standard Practice for Liquid Penetrant Testing for General Industry

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ASTM E340	-	Standard Practice for Macroetching Metals and Alloys
ASTM E566	-	Standard Practice for Electromagnetic (Eddy Current/Magnetic Induction) Sorting of Ferrous Metals
ASTM E1032	-	Standard Test Method for Radiographic Examination of Weldments Using Industrial X-Ray Film
ASTM E1254	-	Standard Guide for Storage of Radiographs and Unexposed Industrial Radiographic Films
ASTM E1417/E1417M	-	Standard Practice for Liquid Penetrant Testing
ASTM E1441	-	Standard Guide for Computed Tomography (CT)
ASTM E1444/E1444M	-	Standard Practice for Magnetic Particle Testing for Aerospace
ASTM E1570	-	Standard Practice for Fan Beam Computed Tomographic (CT) Examination
ASTM E1742/E1742M	-	Standard Practice for Radiographic Examination
ASTM E2033	-	Standard Practice for Radiographic Examination Using Computed Radiography (Photostimulable Luminescence Method)
ASTM E2445/E2445M	-	Standard Practice for Performance Evaluation and Long-Term Stability of Computed Radiography Systems
ASTM E2698	-	Standard Practice for Radiographic Examination Using Digital Detector Arrays
ASTM E2699	-	Standard Practice for Digital Imaging and Communication in Nondestructive Evaluation (DICONDE) for Digital Radiographic (DR) Test Methods
ASTM E2737	-	Standard Practice for Digital Detector Array Performance Evaluation and Long-Term Stability
ASTM E3024/E3024M	-	Standard Practice for Magnetic Particle Testing for General Industry

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ASTM E3052	-	Standard Practice for Examination of Carbon Steel Welds Using Eddy Current Array

(Copies of these documents are available from www.astm.org.)

AMERICAN WELDING SOCIETY (AWS)

AWS A2.4	-	Standard Symbols for Welding, Brazing and Nondestructive Examination
AWS A3.0M/A3.0	-	Standard Welding Terms and Definitions Including Terms for Adhesive Bonding, Brazing, Soldering, Thermal Cutting, and Thermal Spraying
AWS A5.01M/A5.01	-	Welding and Brazing Consumable – Procurement of Filler Metals and Fluxes
AWS A5.4/A5.4M	-	Specification for Stainless Steel Electrodes for Shielded Metal Arc Welding
AWS A5.5/A5.5M	-	Specification for Low-Alloy Steel Electrodes for Shielded Metal Arc Welding
AWS A5.9/A5.9M	-	Welding Consumables-Wire Electrodes, Strip Electrodes, Wires, and Rods for Arc Welding of Stainless and Heat Resisting Steels-Classification
AWS A5.18/A5.18M	-	Specification for Carbon Steel Electrodes and Rods for Gas Shielded Arc Welding
AWS A5.20/A5.20M	-	Specification for Carbon Steel Electrodes for Flux Cored Arc Welding
AWS A5.22/A5.22M	-	Specification for Stainless Steel Flux Cored and Metal Cored Welding Electrodes and Rods
AWS A5.23/A5.23M	-	Specification for Low-Alloy Steel Electrodes and Fluxes for Submerged Arc Welding
AWS A5.28/A5.28M	-	Specification for Low-Alloy Steel Electrodes and Rods for Gas Shielded Arc Welding

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AWS A5.29/A5.29M	-	Specification for Low-Alloy Steel Electrodes for Flux Cored Arc Welding
AWS A5.32M/A5.32	-	Welding Consumables - Gases and Gas Mixtures for Fusion Welding and Allied Processes
AWS B2.1/B2.1M	-	Specification for Welding Procedure and Performance Qualification
AWS B4.0	-	Standard Methods for Mechanical Testing of Welds
AWS B5.1	-	Specification for the Qualification of Welding Inspectors
AWS B5.2	-	Specification for the Training, Qualification, and Company Certification of Welding Inspector Specialists and Welding Inspector Assistants
AWS C1.1M/C1.1	-	Recommended Practices for Resistance Welding
AWS C1.4M/C1.4	-	Specification for Resistance Welding of Carbon and Low-Alloy Steels
AWS C5.4	-	Recommended Practices for Stud Welding
AWS C6.1	-	Recommended Practices for Friction Welding
AWS C6.2/C6.2M	-	Specification for Rotary Friction Welding of Metals
AWS D1.1/D1.1M	-	Structural Welding Code – Steel
AWS D8.1M	-	Specification for Automotive Weld Quality-Resistance Spot Welding of Steel
AWS D8.8M	-	Specification for Automotive Weld Quality-Arc Welding of Steel
AWS D14.3/D14.3M	-	Specification for Welding Earthmoving, Construction, Agricultural, and Ground-Based Material Handling Equipment.
AWS D14.8M	-	Standard Methods for the Avoidance of Cold Cracks
AWS D17.1/D17.1M	-	Specification for Fusion Welding for Aerospace Applications
AWS D17.2/D17.2M	-	Specification for Resistance Welding for Aerospace Applications
AWS QC1	-	Standard for AWS Certified Welding Inspectors
AWS RWM	-	Resistance Welding Manual

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AWS WHB-1.0	-	Welding and Cutting Science and Technology VOLUME 1

(Copies of these documents are available from www.aws.org.)

CANADIAN STANDARDS ASSOCIATION

CSA W178.2	-	Certification of Welding Inspectors

(Copies of this document are available online at http://www.csa.ca.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

ISO/IEC 17025	-	General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of this document are available online at www.iso.org.)

NATIONAL AEROSPACE STANDARD

AIA/NAS NAS410	-	NAS Certification & Qualification of Nondestructive Test Personnel

(Copies of this document are available online at www.aia-aerospace.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Active flux. An active flux is defined as being one that readily oxidizes the alloying elements, or adds significant alloy content to the weld.

3.2 Amplitude Rejection Level (ARL). The horizontal level on the instrument display, such as in ultrasonic inspection, which is established as either a percentage of full-scale height or as a decibel (dB) level based on the peak amplitude of the signal received from the applicable reflective surface in the calibration standard.

3.3 Active supervision. Direct on-site readily available supervision supplied by the certified visual inspector (CVI) (see 5.8.2.1) in the form of necessary instructions to ensure the authorized visual inspectors (AVI) (see 5.8.2.2) can execute specific inspections (see AWS QC1).

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3.4 Contractor. The organization having a direct contract with the United States Government. In this standard, the term contractor refers to the prime contractor.

3.5 Deposition thickness. The depth of weld metal deposited in a weld joint for each process, transfer mode, or filler material classification used. This term is associated with single and multipass welds, complete and partial penetration joints, and multi-process welds. Commonly referred to as t.

3.6 Design engineering activity. The authorized personnel with overall design authority of the system, sub-system, or component.

3.7 Digital radiography (DR). All radiography methods whereby images are in a digital format.

3.8 Entity. All sites of an organization under the same quality assurance management plan and using the same quality manual, performing work to which this document is applicable.

3.9 Entity responsible representative. Any representative of an entity that is specifically authorized to approve equipment, material, or procedures within the scope of this document.

3.10 Essential variables. Essential variables are those elements, either material or process, that are important in establishing a welding procedure. These variables are defined as part of a welding procedure. Changes in these variables, after a procedure has been qualified, will require a change in the written procedure and may require requalification, depending on the element being changed and the magnitude of the change based on the requirements of this document.

3.11 Field repair. Reprocessing of nonconforming material in accordance with approved written repair procedures and operations to address the nonconformity in theater. The purpose of a field repair is to restore partial functionality to an item being repaired on an as-needed basis. Field repairs are distinguished from other repairs, in that the item, after repair, is temporarily excused from completely conforming to all of the applicable drawings, specifications or contract requirements. The system will be brought back into full conformity once the repaired item reaches a facility that is capable of performing a complete repair. This type of repair will not be used in manufacturing of new vehicles or repairs at a facility capable of performing a complete repair.

3.12 Forced cooling. The cooling of welds using any other method greater than 5 mph air movement at ambient air temperature.

3.13 Inadvertent inspection. Unplanned or unintentional observed indications or defects discovered during nondestructive testing (NDT) processing of nearby areas.

3.14 Manufacturer. The entity performing the operations covered by this standard. The contractor may or may not be the manufacturer. The manufacturer may also be known as the sub-contractor or sub-contracting entity.

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3.15 Material Review Board (MRB). A board consisting of contractor and government personnel necessary to review, evaluate, and determine or recommend disposition of nonconforming material referred to it.

3.16 Neutral flux. A neutral flux is defined as being one that does not readily oxidize the alloying elements or add significant alloy content to the weld.

3.17 Nominal thickness. The thickness that is specified on the plans or drawings without application of any allowed tolerance. The abbreviation for nominal thickness is T.

3.18 Procuring contracting officer (PCO). The individual authorized on behalf of the government, and who is responsible for overall procurement under the contract.

3.19 Proofing Projectile Impact Testing (PPIT). This test was previously referred to as the Ballistic Shock Test (BST). This is a high velocity impact test performed with a blunt nose proofing projectile to evaluate the performance of the weld under this type of impact. This test is performed in accordance with TOP 2-2-711, Ballistic testing of armor weldments. This test is not correlated to system level performance specifications or other system requirements.

3.20 Reference standard. A material or object for which all relevant chemical and physical characteristics are known and measurable, used as a comparison for, or standardization of, equipment or instruments used for nondestructive testing.

3.21 Repair. Reprocessing of defective welds, discontinuities, adjacent welding regions, base material, and improperly located components or weldment features; as to subsequently bring into conformance with this document, drawings, specifications, and contract requirements.

3.22 Steel. For the purposes of this document, steel covers all base materials incorporated under this standard, including armor grade steel (see 3.22.1) and non-armor grade steels (see 3.22.2).

3.22.1 Armor grade steel. Any steel that is manufactured and qualified to a military specification for armor grade steel, such as MIL-DTL-12560, MIL-DTL-46100, MIL-DTL-32332 and MIL-A-11356, or approved by the procuring contract officer for use as armor.

3.22.1.1 Cast homogeneous armor. Armor grade steel qualified under MIL-A-11356. Referred to as cast armor.

3.22.1.2 High hardness armor. Armor grade steel qualified under MIL-DTL-46100. Commonly referred to as high hard or HHA.

3.22.1.3 Rolled homogeneous armor**.** Armor grade steel qualified under MIL-DTL-12560. Commonly referred to as RHA.

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3.22.1.4 Ultra-high-hardness armor**.** Armor grade steel qualified under MIL-DTL-32332. Commonly referred to as ultra-hard or UHA.

3.22.1.5 Very high hardness armor**.** Armor grade steel qualified under MIL-DTL-46186. Commonly referred to as very hard or VHA.

3.22.2 Non-armor grade steel. Steel not manufactured and qualified to a specification for armor grade steel (see 3.23.1), covered by a base material group number per table I of this document, NAVSEA S9074-AQ-GIB-010/248, or the Welding Procedures Qualified by Employer section of AWS B2.1/B2.1M and Annex C, or the Ferrous and Nonferrous P-Numbers Grouping of Base Metals Table of ASME BPVC Section IX.

3.22.2.1 High strength steel. For the purposes of this document, high strength steel is defined as any S-3, S-4, S-5 or S-11 steel with a yield strength greater than 90ksi (620MPa). This includes some grades of high strength low alloy (HSLA) steels, high yield strength (HY) steels, as well as many quenched and tempered steels.

3.22.3 Sheet steel. Sheet steel is any steel article with a thickness of 0.1875 inches (4.8 mm) or less.

3.23 Standard welding terms and definitions. All standard welding terms and definitions are located in the latest edition of AWS A3.0M/A3.0.

3.24 Technical authority (TA). Entity responsible to establish, approve, and assess conformance to technical, safety, and certification requirements and policy for both products and processes within this document. The term TA term can refer to the overall TA for a program or the TA for a specific engineering or technical discipline or design consideration. For example, the government welding engineer is the TA for welding procedure qualification review and the design engineer is the TA for joint design classification.

3.25 Validation test plate. A test plate weldment which is nearly identical to the original Procedure Qualification Record (PQR) test plate, used to confirm procedure parameters and robustness of a process and practices, to authorize the transfer of a Class I WPS.

3.26 Weld repair procedure. A unique document that works in conjunction with a qualified welding procedure specification (WPS) to provide a detailed plan for each area that requires repair.

3.27 Welder qualification continuity record. A document or database that contains the record of the welder or welding operators training and usage of welding processes and procedures they are qualified to perform. The record will show continuity of usage of the qualified process for each welder or welding operator and will note the date the process must be used before expiration of qualification.

3.28 Welding engineer. The Government or entity’s engineer assuming technical authority for welding engineering responsibilities, including but not limited to, weld design,

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quality, procedure and personnel qualification and development. For full descriptions of responsibilities, see AWS D1.1/D1.1M, and for qualifications and experience see AWS D14.3/D14.3M.

3.29 Welding equipment. All non-consumable equipment used for, or in support of, welding, and that provides instantaneous information to the user about the welding process. Information can be in the form of data output such as the amperage reading on the welding machine, or temperature measuring devices or gauge readings such as flow meters for shielding gases.

3.30 Welding symbols. Welding symbols are to be those shown in the latest edition of AWS A2.4.

4. GENERAL REQUIREMENTS

4.1 Qualification requirements. All entities shall qualify procedures and personnel in accordance with this standard and submit applicable documents (see 5.3, 5.5, 5.6, and 5.8) to the procuring contract officer or technical authority upon request, prior to welding any production weldments (see 6.2). This document shall serve as the fabrication or construction document, when a fabrication or construction document is referenced in any other document, unless otherwise required by the procuring contract officer.

4.1.1 Submission format. Procedure Qualification Records (PQRs), Welding Procedure Specifications (WPSs), and Welder Qualification Records (WQRs) shall be in an organized and efficient packet. Guidance and overview of the essential variables in the form of a supplementary table of contents, outlines, tables, charts, or matrices are encouraged. The quantity of each document type should be identified in the submission along with number of pages. Illegible or blurry submissions are unacceptable and shall be re-produced if clarity cannot be met.

4.1.1.1 PQR submission format. Each PQR shall be uniquely identified with a traceable identifier and submitted to the procuring contract officer or technical authority with all applicable supporting documentation upon request. Supporting documentation for PQRs includes but is not limited to: nondestructive and destructive test data and results, material certifications, lot conformance testing results, shielding certifications, shot records, and any modification of the PQR (see 5.3, 5.4, 5.7 and 5.9). PQR and supporting documentation shall be put into a packet and delivered for all applicable requests.

4.1.1.2 WPS submission format. WPSs shall be uniquely identified with a traceable identifier and submitted to the procuring contract officer or technical authority with all applicable supporting documentation upon request. Supporting documentation for WPSs includes but is not limited to supporting PQRs, shot records, and testing data/results (see 5.4, 5.5, 5.7, and 5.9). Each WPS and supporting documentation shall be put into a packet and delivered for all applicable requests.

4.1.1.3 WQR submission format. Each WQR shall be uniquely identified with a traceable identifier and submitted to the procuring contract officer or technical authority with all

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applicable supporting documentation upon request. Supporting documentation for WQRs includes but is not limited to nondestructive testing results, WPS used for WQR Test, and destructive testing results (see 5.6, 5.7, and 5.9). WQR and supporting documentation shall be put into a packet and delivered for all applicable requests.

4.2 Qualification applicability. Qualifications performed and approved in accordance with this document are intended to be based on material type and weld classification, and are applicable to any weldment that aligns with the scope of the qualification.

4.3 Responsibility. When this document is specified by the contract or drawing, each entity qualifying procedures under the requirements of this document shall prepare written welding procedures and perform the required tests to qualify these procedures. Prior to any fabrication, the entity shall produce all procedure qualification records, welding procedure specifications, and welder qualification records for all welds in accordance with this document and shall be delivered to the procuring contract officer upon request. Examples of a blank PQR and WPS are located in figure B-1 and figure B-2 respectively.

4.4 Previous qualification to MIL-STD-3040 revisions. Procedures and personnel previously qualified to earlier revisions of this document, for which records are available and qualifications have been maintained (see 5.3.8, 5.5.7, and 5.6.9), are acceptable for use to this document. Table II shall be used as a reference to determine the new weld classification (see 4.6.1) from the earlier revision of this document. All previously approved welding procedures and personnel qualifications, and their supporting documentation shall be submitted to the procuring contract officer and the technical authority for approval for each scope of work upon request (see 6.2). The technical authority may require requalification to this document on a case-by-case basis where any of the following apply:

a. There are specific concerns about the adequacy or accuracy of the welding procedures.

b. There are specific concerns about the ability of personnel to produce acceptable work which meets the requirements of the applicable fabrication document.

c. The existing qualification does not meet the requirements of the earlier revision of this document originally qualified to.

d. Any exceptions made to a previous procedure that does not meet the requirements of the earlier revision of this document originally qualified to.

4.4.1 Previous qualification to other standards. Procedures and personnel (5.6.6) previously qualified to other standards or documents, for which records are available and qualifications have been maintained (see 5.3.8, 5.5.7, and 5.6.9), are acceptable for use to this document. Table II shall be used as a reference to determine the new weld classification (see 4.6.1) from the earlier revision of this document. Previously approved welding procedures with incomplete records may submit a supplemental PQR (see 5.3.4) to re-attain qualification in accordance with the requirements of this document. All previously approved welding

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procedures and personnel qualifications, and their supporting documentation shall be submitted to the procuring contract officer and the technical authority for approval for each scope of work upon request (see 6.2). The technical authority may require requalification to this document on a case-by-case basis where any of the following apply:

a. There are specific concerns about the adequacy or accuracy of the welding procedures, the ability of the personnel to produce acceptable works, or works that are not able to meet the requirements of the applicable fabrication document.

b. The existing qualification does not meet the requirements of its original qualification document.

c. Any exceptions made to a previous procedure that does not meet the requirements of the original qualification document.

4.5 Safety precautions. This standard may involve hazardous materials, operations, and equipment. This standard does not address any of the safety issues associated with their use. It is the responsibility of the user to establish appropriate health and safety practices.

4.6 Joint design. The weld joint design shall be in accordance with engineering best practices and shall be the responsibility of the entity's design engineering activity with the approval of the procuring contract officer for Class I and II welds upon request. For specific joint design recommendations and calculations, see the latest version of MIL-STD-22 and NAVSEA S9074-A1-GIB-010/1628, as applicable. All joint designs and joint design calculations, including finite element analysis (FEA) results, shall be delivered to the PCO for review upon request. Square groove, plug, or slot welds shall not be used for Class I and Class II joint designs without adequate supporting data, including a test plan for demonstrating weld soundness, which shall be submitted to the PCO for approval prior to use.

4.6.1 Weld joint classification requirements. Each weld made using this standard shall be classified based on the service conditions. The classification governs the extent to which quality assurance provisions are applied to the weld joint. The design engineering activity shall determine the classification of welds in accordance with this document and table III, with final weld classification approval by the procuring contract officer. Factors to consider when classifying welds include: high-strain-rate events, static loading vs. dynamic loading, cyclic loading, corrosion, temperature, material characteristics, probability of failure, consequences of failure.

Where a weld can be assigned to more than one class, the more stringent class shall be selected. Where welds intersect or overlap another weld of a higher classification, the intersection point and 2 inches in all directions from the intersection point shall meet the quality assurance requirements of the higher classification (see figure 1). Note that weld joint classification is not related to base material classes for armor grade steels, system level performance specification requirements or NDT acceptance criteria. Weld classes shall be chosen on the basis of the following definitions:

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a. Class I welds. Welds constructed of armor grade steels welded to armor grade steels in critical load bearing elements in the primary loading path. Class I welds are typically used in primary load bearing connections and survivability. Square groove, plug, or slot welds shall not be used for Class I joint designs without adequate supporting data, including a test plan for demonstrating high strain rate soundness, which shall be submitted to the procuring contract officer and technical authority for approval prior to use. Failure of a Class I weld in service would be catastrophic and would result in the loss of life, system(s), control, major components, unintentional release of critical stores, or a mission-critical failure. A Class I Weld shall be categorized as "High" on the weld classification decision matrix. Possible examples of Class I welds include but are not limited to: turret structure, armor solution, monocoque armored hull, and exterior components and assemblies.

b. Class II welds. Welds constructed of nonarmor grade steels, in load bearing elements that are either in the primary or secondary loading path. Class II welds are typically used in primary load bearing connections. Square groove, plug, or slot welds shall not be used for Class II joint designs without adequate supporting data, including a test plan for demonstrating weld soundness, which shall be submitted to the procuring contract officer and technical authority for approval prior to use. Failure of a Class II weld in service would be catastrophic and would result in the loss of life, system(s), control, major components, unintentional release of critical stores, or a mission-critical failure. A Class II weld shall be categorized as "High" on the weld classification decision matrix. Possible examples of Class II weld include but are not limited to: structural welds, drivetrain components, suspension components, steering components, highly stressed carriage components, ammunition boxes, lifting lugs, and critical components.

c. Class III welds. Welds constructed of any combination of steels in load bearing elements. Class III welds are typically used in secondary load bearing or shared load connections. Failure of a Class III weld in service would reduce the overall efficiency of the system, and may result in the loss of life, system(s), control, major components, or a mission-critical failure. A Class III Weld shall be categorized as "Medium" on the weld classification decision matrix. Possible examples of Class III welds include but are not limited to: interior components and attachments, door or hatch components, leak-proof containers, ammunition racks, and moderately critical components.

d. Class IV welds. Welds constructed of any combination of steels in minor load bearing and noncritical elements. Failure of Class IV welds would have minor to no effect on the efficiency of a system and endangerment to personnel. Class IV welds are typically used in connections where any expected load transfer at the weld would be secondary or tertiary load bearing, shared load connections or negligible. Class IV welds shall be categorized as "Low" on the weld classification decision matrix. In any case, Class IV shall not be specified for welds used for making connections onto critical or primary load path elements, such as lift points, or elements directly related to personnel supporting activities, regardless of the loading condition or direction. Possible examples of Class IV welds include but are not limited to: ladders, handrails, stowage boxes, exterior attachments, brackets, fenders, heaters, air ducts, boiler casings, air vents, and most types of sheet metal assemblies, and non-critical components.

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4.6.1.1 Armor welds without class designations. When a weldment is constructed of an armor grade steel welded to an armor grade steel, or to a steel approved for use as an armor solution, and does not have a weld classification, the technical authority shall assign the weld classification. The procuring contract office shall, in its best effort, contact the original design activity to define the susceptibility and severity of weld failure for this particular weldment. If the original design activity is unavailable, the technical authority shall work with the design engineering activity to define weld classification for this particular weldment. When a weldment consists of armor grade steel welded to non-armor grade steel, or non-armor grade steels welded together, that does not have a weld classification, the design engineering activity shall determine the classification of welds in accordance with this document.

4.6.1.2 Welding symbol requirements for classification. To minimize fabrication costs by avoiding over-inspection and unnecessary repair, all welds on a drawing shall be classified individually. This can be accomplished by including a note on the engineering drawing with the general format shown in figure 2 which specifies only the DoD nomenclature. The weld class shall then be indicated by one the following:

a. Detailing the specific weld class within the tail of welding symbol for each individual weld joint or,

b. By using specific flag notes with the welding symbol for the individual weld joints (see figure 3) or,

c. By using detailed drawing notes, provided that all of the welds on each detail drawing are the same weld class.

4.6.2 Minimum fillet weld size. Unless specified in drawings or contract requirements, all fillet welds shall conform to the requirements listed in NAVSEA S9074-A1-GIB-010/1628.

4.6.3 Intermittent fillet welds. Intermittent fillet welds shall be allowed for Class III and IV welds only. Length and pitch specified shall be a result of design calculations. The minimum fillet weld length shall be no less than 2 inches (50.8 mm).

4.7 Welding processes.

4.7.1 Acceptable welding processes. The following welding processes are acceptable for use in accordance with this document without restriction.

a. Shielded Metal Arc Welding (SMAW)

b. Gas Metal Arc Welding (GMAW) spray transfer mode (GMAW-Spray), and pulsed spray transfer mode (GMAW-P) only.

c. Gas Tungsten Arc Welding (GTAW)

d. Flux Cored Arc Welding (FCAW) (see 5.2.3.1.1)

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4.7.2 Restricted welding processes. The following welding processes shall only be used within the limitations described herein. Any use outside of the limitations listed below shall require approval from the technical authority with supporting data.

a. GMAW utilizing a globular transfer mode (GMAW-G) shall be restricted to Class III and IV welds in the flat or horizontal position only.

b. GMAW utilizing a short circuit transfer mode (GMAW-S) shall be restricted to Class IV welds only, and with both material thicknesses being of 3/16" (4.8 mm) and less.

c. Submerged Arc Welding (SAW) shall be restricted to the flux-electrode classifications in combinations as qualified. SAW shall require approval from technical authority prior to use for Class I and Class II joints.

d. Metal Cored Arc Welding (MCAW) shall require approval from the technical authority prior to use for Class I and Class II joints. No prior approval is required for Class III or IV joints.

4.7.3 Additional welding processes. Additional welding processes not covered by this document shall require qualification in accordance with the document listed within this section for each process below. Each welding process shall be identified on the engineering drawing.

a. Stud Welding (SW) – AWS B2.1, AWS C5.4, or AWS D1.1.

b. Resistance Welding (RW) – AWS C1.1M/C1.1, AWS C1.4M/C1.4, AWS D17.2/17.2M, or AWS RWM.

c. Resistance Spot Welding (RSW) – AWS D8.1M.

d. Friction Welding (FW) – AWS C6.1 or AWS C6.2/C6.2M.

e. Other Welding Processes – Unlisted welding processes shall qualify via applicable industry standards. Other welding processes may be used with approval of the procuring contract officer and the technical authority (see 6.2). Additional welding and workmanship requirements may be required, depending on the process to be approved.

4.8 Weld positions. For descriptions of all welding positions and position limitations, see the most recent version of AWS A3.0M/A3.0.

4.9 Quality control. It shall be the full responsibility of the entity to institute and maintain a quality program and the inspection standards necessary to ensure that the part, assembly, sub-assembly, and end product is produced and meets the requirements of the contract, drawings, applicable regulations, system relevant technical manuals and bulletins, and this document. The quality program shall also ensure that measuring, test and manufacturing

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equipment are properly maintained, and that appropriate actions are taken for nonconforming materials.

4.9.1 Quality assurance system. Each entity shall maintain a quality assurance system that is adequate to assure the procuring contract officer that all of the requirements of this document have been and are continuously being met. Written procedures shall be prepared to assign responsibility and provide accountability for performing work and inspections. This system shall be defined in a document such as the entity's quality document and shall be made available for review to the PCO upon request.

4.9.1.1 WPS transfer requirement. A section in the quality manual or similar document specifically addressing the transfer of qualified procedures from one site of an entity to another site within the same entity. For requirements to transfer from entity to sub-entity, see 5.5.4.2.

4.9.2 Welding equipment calibration and verification. Equipment calibration shall be established and verified to a calibrated meter at a minimum of an annual basis. The entity shall develop and maintain an equipment verification program for all welding equipment. All equipment verifications shall be performed with a calibrated meter that is NIST certified, and documented within the entity's quality assurance system.

4.10 Welding prohibitions. Unless specifically approved by the technical authority, the following are prohibited:

a. Welding in the vertical down position in Class I, II, or III welds.

b. Oxyfuel welding and gouging for use on all S-11 and S-15 steels (see table I).

c. Welding over coated surfaces, such as galvanization, primers, and paint (see 5.11.1).

d. Anti-spatter dip or spray.

e. Forced cooling of welds

f. Welding on material when water, oil, or similar materials are in contact with the surface opposite the side to be welded and less than 1/4 inch of base material is between the weld area and the surface in contact with the liquid.

4.11 Company name change. Properly documented WPSs and PQRs qualified under the provisions of this standard by a company that later has a name change due to voluntary action or consolidation with a parent company may utilize the new name on its WPS documents while maintaining the supporting PQR qualification records with the old company name.

5. DETAILED REQUIREMENTS

5.1 Base material.

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5.1.1 Base material certification. The material certification from the manufacturer shall be verified by the contractor against the requirements of the applicable specification prior to its usage. All certification documentation for armor steel and high strength steel base material, as outlined in this section, shall be accessible by the entity for the period of performance of the contract, at a minimum. See 5.1.7 for record retention requirements.

5.1.1.1 Base material control requirements. The identification of the material shall be maintained to the point of initial fabrication in accordance with a written procedure to ensure traceability. The identification of the material shall be verified at the point of initial fabrication as meeting the requirements of the drawing.

5.1.2 Armor base material. All accepted armor grade steel shall be limited to materials qualified under MIL-A-11356, MIL-DTL-12560, or MIL-DTL-46100. Fabrication using MIL-DTL-32332 or MIL-DTL-46186 shall require use of special techniques (see 5.1.2.1) and shall require approval by the procuring contract officer and the technical authority prior to use (see 6.2). Any other steel utilized as armor shall be approved by the PCO prior to use (see 6.2), and shall then be subject to the guidelines of armor materials for procedure qualification (see 5.3 and 5.6). Unless otherwise specified, armor base material shall be welded in the fully quenched and tempered condition.

5.1.2.1 Very and ultra high hard armor restrictions. VHA and UHA are considered unweldable by standard techniques due to the elevated carbon equivalence. Any welding, cutting or bending procedure involving MIL-DTL-46186 or MIL-DTL-32332 shall be specifically approved by the technical authority prior to use (see 6.2). Special techniques may include but are not limited to maintaining a 200 °F minimum, and a 300 °F maximum preheat and interpass temperature, restricting heat input from 25kJ/in to 45kJ/in, peening in accordance with 5.10.10, distortion control methods listed in 5.11.6, and controlled or slow cooling after welding is completed.

5.1.3 Base material grouping. Base materials are categorized in groups for qualification purposes based on similar metallurgical characteristics and weldability. Material groups shall be in accordance with table I of this document, NAVSEA S9074-AQ-GIB-010/248, or the Welding Procedures Qualified by Employer section of AWS B2.1/B2.1M and Annex C, or the Ferrous and Nonferrous P-Numbers Grouping of Base Metals Table of ASME BPVC Section IX for Class II, III and IV welds. Base material grouping of armor steels is permitted in accordance with table I of this document and its notes.

5.1.4 Backing material.

5.1.4.1 Permanent backing material. Permanent backing for Class I welds shall not be permitted unless approved by the technical authority. For permanent steel backing material used in Class II, III and IV welds, the thickness of the backing material shall be equal to the thickness of the thinnest base material used with a maximum thickness of 0.375 inches (9.5 mm). If base materials are not identical, then the backing material shall be the same MIL-DTL or S-number of the lesser tensile strength base material in accordance with material specifications.

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5.1.4.2 Removable backing material. Metallic and non-metallic backing material shall be allowed for all classes of welds at the discretion of the entity. If using metallic backing material, when both base materials are identical, the backing material shall be identical with the base material. If using metallic backing material when base materials are not identical, the backing material shall be the same MIL-DTL or S-number of the lesser tensile strength material. Copper backing material is prohibited for use on any stainless steel attachment welds, or any welds involving S-3, S-4, S-5, S-11, or S-15 steels.

5.1.5 Run off tab material. Run off tab material shall be removed at the completion of welding before NDT or visual inspection. Run off tabs are acceptable for use with all classes of welds.

5.1.6 Material record retention. Material certifications for all steel used in Class I, Class II or Class III welds (see 4.6.1), as provided by the steel mill in accordance with the applicable standard shall be accessible for the period of performance of the contract at a minimum and be delivered to the procuring contract officer upon request. All certifications shall be maintained for the mechanical and ballistic requirements of the applicable base material specification for each melt/heat treatment lot for seven years after final contract delivery. For all weldments involving armor using multiple lots of material, this shall include all material certifications utilized for the construction of that end item.

5.2 Welding materials.

5.2.1 Identification. All welding filler and shielding materials shall be identified up to the point of usage.

5.2.1.1 Electrodes. Each spool or coil of electrodes shall carry an identifying label. Each piece of filler metal shall have distinguishable color code, type designation, or classification number marking.

5.2.1.1.1 Covered electrodes. Each covered electrode shall have distinguishable color code, type designation, or classification number marking. If markings are destroyed by baking or other means, electrodes shall not be used until identification is restored.

5.2.1.2 Granular flux. Each container of granular flux shall be labeled as to type, specification, and granular mesh size.

5.2.1.3 Shielding gas. Each shielding gas line shall be identified near the welding machine as to type and composition.

5.2.2 Filler materials.

5.2.2.1 Requirements. Table IV contains a list of filler materials that may be used for the corresponding welding processes. The use of these filler materials does not guarantee successful process qualification testing. Each filler material listed in table IV has the military specification

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and an AWS specification acceptable substitution listed. Any filler material not listed in table IV may be used if all procedure qualification criteria is met in accordance with this standard. When welding non-armor grade steel to armor or non-armor grade steel, the strength of the filler metal used in the joint shall not be less than the filler metal minimum tensile strength classification of that specified on the engineering drawing. When an engineering drawing does not have the filler metal minimum tensile strength classification specified, then the filler metal minimum tensile strength classification used in the joint shall be not less than that of the lowest base metal minimum tensile strength (see 5.7.2.1). Electrodes with diffusible hydrogen designations higher than H8 are prohibited.

5.2.2.2 Lot conformance testing for Class I and Class II welds. Except when purchased according to a military specification included in table IV, welding filler materials that are used for Class I and Class II welds (see 4.6.1) shall be subject to lot conformance testing. This shall include testing for mechanical properties, toughness, weldability, chemical composition, and cleanliness. If the AWS acceptable substitutions listed in table IV does not have this type of testing, the entity shall be responsible for ensuring all welding filler materials have been subjected to this type of testing. Conformance test requirements for procurement of filler materials shall be submitted along with the welding procedure and associated qualification data (see 4.1.1). As a minimum, the requirements of AWS A5.01M/A5.01 for schedule I testing to the applicable filler material specification shall be done. Flux cored electrodes shall at minimum meet the requirements of schedule J.

5.2.2.3 Filler material control requirements. The entity shall have a consumables management procedure including but not limited to; identification, storage, handling, preservation, inventory control, and distribution.

5.2.2.3.1 Filler material handling. Welding electrodes shall be handled carefully to prevent damage to their coatings, if coatings exist. Damaged containers shall have their contents examined for excessive moisture content, cracked coatings or other damage to the electrodes. Filler spools or drums shall be examined for distorted spools, entwined winding on spooled wire, or surface contaminates. Filler material damaged to the extent that it does not meet specification requirements (see table IV) shall not be used for production welding.

5.2.2.3.2 Filler material inventory control. All welding filler materials and the flux shall be identified up to the point of usage, see 5.2.1. Returned electrodes. Electrodes turned in after 9 hours orless shall be returned to the holding ovens for 8 hours prior to reissue. Electrodes turned in after more than 9 hours exposure shall be rebaked in accordance with 5.2.2.3.4.

5.2.2.3.3 Filler material storage. Bare filler material, flux core wire, and neutral granular flux shall be stored in a dry area. Low hydrogen covered electrodes shall be stored in accordance with 5.2.2.3.3.1.

5.2.2.3.3.1 Low hydrogen electrode storage. Any low hydrogen covered electrode not qualified in accordance with a military specification shall be stored in accordance with applicable industry standards or manufacturers' suggested storage and environmental exposure recommendations. Covered electrodes qualified in accordance with military specifications are

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MIL-7018-M, MIL-8018-C3, MIL-8018-C3R, MIL-9018-M, MIL-10018-XXI, MIL-11018-XXI, MIL-12018-XXI, or their AWS acceptable substitutions that includes the "M" designator as listed in table IV. Covered electrodes which have been in physical contact with water shall not be used. Covered electrodes shall be used within 9 hours of removal from hermetically-sealed, manufacturer packaging or removal from a vented holding oven maintained at 225 to 300 °F (107 to 149 °C). Electrodes not used within 9 hours after issue or otherwise exposed to ambient conditions shall be held for at least 8 hours in a vented holding oven maintained at 225 to 300 °F (107 to 149 °C) prior to reissue.

5.2.2.3.3.2 Exposure of ferritic covered electrodes. Exposure of MIL-10018-M1, MIL-10718-M, MIL-11018-M, MIL-12018-M2 electrodes shall be controlled. Upon removal from the manufacturer's hermetically sealed containers or holding ovens, electrodes shall not be used after being exposed to the atmosphere for a total period of more than 9 hours. During exposure, suitable means shall be provided to protect electrodes from inclement weather.

5.2.2.3.4 Electrode baking/re-baking requirements. Covered electrodes exposed to ambient conditions for longer than 9 hours may be baked to remove moisture content or disposed of. Only one baking of any covered electrode shall be permitted unless the requirements of 5.2.2.3.5 are met. Baking of covered electrodes shall be in accordance with the following conditions:

a. Electrodes shall be placed in the oven for baking before the temperature of the oven exceeds 300 °F (149 °C).

b. During baking, the temperature of the oven shall not be raised more than 300 °F (149 °C) per hour when the oven temperature is 500 °F (260 °C) or above.

c. The total elapsed time at oven temperatures above 500 °F (260 °C) shall be kept to a minimum.

d. Covered electrodes shall be baked at a temperature of 800±25 °F (427±14 °C) for a total time of 30 minutes to 1 hour.

e. Electrodes shall be spread on suitable trays in the oven. Electrodes may also be placed into open aluminum cans and held in the vertical position. Trays or cans shall not be fabricated using any solder or joint filler which liquefies at less than 1000 °F (538 °C).

f. Automatically controlled, forced convection or circulation ovens which bake electrodes in accordance with the requirements of this paragraph shall be used.

g. Oven controls and recording instruments shall be checked at periodic intervals in accordance with an established calibration program.

5.2.2.3.5 Storage in holding ovens. Electrodes which have been baked shall be transferred to holding ovens without being allowed to cool below 150 °F (66 °C). Transfer shall be accomplished with the electrodes protected from inclement weather. Electrodes shall be

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stored in vented holding ovens after manufacturer containers have been opened. Holding oven temperature shall be 225 to 300 °F (107 to 149 °C). Holding oven shall be checked at periodic intervals in accordance with an established calibration program.

5.2.2.3.6 Rebaking limitations. Electrodes shall not be baked by the entity more than once unless each electrode brand is initially tested after a maximum proposed number of baking cycles and shown to meet electrode specification requirements for moisture, usability, all-weld metal tensile mechanical properties and chemistry. Recertification of each electrode brand shall be performed annually.

5.2.2.4 Filler material record retention. Filler material certifications shall be kept for the period of performance of the contract at a minimum, and delivered to the procuring contract officer upon request.

5.2.3 Shielding.

5.2.3.1 Shielding gas. All shielding gas or gas mixtures shall be in accordance with AWS A5.32M/A5.32, and shall be certified by the gas manufacturer. A gas or gas mixture used for shielding shall be of a welding grade and shall have a dew point in accordance with the "Minimum requirements on purities and moisture contents of gases and gas mixtures" table of AWS A5.32M/A5.32. When shielding gas mixers are required, calibrated mass flow meters shall be used for proportioning the percentages of gases to conform to the requirements of the WPS (see 4.9). The procuring contract officer may conduct reviews of the calibration and certification documentation to assess conformance to this standard.

5.2.3.1.1 Shielding gas required for FCAW. FCAW shall require the use of a shielding gas in combination with the shielding provided by the flux. Self-shielded filler material shall not be permitted for the FCAW process.

5.2.3.2 Purging gas. All purging gas or gas mixtures shall be in accordance with shielding gas requirements (see 5.2.3.1).

5.2.3.3 Granular flux.

5.2.3.3.1 Responsibility. The activity shall establish and maintain written flux control procedures which will ensure that flux moisture levels in the production welding process do not exceed filler material procurement specification requirements.

5.2.3.3.2 Conditioning and maintenance of flux. Flux may be used directly from manufacturer's hermetically sealed containers without baking prior to its initial issue. Subsequent uses of flux shall meet the requirements of 5.2.3.3.3. Where the moisture requirements of NAVSEA T9074-BC-GIB-010/0200 are exceeded or where deemed necessary by the activity concerned, baking followed by storage in holding ovens shall be accomplished in accordance with the following:

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5.2.3.3.2.1 Baking. Flux shall be baked or reconditioned to the manufactures recommendations. If the manufacturer does not provide instructions, flux shall be conditioned in accordance with the instructions of NAVSEA T9074-BC-GIB-010/0200 to restore the moisture content to the requirements.

5.2.3.3.2.2 Storage in holding ovens. Upon removal from the manufacturer's hermetically sealed container or after baking in accordance with 5.2.3.3.2.1, flux shall be stored in holding ovens at a minimum temperature of 250 °F. Flux which has been baked should be transferred to holding ovens while still hot without being allowed to cool below 225 °F. Transfer shall be protected from inclement weather. As flux can absorb moisture at elevated temperature over prolonged periods, indefinite storage in holding ovens should be avoided. The storage period should be established by the using entity.

5.2.3.3.3 Reuse. Unfused granular flux may be reused subject to the following conditions:

a. Flux shall be collected from clean, dry work pieces which have been maintained at the 125 °F (52 °C) minimum preheat/interpass temperature and must be at least warm to the touch at the time of reuse.

b. If flux is not collected as required above, it shall be discarded or reconditioned as described in 5.2.3.3.2.1 and stored at a minimum temperature of 250 °F (122 °C).

5.2.3.3.4 Exposure of flux. Exposure of flux shall be limited to ensure that the reduced moisture requirements of NAVSEA T9074-BC-GIB-010/0200 are not exceeded. An appropriate exposure period should be established by the using entity, however, exposure shall not exceed nine hours. Upon removal from the manufacturer's hermetically sealed containers or holding ovens, flux shall not be used after being exposed to the atmosphere for a total period of more than nine hours. During exposure, suitable means shall be provided to protect flux from inclement weather.

5.2.3.4 Shielding record retention. All shielding certifications from the manufacturer shall be kept by the entity for the period of performance of the contract at a minimum, and be delivered to the procuring contract officer upon request.

5.3 Procedure Qualification Records (PQRs).

5.3.1 Requirements**.** This section describes the general requirements for developing and qualifying a PQR. A welding PQR shall be prepared in accordance with the requirements of this document prior to any welding, except for welding required to support the development of the PQR. The PQR shall list applicable procedure qualification variables specified in table V for the class of weld being qualified. Data entered on the PQR shall be the actual item description or numerical value recorded during the welding of the procedure qualification test assembly. All observed values shall be recorded as single data points for each variable used during welding. For inherently fluctuating variables, use the mean value observed during welding. The PQR shall be developed and serve as the basis for the welding procedure specification (WPS). The PQR

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shall include, as a minimum, welding parameters used for the welding of the qualification test assembly, destructive testing (see 5.7) results, and NDT examination (see 5.8). All supporting testing records shall be included with the PQR. Any PQR that supports a WPS that is planned for submission, shall be submitted to the procuring contract officer upon request (see 6.2). PQRs shall be signed by the entity's responsible representative stating that the PQR and all testing results conform to the requirements of this standard. Requirements for qualification of various classes and types of welds are specified in 5.3.1.1 through 5.3.1.6. For all classes of welds, base metals shall be grouped as specified in table I.

5.3.1.1 PQR requirements for Class I. All Class I PQRs shall require a qualification test plate to be fabricated in accordance with figure 4 for destructive testing in accordance with table VI, table VII, and 5.7, and an H-plate in accordance with table VII, table A-I, and figure A-1 for PPIT. All applicable testing and essential variables of table V shall be met for the PQR to be qualified. A Class I welding procedure qualification performed in this manner qualifies the procedure for all weld classes.

5.3.1.2 PQR Requirements for Class II, III, and IV. All Class II, III and IV PQRs shall require a qualification plate (see figure 4 or figure 5) for destructive testing in accordance with table VI, table VII, table VIII, and 5.7. If armor grade steel is not used then the PQR shall be qualified in accordance with 5.3.1.5. All applicable testing and essential variables of table V and 5.4 shall be met for the PQR to be qualified. A Class III PQR, comprised of armor steel to armor steel configuration, can be developed into a Class I PQR in accordance with 5.3.4, after developing and successful testing of a supplemental H-Plate in accordance with table VII, table A-I, and figure A-1 for PPIT.

5.3.1.3 Armor to armor welds. All armor to armor welds shall be qualified in accordance with this document.

5.3.1.4 Armor to non-armor welds. All armor to non-armor welds shall be qualified in accordance with this document. Other base metals that are in the same material grouping in the following standards as the non-armor base metal qualified are also qualified:

a. Table I, of this document.

b. The M-number and Group numbers in AWS B2.1/B2.1M.

c. The P-number and Group number in ASME BPVC Section IX.

This qualification shall also support a WPS for welding of non-armor base materials within the same material grouping to themselves. If material of similar chemistry and mechanical properties not listed in these documents is to be used, it should be submitted (see foreword) for consideration for addition to table I as part of a group.

5.3.1.5 Non-armor to non-armor welds. All applicable non-armor to non-armor welding procedures qualifications involving carbon molybdenum steels (S-3), manganese molybdenum steels (S-4), chromium molybdenum steels (S-5), quenched and tempered steels (S-11), or age

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hardening steels (S-11), see table I, shall be qualified in accordance with this document. Alternatively, procedures involving materials assigned in table I as carbon steel (S-1) or quench and tempered carbon steels (S-2), carbon molybdenum steels (S-3), manganese molybdenum steels (S-4), chromium molybdenum steels (S-5), quenched and tempered steels (S-11), or age hardening steels (S-11) may be qualified in accordance with AWS B2.1/B2.1M, ASME BPVC Section IX, or NAVSEA S9074-AQ-GIB-010/248 providing qualification includes toughness testing. Procedures involving materials assigned to S-1 or quench and tempered carbon steels (S-2) may also be qualified in accordance with AWS D1.1/D1.1M, providing qualification includes toughness testing. Base metals qualified using a material assigned to one material grouping (M-number and Group Number or P-number plus Group number) qualifies all materials in that material grouping. All applicable non-armor to non-armor welds shall comply with the inspection requirements (see 5.9), welding requirements (see 5.10), workmanship requirements (see 5.11), and repair requirements (see 5.12) according to class of weld.

5.3.1.6 PJP, fillet and flare bevel groove weld PQRs. Flare bevel groove welds and PJP groove welds shall be qualified independently. Any PJP qualification, including fillet and flare bevel grooves, involving non-armor to non-armor materials shall be qualified either in accordance with 5.3.1.5 or the following requirements. For qualification of PJP, fillet and flare bevel groove welds to this document, see the following requirements:

5.3.1.6.1 PJP and flare bevel groove welds when a CJP PQR is present. Any supporting CJP PQR shall serve as the base mechanical testing qualification for any associated PJP and flare bevel groove welds. In addition, three macrographs shall be required for each groove weld type to validate sufficient penetration depth, weld size, and proper fusion has been achieved in the PJP configuration, see 5.7.4.1. If a Class I PJP is desired, the PQR shall be qualified in accordance with 5.3.1.1 in addition to the requirements of this paragraph.

5.3.1.6.2 PJP and flare bevel groove welds when a CJP PQR is not present. If a supporting CJP was not performed, then the PJP welds shall be qualified in accordance with 5.3.1.2. In addition three macrographs shall be required for each groove weld type to validate sufficient penetration depth, weld size, and proper fusion has been achieved in the PJP configuration. Macrographs shall be performed prior to preparation for mechanical test specimens. Only then shall the root side of the PJP be machined to the minimum achieved weld size and the required tension, bend, and other test specimens be prepared and tested, as applicable in accordance with 5.7. If a Class I PJP is desired, the PQR shall be qualified in accordance with 5.3.1.1 in addition to the requirements of this paragraph.

5.3.1.6.3 Fillet welds. Fillet weld qualification shall require both single pass and multiple pass fillet welds to determine maximum single pass weld size and minimum multipass weld size in accordance with table IX. Both macrograph and fillet weld break tests shall be required in accordance with table VI to validate root penetration, root fusion, leg fusion, weld leg size, and actual throat achieved in the fillet configuration.

5.3.1.7 Multiple welding processes used in PQR. A PQR test weld assembly that requires multiple passes by way of using different welding processes shall have all essential variables of each process listed in the PQR data sheets. If multiple processes are intended for use

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in production of a single joint application, then all of the processes shall be qualified in a single PQR test assembly. The deposition thickness (t) of each process to be qualified in a single joint shall be based on the thickness deposited with each process, and qualified thickness for all processes combined in one joint shall be based on the PQR test assembly thickness (T) (see table X). A single process may be utilized from this type of qualified procedure, provided that all limitations of that process are observed. However, the root layer(s) up to a thickness of 3/16 inch shall be deposited with the same process used in the qualification of the root layer(s).

Each process shall be included in all destructive test specimens. Where specimens are too small to include representative weld metal from all processes (e.g., Charpy specimens), all required test specimens shall be taken from weld metal from each process. An exception shall be made for a process subject to the following limitations:

a. The weld process involves no more than two layers in qualification and production welds (e.g., the root plus one layer, etc.), and

b. The test specimens involved cannot be removed so as to include this limited weld region.

Example – when welding a 1” thick PQR test plate (T), GTAW was used for the root with GMAW balance used for the remaining fill and cover passes. 0.25” (t) GTAW and 0.75” (t) GMAW were deposited. Therefore 0.5” (2t) for GTAW and 1.5” (2t) for GMAW are qualified (see table X). However, GMAW is not qualified to weld the root of the joint because it was not qualified in the PQR for root pass welding.

5.3.1.8 PQR requirements for 120ksi and greater wires. Filler materials of 120ksi and greater strength shall require two mechanical test plates: one slow-cooling plate representing the maximum heat input and highest preheat/interpass temperature on the thinnest section and one fast-cooling plate produced with the lowest heat input and lowest preheat/interpass on the thickest section (not to exceed 2 inches) allowed by the procedure. Preheat/interpass temperature and specific welding parameters, including arc voltage, welding current, wire feed speed (for all GMAW and SAW), and travel speed, shall be recorded for each pass deposited. The following steps shall be observed:

a. A postheating between the minimum preheat and the maximum interpass temperature for a minimum hold time of 4 hours

b. and a final hold time of 24 hours following the completion of welding is required prior to destructive testing (see table VI).

This shall be performed during the procedure qualification development as it would then be indicative of the production welding processing activities. Flame cutting shall not be used for the removal of test coupons.

5.3.2 PQR cross qualification. Table XI may be used to extend procedure qualifications to base material combinations other than those specifically welded and tested. The essential variables (see table V) for welding each base material and filler material combination shall be

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incorporated into the welding procedure and be in accordance with the requirements of this document.

5.3.3 PQR revision. Revision of PQRs shall be prohibited, except for minor spelling and grammatical errors and the repair of H-plates in accordance with A.7.4.

5.3.4 Supplemental PQRs. A supplemental PQR is a PQR that is developed to support or expand the parameters of a WPS. Supplemental PQRs shall be allowed to be created and incorporated as additional supporting PQRs to an existing qualified WPS for all classes of welds. To create a supplemental PQR, a test plate shall be created in accordance with the requirements of 5.3.1 and its applicable subparagraphs. The welded supplemental test plate shall then be subjected to all of the required nondestructive testing (ses 5.9) and the lacking destructive testing (see 5.7) as required to obtain the qualification. Dependent on the weld classification a PPIT (see A.1) may also be required to obtain qualification. All required test results shall be listed on the supplemental PQR. A supplemental PQR may include the following items absent from the original PQR:

a. Qualification data that is required by this document such as destructive testing results as required by 5.3, 5.4, table VI, and the required supporting documentation.

b. For upgrading to Class I welds, PPIT results performed in accordance with A.1 and the required supporting documentation.

c. All essential variables listed within this standard not present on the original PQR (see 5.3, 5.4, and table V).

A supplemental PQR may allow for a previously qualified WPSs of Class III and Class IV welds to now attain Class I qualification. Previously qualified procedures are addressed in 5.3.5. For procedures where proofing projectile impact testing data is available, but exceptions were granted from the original qualification document, a supplemental PQR shall be developed to attain qualification in accordance with this document.

5.3.5 Previously qualified welding procedures. The previously qualified welding procedure shall meet all applicable essential variables of the original qualification document. If the previously qualified welding procedure essential variables do not meet the requirements of the original qualification document, or any of the other requirements of 4.4.or 4.4.1, then a new PQR (see 5.3.4) shall be created, in accordance with this document. If the essential variables are met but complete testing results are not available, a new or supplemental PQR shall be created in accordance with this document.

5.3.6 Bead width limitation for weaving. To qualify the use of a weaving technique, a PQR using a weaving technique shall be generated. The maximum weave width to be used in the WPS shall be limited to the maximum weave width qualified in the PQR test plate.

5.3.7 Base material repair qualification.

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5.3.7.1 Armor base material repair applications. For repair qualifications of armor base material, a simulated repair plate shall be prepared using a qualified WPS for each position of welding in which the procedure application is intended. The dimensions of the plate shall be 24 in. x 24 in., at a minimum, but shall be of sufficient size to allow for a minimum of 5 simulated repairs. The thickness of the plate shall be of the same thickness as the intended repair for production application. In addition, the condition of the final repair weld surface should be the same as that intended for production application, i.e., as contoured by welding or mechanical finishing. The simulated repair plate shall then be subjected to ballistic testing in accordance with the applicable MIL-DTL ballistic tests section to determine compliance with the requirements of the minimum required ballistic limits. The simulated repair shall meet or exceed the required ballistic limits of the MIL-DTL to be qualified as an armor base material repair procedure. All repair qualifications of cast homogeneous armor shall be in accordance with MIL-A-11356. Procuring contract officer approval of the listed qualification documents and a proposed detailed repair procedure (see 5.12.2) shall constitute approval for repair.

5.3.7.2 Non-armor base material repair applications. A CJP groove welding procedure qualification involving the same base material group or groups to be repaired, shall constitute approval for repair and build-up. The deposition thickness qualified in the WPS shall limit the maximum qualified thickness to be deposited for the repair. The filler material to be used for the repair shall match the mechanical properties of the base material to be repaired. Under matching filler material is prohibited.

5.3.8 PQR record retention. All PQRs shall be identified in a manner to allow permanent traceability. PQRs shall be retained as long as the PQR is applicable. Documentation of the test results detailed in the procedure qualification shall be kept by the entity as long as the PQR is applicable. All PQRs and testing data shall be delivered to the procuring contract officer upon request.

5.4 Essential variables.

5.4.1 Essential variables. This section describes the essential variables for PQRs and WPSs. The PQR essential variables are detailed in this section and table V. Each applicable variable shall be determined for each class of weld, recorded as a single variable on the PQR data sheet for each weld pass, and meet the requirements of this section and table V. See 5.3 for further details on PQR development. The WPS essential variables and their ranges are detailed in this section and table V. These essential variable ranges should be applied to the WPS (see 5.5).

5.4.2 Base material essential variables.

5.4.2.1 Base material specification for armor. A change in the armor grade steel shall be cause for requalification. This includes a change from one armor base material qualified under a military specification to another armor base material qualified under a different military specification, such as a change from MIL-DTL-12560 to MIL-DTL-46100. This also includes a change from a material qualified under a military specification to an armor that has been approved by the procuring contract officer (see 5.1.2). Material groupings for welding procedures in accordance with 5.1.3 and table I shall be observed.

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5.4.2.2 Base material specification for non-armor. A change in base material specification from one group to another base material group in accordance with the document used for qualification (see 5.3.1.5) shall require requalification. Any ungrouped base material specification shall be qualified separately. Materials may be submitted (see foreward 5 for contact information) for consideration for addition to table I. For base material groupings, see 5.3.2 and table I.

5.4.2.3 Base material thickness (T). A change in base material thickness outside the limits for which the procedure was qualified shall be a cause for requalification (see table V). For Class I qualification limitations, see the requirements in table A-I. For Class II, III, and IV qualification limitations, PJP, and flare bevel welds see the requirements of table X. For all fillet welds qualified in accordance with 5.3.1.6, use table IX for thickness limitations.

5.4.2.4 Deposition thickness (t). The thicknesses of each process, transfer mode, or filler material classification used shall be recorded in the PQR. A change exceeding the qualified maximum weld deposit thickness is a cause for requalification.

5.4.2.5 Backing material. A change of material group (see table I) of the backing material shall be cause for requalification.

5.4.3 Filler material essential variables.

5.4.3.1 Electrode classification. A change in electrode classification from one classification to another, except as permitted by table IV, shall be cause for requalification. A change in the chemical composition of the electrode, except as permitted by table IV, shall be cause for requalification.

5.4.3.1.1 Change in 120ksi or greater electrode. Filler materials of 120 ksi and greater shall be qualified under a military specification. A change from one manufacturer or trade name to another shall be cause for requalification.

5.4.3.2 Low hydrogen electrodes. A change from low hydrogen electrodes to an electrode without a low hydrogen designation shall be cause for requalification. A decrease in H number of electrode classification shall be cause for requalification.

5.4.3.3 Number of electrodes. Any change in the number of electrodes used shall be cause for requalification. This includes autogenous welds, made without filler material.

5.4.3.4 Cold wire and hot wire. A change from cold wire to hot wire or vice versa shall be cause for requalification.

5.4.3.5 Electrode diameter. A change in electrode diameter outside of the following limitations:

a. For GMAW or MCAW, any increase or decrease

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b. For FCAW, any increase

c. For GTAW, any increase or decrease of 1/16 inches (1.6mm)

d. For SMAW, any increase of 1/32 inches (0.8 mm)

e. For SAW, any increase of 1/32 inches (0.8 mm)

5.4.3.6 Powder or granular filler material. The change in the amount of powder or granular filler material greater than 10% shall be cause for requalification. If the alloy content of the weld metal is largely dependent on supplemental powered filler metal, any WPS change that results in a weld deposit with the important alloying elements not meeting the WPS chemical composition requirements shall be cause for requalification.

5.4.4 Shielding.

5.4.4.1 Gas. A change from no gas to any gas or vice versa shall be cause for requalification

5.4.4.1.1 Flow rate. A change in gas flow rate with an increase of 50% or a decrease of 20% shall be cause for requalification.

5.4.4.1.2 Composition. A change from one single gas to another single gas or to a gas mixture shall be cause for requalification. Any change in the specified nominal percentage of a gas mixture shall be in accordance with AWS A5.32M/A5.32. A change in gas mixture outside of AWS A5.32M/A5.32 shall be cause for requalification.

5.4.4.2 Flux. A change in flux type used in qualification to another flux type is a cause for requalification. A change in flux-electrode classification to another flux-electrode classification is a cause for requalification. A change from neutral to active flux in multilayer welds, or vice versa, is a cause for requalification.

5.4.5 Joint design.

5.4.5.1 Joint design limitations for Class I welds. Qualification of a Class I welding procedure to Appendix A shall allow for PJP welds without full requalification, as long as all the provision of 5.3.1.6 are met. Qualification using a CJP butt joint design qualifies all standard joint design tolerances to be used in fabrication under MIL-STD-22.

A change in groove type (a V-groove to a U-groove, or double-V to double-U, or single-bevel to double bevel) does not require full requalification, as long as all the provisions of 5.3.1.6 are met. However, technique and joint accessibility with respect to weld quality for different joint design configurations should be considered.

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A change made from a weld qualification made with backing (single sided) or with backgouging (two-sided) to an open root single sided weld (with no backing or no backgouging) shall require requalification.

5.4.5.2 Joint design limitations for Class II, Class III and Class IV. Qualification using a CJP butt joint design qualifies all standard joint designs of MIL-STD-22, excluding the requirements of 5.3.1.6. For all Class II and Class III procedures, a change made from a weld qualification made with backing (single sided) or with backgouging (two-sided) to an open root single sided weld (with no backing or no backgouging) shall require requalification.

5.4.5.3 Joint soundness. A change from PJP welds to CJP or CJP to PJP welds shall be a cause for requalification (see 5.3.1.6). If a joint design used in production inhibits accessibility or has repeated weld quality issues related to fusion, then it should be changed to a more conservative design that allows for accessibility and ease of weld deposit. Any joint design change shall comply with the requirements of 4.6.1 and 5.4.5.1 for Class I welds or 5.4.5.2 for Class II, Class III, and Class IV welds.

5.4.5.4 Joint details for PJP and flare bevels. Any of the following shall be cause for requalification:

a. Any reduction in the root opening from that qualified shall be cause for requalification.

b. Any decrease in bevel or groove angle from that qualified shall be cause for requalification.

c. Any increase in depth of groove and resultant increased weld size from that qualified shall be cause for requalification as well.

d. Any reduction below the qualified radius of flare bevel joints shall be cause for requalification

Therefore, any increase in the root opening and bevel or groove angles, along with any lesser joint thickness or depth of bevel below that qualified shall not be cause for requalification.

5.4.5.5 Backing or spacer strip. The exclusion of, but not inclusion of the backing shall be cause for requalification. A change made from a weld made with backing (single sided) or backgouging (double sided) to an open root weld (single sided or double sided) excluding backing or backgouging shall be cause for requalification.

5.4.5.5.1 Restrictions for using adhesive tape with ceramic backing materials. Unless specifically approved by the technical authority, the use of adhesive coated tape to hold ceramic backing material in place is prohibited when welding with MIL-140 series filler materials, and shall not be applied closer than 1/2 inch from the bevel edge or weld zone, at the time of welding when welding with MIL-120 series filler materials.

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5.4.5.6 Square groove limitation. A change from any groove type to a square groove and vice versa shall be cause for requalification.

5.4.6 Electrical parameters.

5.4.6.1 Amperage. A change outside of the following limitations or a change that causes heat input to exceed the as qualified heat input limitation shall be cause for requalification:

a. For GMAW, FCAW, and SAW, an increase or decrease of 10%

b. For GTAW and SMAW, an increase or decrease of 25%

5.4.6.2 Voltage. A change outside of the following limitations or a change that causes heat input to exceed the as qualified heat input limitation shall be cause for requalification:

a. For GMAW, an increase or decrease of 7%

b. For FCAW, an increase or decrease of 7%

c. For SAW, and increase or decrease of 7%

5.4.6.3 Travel speed. A change outside the following limitations or a change that causes heat input to exceed the as qualified heat input limitation shall be cause for requalification:

a. For GMAW and FCAW, an increase or decrease of 25%

b. For GTAW and SMAW, an increase of decrease of 50%

c. For SAW, an increase or decrease of 7%

5.4.6.4 Heat input. A change in heat input outside the as-qualified maximum value qualified in the PQR, shall be cause for requalification. The WPS electrical parameter ranges, and the high heat input values shall be limited to that which was qualified in the supporting PQR(s). Creation and implementation of a minimum travel speed table should be listed in the entities WPS (see 5.5.3). Due to the effects of heat input on strength, elongation, toughness, and hardness in the heat affected zone, heat input shall be limited to the maximum value qualified in the PQR. See 5.10.6 for heat input calculations.

5.4.6.4.1 Bead width limitation for weaving. Exceeding the maximum weave width qualified in accordance with 5.3.6 shall be cause for requalification. Maximum heat input governs bead width for weaving, if maximum heat input is exceeded, it shall be a cause for requalification.

5.4.6.4.2 Heat input requirements for 120ksi or greater filler materials. Filler materials of 120ksi and greater shall have a minimum and a maximum heat input for production, to be defined by the two PQR plates (see 5.3.1.8). Guidance for heat input is presented in table XII.

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Heat inputs should be maintained within ±2 kJ/in of the target heat input for each weld pass for GMAW or SAW processes and ±4 kJ/in for each SMAW weld pass. If a weld pass or bead falls outside the acceptable heat input tolerance, it should be removed by grinding and recorded as such.

5.4.6.5 Current type. A change from alternating current (AC) to direct current (DC) or vice versa shall be cause for requalification.

5.4.6.6 Polarity type. A change from direct current electrode negative (DCEN) to direct current electrode positive (DCEP) or vice versa shall be cause for requalification.

5.4.6.7 Output type. A change from Constant Current (CC) to Constant Voltage (CV) or vice versa shall be cause for requalification.

5.4.6.8 Pulsed current. A change from pulsed current waveform to CC/CV waveform (non-pulsed) or vice versa shall be cause for requalification.

5.4.6.9 Transfer mode. A change in transfer mode shall be cause for requalification.

5.4.6.10 Tungsten electrode. A change in electrode classification shall be cause for requalification. A change in diameter outside the allowable range of table V shall be cause for requalification.

5.4.7 General procedure parameters.

5.4.7.1 Welding process. A change in welding process shall be cause for requalification.

5.4.7.1.1 Multiple welding processes used in PQR. A change in excess from what is allowed in 5.3.1.7 shall be cause for requalification. A root layer welded with any process not qualified to weld the root layer, as described in 5.3.1.7, shall be cause for requalification.

5.4.7.2 Welding method of application. A change in method of welding process application shall be a cause for requalification. A change between a manual or semiautomatic process to a mechanized, automatic, and robotic process application method shall be cause for requalification. A change in method of application for a previously qualified Class I procedure from manual or semiautomatic to mechanized, automatic or robotic shall be treated as a procedure transfer (see 5.5.4.2) and require validation (see 5.5.4.2.1). See AWS A3.0M/A3.0 for definitions and “Method of Applying Welding Processes” Figure in AWS WHB-1 for further clarity of definitions.

5.4.7.2.1 Manual and semiautomatic applications. A change from a manual application to a semiautomatic or vice versa shall be cause for requalification.

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5.4.7.2.2 Mechanized, automatic, and robotic. Qualification using a mechanized, automatic, or robotic process application shall qualify the use of mechanized, automatic, and robotic process applications.

5.4.7.2.2.1 Adaptive control. Applications that use an adaptive control process shall be considered unique and shall each be qualified separately. The addition of, subtraction of, or change in adaptive controls shall be cause for requalification.

5.4.7.3 Position. A change in position exceeding the allowances of table VII or table VIII shall be cause for requalification.

5.4.7.4 Preheat temperature. A change in preheat temperature outside of the limitations in table XIII shall be cause for requalification. Alternative as-qualified lower temperatures may be considered with technical authority approval provided supporting data is submitted for review of acceptability. The WPS preheat temperature shall in no case be reduced below the as-qualified minimum PQR preheat temperature. Level of restraint during fabrication and in-service should always be considered. Suggested examples of alternative weldability testing are outlined in the Weldability Testing section of AWS B4.0, AWS D14.8M, or AWS D1.1/D1.1M Annex B.

5.4.7.5 Interpass temperature. A change in interpass temperature outside of the limitations in table XIII shall be cause for requalification.

5.4.7.6 Mulitpass and single pass welds. A change from a multiple pass weld to a single pass weld, or vice versa, shall be cause for requalification. A change outside the limits of the maximum single pass weld size, or the minimum multipass weld size shall be cause for requalification.

5.4.7.7 Backgouging. The exclusion of, but not inclusion of backgouging shall be a cause for requalification. A change made from a weld made including backing (single sided) or backgouging (double sided) to an open root (single sided or double sided) weld excluding backing or backgouging shall be cause for requalification.

5.4.7.7.1 Method of backgouging. A change from mechanical backgouging to thermal backgouging shall be cause for requalification.

5.4.7.8 Post Weld Heat Treatments (PWHT). The addition of or change to PWHT, change of cooling rate above 600 °F by ± 90 °F (50 °C) per hour in the thickest section, or temperature of post weld heat treatment process by ± 90 °F (50 °C) in the thickest section, or duration by ±0.5 hours/in shall be cause for requalification.

5.4.7.9 Cleaning method. A change from mechanical cleaning to chemical cleaning, or vice versa shall be cause for requalification.

5.5 Welding Procedure Specification (WPS).

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5.5.1 Requirements. Prior to any fabrication, a qualified PQR (see 5.3) shall be used to generate a welding procedure specification (WPS) in accordance with this document. The WPS shall be delivered to the procuring contract officer for approval upon request (see 6.2). The values listed on the WPS shall be determined by the allowable ranges in section 5.4 using the actual value recorded from the PQR as the reference point. Any change to an essential variable that falls outside the WPS ranges shall require a new PQR to create or to supplement a WPS. WPSs shall be signed and dated by the entity responsible representative stating that the WPS and supporting documentation conform to the requirements of this standard.

5.5.1.1 Supporting PQRs used for WPSs. A WPS may have more than one supporting PQR. One PQR or multiple PQRs may support more than one WPS. Supporting PQRs shall be based on weldability and compatibility to the original qualification intent and observe the limitation of each essential variable when growing the WPS ranges for applicability.

5.5.2 WPS essential variables. All WPS essential variables shall be in accordance with 5.5.1 and table V. Any change to an essential variable in excess of the WPS ranges listed in 5.4 shall require requalification.

5.5.3 Minimum travel speed tables. It is recommended that minimum travel speed tables be outlined within the WPS. This table should display the amperage vs. the voltage values, resulting in the minimum travel speed that must be met to not exceed the maximum heat input value for each amperage and voltage settings. Table XIV shows an example of a minimum travel speed table.

5.5.4 WPS transfer requirements.

5.5.4.1 Procedure transfer within an entity. Transfer of qualified procedures between sites of an entity shall be permissible providing the requirements stated in the entity's quality assurance manual and this document are met. The entity shall notify the procuring contract officer of the transfer prior to use at the entity's new site. Field service welding repair (see 3.11) activities shall be considered as the same entity, provided that both 1) personnel performing the work shall be from the same entity and 2) shall be guided by the same quality assurance manual as the original work was performed under.

5.5.4.2 Transferring of Class I WPS from an entity to a sub-entity. A WPS for Class I welds shall only be transferred from an entity to their sub-entity under an active contract. The contracted sub-entity shall be known hence forth as "the transferee". The WPS and its supporting PQR(s) including all test results shall be provided to the transferee. The transferee shall pass a validation test (see 5.5.4.2.1) before using that WPS. The WPS, its supporting PQR(s), the validation PQR, and the test results shall be submitted to the procuring contract officer prior to use (see 4.1). A WPS that is transferred shall meet the following requirements:

a. All required testing has been performed in accordance with table VI, 5.7, and 5.9.

b. A successful PPIT has been performed in accordance with the requirements of Appendix A.

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c. All the required nondestructive and destructive testing results are listed on the original PQR(s) and any applicable supplemental PQR(s) for the desired WPS to be transferred.

d. All essential variables of this standard are met and recorded in the original PQR, any applicable supplemental PQR(s), and the Class I WPS to be transferred.

NOTE: If the original PQR for the Class I WPS to be transferred is missing any of these requirements, a supplemental PQR shall be prepared in accordance with 5.3.4 by the original entity.

5.5.4.2.1 Validation of a WPS to be transferred. The transferee shall prepare a validation test plate within the welding parameters recorded on the original PQR(s) and any applicable supplemental PQR(s), within +/- 3 kJ/inch of the original heat input. The welded validation plate shall then be subjected to all of the required nondestructive (see 5.9) and destructive testing (see table VI and 5.7) required by this document, excluding the PPIT. All destructive test results shall be not less than 5% below the results recorded for the destructive test results shown on the original PQR(s) and any applicable supplemental PQR(s). All the validation testing results shall be listed on the validation PQR, which shall be differentiated from the original PQR and any applicable supplemental PQR(s).

The validated Class I WPS shall be certified by the transferee entity responsible representative. The validated WPS shall be permanently marked, including the addition of the validation PQR, and shall be uniquely identified for traceability. The validation PQR shall include its test results, along with the names, signatures, and dates of those involved in the validation testing process. The transferee validated WPS and all supporting documentation shall be delivered to the procuring contract officer upon request, see 5.5.4.2.

5.5.4.3 Class II, III and IV procedure transfers. Transfer of any procedure between any entities, including sub-entities, for Class II, III, or IV welds shall be prohibited. It is the responsibility of each entity to qualify all Class II, III, and IV welds in accordance with this document.

5.5.5 Contracted sub-entity qualification. It shall be the responsibility of the entity to ensure that their contracted sub-entities have their own qualified procedures that meet all the requirements of this document. For WPS transfer to a contracted sub-entity, see 5.5.4. A failure at the contracted sub-entity level for welding related issues that indicate significant quality concerns may include: Corrective Action Request (CAR) producing a Major or Critical Defect, systemic weld failures, or welding or welder negligence. These types of issues will require Government audits of the entity's welding quality system to determine conformance to this document or the contract. This could include but is not limited to: welding supplier quality audits, qualification approval process audits, or related inspection audits.

5.5.6 WPS revision. Any changes made to a WPS shall be documented on a revised WPS, then authorized and dated by the entity responsible representative. The entity shall notify

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the procuring contract officer that a change was made. The revised WPS and any supporting documentation shall be delivered to the PCO upon request (see 6.2).

5.5.7 WPS record retention. All WPSs shall be identified in a manner to allow permanent traceability. WPSs shall be retained as long as the WPS is applicable. Documentation of the supporting PQRs, and test results detailed in the welding procedure shall be kept by the entity as long as the WPS and the supporting PQR(s) are applicable. All WPSs, PQRs, and associated testing data shall be delivered to the procuring contract officer upon request.

5.6 Welder performance qualification.

5.6.1 Requirements. The type and number of destructive test specimens required for performance qualification by both weld class and type are identified in table XV. All completed welds shall pass visual testing (VT) in accordance with 5.9.2.1 and table XX. Each entity shall establish that each welder and welding operator to be employed for welding has been qualified by demonstrating their ability to produce welds which meet the requirements of this document. The entity shall establish and maintain valid qualifications for each welder or welding operator, in accordance with this document, prior to assigning any welder or welding operator to weld any contract deliverable component(s).

5.6.2 Welder Qualification Records (WQR). The entity shall document the results of the welder or welding operator qualification test in a WQR for every welder or welding operator. WQRs shall be delivered to the procuring contract officer for review and approval upon request. WQRs shall be maintained for the entirety of the welder or welding operator's term of employment, or period of performance of the contract, whichever is greater. A WQR example is shown in figure B-3. The WQR shall include, at a minimum:

a. Entity name.

b. Welder or welding operator identification.

c. Date of test.

d. WPS used for WQR.

e. Essential variables listed in table XVIII.

f. NDT inspector information.

g. Results of applicable qualification test(s).

h. Entity responsible representative signature.

5.6.3 Welder and welding operator performance qualification. A qualified WPS (see 5.5) shall be used for each welder or welding operator qualification plate. Qualifying as a welder shall not qualify the individual as a welding operator or vice versa. A welder's performance

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qualification is measured in their ability to create a test plate which meets the acceptance criteria of this standard. A welding operator's performance qualification is measured in their ability to successfully operate the welding equipment and produce a test plate which meets the acceptance criteria of this standard. A person tending a robotic work cell loading and unloading parts shall not require qualification. Robotic qualification is required for the Programmer (operator) who has set up, input the arc motion and work motion devices, torch path, and all associated welding parameters into the robot controller teach pendant and specific weldment part program. If the duties of the robot tender include additional responsibilities such as visual inspection and process monitoring, the robot tender shall have appropriate training for that task.

5.6.3.1 Class I welds. Prior to welding a Class I weld, the welder or welding operator shall qualify in accordance with this document. All required qualification test plates shall be fabricated in accordance with the test plate assemblies shown in figure 6 and figure 7 and shall be complete joint penetration welds (CJP). Both test plates shall utilize back gouging to be considered CJP and be required for qualification. Qualification to figure 6 shall qualify the welder or welding operator for 0.125-in to 0.75-in and qualification to figure 7 shall qualify the welder or welding operator for 0.125-in to unlimited thickness. Welders or welding operators that qualify to figure 7 shall also be qualified to figure 6 and shall meet the requirements of 5.6.3.2. The usage of temporary backing material shall be permitted during qualification. Qualification positions shall be in accordance with table XVI. All completed test plates shall pass a volumetric inspection (see 5.9) or destructive testing in accordance with table XV and 5.7.

5.6.3.2 Class II, III, and IV welds. Prior to welding a Class II, III, or IV welds, the welder or welding operator shall qualify in accordance with this document. All required qualification test plates shall be fabricated in accordance with the six test plate assemblies shown in figure 8 through figure 13, based on type and position of qualification. These joint designs, or similar joint designs found in MIL-STD-22, or those found in the qualified WPS for which the welder is testing to, may all be used as-is or in combination to facilitate the typical encountered joints and practices observed in production. Test plates shall utilize backing or backgouging to be considered CJP and be required for qualification. All backing shall be removed prior to testing, except permanent backing 3 inch (76mm) wide or greater may remain during NDT as applicable. Qualification positions shall be in accordance with table XVI and table XVII. Qualification of Class I WQR shall qualify the welder or welding operator to perform all classes of welding. However a Class II, or III, or IV WQR shall only qualify the welder or welding operator to perform Class II, III, or IV welds.

All groove and fillet welds shall be destructively tested in accordance with the requirements of table XV and 5.7. Volumetric inspection may be performed in place of bend testing. (see 5.9) Permanent backing material shall be removed only after volumetric inspection, or before destructive testing. CJP groove welds shall qualify plug and slot welds. For plug and slot welds only (see figure 11), macroetch for plug and slot weld qualification shall have no cracks, and shall have sound fusion to the backing and joint root.

5.6.3.2.1 External qualification for welding non-armor grade steel. If welding class II, III, or IV welds that involve non-armor to non-armor combinations, the qualification shall be in accordance with an applicable qualification document listed in 5.3.1.5. All qualification test

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plates shall be fabricated in accordance with the test plate assemblies listed in the applicable qualification document. A welder or welding operator qualified to a document listed in 5.3.1.5 shall not be considered qualified to weld in accordance with this document unless the requirements of 5.6.9 are also met.

5.6.3.3 Welder and welding operator performance qualification using 120ksi or greater filler material. For procedures using MIL-120 or MIL-140 series filler materials, the following additional welder and welding operator performance qualification requirements apply:

a. Initial performance qualification is required using the same filler material type to be used in production (e.g., MIL-120S, MIL-12018 or MIL-140S, as applicable), regardless of welder qualification status with other A5 filler materials, shall be done in accordance with 5.6.3.1 or 5.6.3.2 respectively. Welder performance qualification using MIL-120S (for GMAW) shall qualify MIL-140S and vice versa. Welder performance qualification using MIL-12018 (for SMAW) shall qualify MIL-14018 and vice versa.

b. Welders and welding operators shall undergo an established training program and demonstrate by examination their understanding of all aspects of the welding procedure and techniques including preheat/interpass temperature control, heat input, and hydrogen removal methods prior to performing production welding.

5.6.4 Welder and welding operator essential variables. Each welder or welding operator qualification shall include the information from table XVIII for each qualification required. The applicable information from table XVIII shall be included in the WQR.

5.6.4.1 Welder and welding operator qualification base material. When destructive testing is used for evaluation, the base material shall be compatible to the filler material with regard to weldability and comparable strength. When volumetric inspection is used in lieu of destructive testing, the base material may be carbon steel. For NDT requirements (see 5.9).

5.6.4.2 Welder and welding operator qualification filler metal requirements. Filler metal used in a welder or welding operator qualification shall be in conformance with table IV. Welder and welding operator qualification within the filler metal categories shown in table XIX may be applied for welding similar, as well as dissimilar base metal combinations with qualified welding procedures. Any filler material not listed in table IV may be used if all procedure qualification criteria is met in accordance with this standard.

5.6.4.2.1 Qualification filler materials performed to external documents. For applicable filler materials related to welder and welding operator qualifications performed outside of this document (see table IV). If the filler material used to qualify the welder or welding operator is listed in table IV, then see table XIX for additional filler material groups that the welder or welding operator would be qualified to use. If the filler material used for qualification is not in accordance with table IV or 5.2.2.1, then the welder or welding operator shall requalify.

5.6.4.3 Welder and welding operator qualification positions. Each welder and welding operator shall be qualified in the position or positions (see table XVI and table XVII) which shall

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be welded in accordance with the applicable WPS (see 5.5). Welder and welding operators who qualify in one or more positions shall be considered qualified in other positions in accordance with (see table XVI and table XVII). For detailed descriptions of positions, see AWS A3.0M/A3.0.

5.6.4.3.1 Qualification positions performed to external documents. Any welder or welding operator qualified to an applicable qualification document shall use table XVI and table XVII for the applicable position or positions qualified in accordance with this document.

5.6.5 Qualifying welder or welding operator through PQR. Any welder or welding operator that performs the welding of a PQR test plate, which passes all the qualification requirements, shall be qualified as a welder or welding operator in accordance with this document. All the applicable essential variables listed in table XVIII shall apply as limitations for this type of welder qualification and shall be recorded on the WQR.

5.6.6 Previous welder or welding operator qualification. Previous qualification tests by welders or welding operators that are properly documented and meet the requirements of 5.6.9 are acceptable (see 4.4) with the approval of the procuring contract officer and technical authority (see 6.2).

5.6.7 Transfer of qualification. Performance qualifications shall not be transferred between entities. Transfer of performance qualification from one entity location to another location of the same entity is not permitted unless the welder's qualification meet the requirements of 5.6.3 and 5.6.9.

5.6.8 Renewal of qualification. The welder or welding operator shall requalify in accordance with this document for the following reasons:

a. Has not welded with the qualified process for a period exceeding six months.

b. A continuity record has not been maintained to provide traceability of process use.

c. As requested by entity or procuring contract officer due to a specific reason to question the abilities, performance, or quality of work of the welder or welding operator.

5.6.9 Maintenance of qualification. It shall be the responsibility of the entity to maintain a traceable continuity record for each welder or welding operator. Evidence of maintenance of qualification shall consist of at least one verification of process use for each qualified welding process during each six month period, and shall be provided to the procuring contract officer upon request. If a continuity record cannot be produced when requested by the PCO or the entity, the welder or welding operator's qualification shall be considered expired and requalification shall be required. This continuity record shall be maintained through the entirety of each welder or welding operator's employment with the entity or the life of the contract, whichever is greater.

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5.6.10 Retest requirements. If a performance qualification test assembly fails to meet the applicable requirements, the initial test failure must be documented on the WQR. Retests may be allowed under the following conditions:

a. Two retest assemblies shall be welded without further training for each test assembly in each position which failed. Both test assemblies must be determined acceptable for the welder to qualify. Or

b. Subsequent retests may be welded for each test assembly that failed to meet the requirements of this document, if the welder or welding operator has had at least 4 hours of documented training or practice designed to correct the causes for previous failures. This training or practice and the initial test failure must be documented on the WQR.

5.7 Destructive testing of welds.

5.7.1 Requirements. The type and number of destructive test specimens required for procedure qualification by both weld class and type are identified in table VI. All destructive testing results shall be included in the PQR or WQR. Entities conducting destructive testing shall be ISO/IEC 17025 accredited, or have an internal quality document for destructive testing. The internal quality document shall include procedures for all equipment used for testing and calibration records, documenting appropriate equipment calibration prior to use and in accordance with the applicable testing standard. If an internal quality assurance document is used, the document shall be delivered to the procuring contract officer for approval prior to use (see 6.2). Any entity that sub-contracts destructive testing shall ensure the test facility complies with this section.

5.7.1.1 Destructive testing personnel qualification. Entities that complete destructive testing internally shall develop a training plan for destructive testing personnel. The training plan shall be made available to the procuring contract officer upon request. Entities that sub-contract destructive testing shall use ISO/IEC 17025 accredited test facilities. Visual inspection of bend, macroetch, PPIT, and fillet weld break test samples must be complete by a certified visual inspector (see 5.8.2.1).

5.7.2 Tension/reduced-section tension testing. Tensile test specimens shall be extracted from the test plate as shown in figure 4. Tensile test samples shall be prepared and tested in accordance with AWS B4.0.

5.7.2.1 Tension/reduced-section tension acceptance criteria. For acceptance of tensile test specimens in non-armor and non-armor to armor welding procedure test coupons, the tensile strength shall be greater than or equal to the specified tensile strength of the weaker base metal. In armor to armor welds, including those steels approved by the procuring contract officer for use as armor (see 5.1.2), or in procedures approved by the PCO using undermatching filler material, the filler material tensile strength shall be used as the acceptance criteria.

5.7.3 Bend testing. Bend testing shall only be performed on welds involving non-armor to non-armor base materials. Bend test samples shall be prepared and tested in accordance with

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AWS B4.0. When guided bend tests are specified in table VI, joints shall be tested by longitudinal or transverse bends. For test plate thickness (T) ≥ 3/8 inch, side bend specimens prepared to 3/8 inch wide should be used in place of face and root bend test specimens. The total specified quantity shall be tested regardless of the type of bend test specimen.

5.7.3.1 Bend testing test acceptance criteria. For acceptance, test specimens after bend testing shall have no single discontinuity greater than 1/8 inches. Cumulative discontinuity length shall not exceed 3/8 inches for all discontinuities greater than 1/32 inches. Cracks occurring on the corners of the specimen during testing shall not be considered unless there is definite evidence that they result from fusion related defects, where then the same 1/8 inches maximum shall apply. Openings in the base metal outside the weld material and heat affected zone shall not be cause for rejection.

5.7.4 Macroetch testing. Cross sections of weldments for macroetch testing shall be taken 1 inch (25.4 mm) or greater from the start or stop of any weld. Macroetch samples shall be prepared and tested in accordance with ASTM E340. Macroetch of the weld cross section shall be recorded as a digital image with a reference scale embedded in the image. The digital image shall be attached to the PQR. See table VI for number of specimens required. The quantity and location of specimens shall be taken equidistant about the weld length. Macroetch samples are required for all PQRs, in accordance with table VI.

5.7.4.1 Macroetch acceptance criteria. For acceptance, the macroetch samples shall meet the requirements of 5.9.2.1.3 and table XX. Additionally, for acceptance, the weld metrics to define weld size shall not be less than the corresponding metric on the applicable fabrication document. If no weld size for PJP welds is specified by the drawing, procedure, or the applicable document, then the weld size shall not be less than the depth of the groove for acceptance.

5.7.5 Hardness testing. Base material hardness testing of armor grade steel shall be in accordance with the relevant armor specification. The welded test plate shall be subjected to hardness surveys in accordance with ASTM E92, with a recommended load of 10kg. Conversions from one hardness scale to another scale shall be in accordance with ASTM E140. Hardness testing for non-armor grade steels shall be in accordance with ASTM E10, ASTM E18 or ASTM E92. Hardness testing report generation and testing locations shall be in accordance with AWS B4.0. Macrograph (scaled photograph) clearly showing the weld metal and Heat-Affected Zone (HAZ) regions shall be produced. Hardness indentions and corresponding values shall be clearly visible in the macrograph.

5.7.5.1 Hardness testing locations. The survey should start in the base material on one side of the joint, continuing through the HAZ region and across the weld metal, through the other HAZ region and into the base material at the opposite side of the joint. A minimum of 3 points shall be taken in the base material region of the specimen, between 5/8 inches (16 mm) and 7/8 inches (22 mm) from the weld interface, taken 0.08 in. (2mm) from the surface, parallel to a line drawn between the toes. A minimum of 3 points shall be taken within the heat affected zone. A parallel traverse should be taken at the root of the weld. For double bevel welds, a third parallel traverse should be taken 0.08 in. (2mm) from the second surface as described above. Examples

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of these testing locations are available in AWS B4.0. A minimum of 15 points per traverse is recommended.

5.7.5.2 Hardness testing sample preparation. Hardness sample preparation shall be in accordance with the ASTM testing standard selected, with a final surface finish adequate for a clear macrograph, as required for the hardness report (see 5.7.5 and ASTM E340). Thermal cutting shall not be used to prepare samples for hardness testing. The hardness traverse may be performed on the reverse of the macroetch sample (see 5.7.4 and figure 4).

5.7.5.3 Hardness testing acceptance criteria. The converted microhardness values obtained from a minimum of 3 points of the base material region of the specimen, taken between 5/8 inches (16 mm) and 7/8 inches (22 mm) from the weld interface, shall be within the correlating hardness value and range as applicable to the base material standard, within a +/-5% margin permitted for measurement errors. A minimum of 3 points taken within the heat affected zone shall be reported to monitor for any reduction in hardness caused from welding. If there is no applicable hardness requirement for the base material standard, hardness of the base material tested prior to any welding operations including preheat, or conversion from ultimate tensile strength shall be permitted.

5.7.6 Fillet weld break test. Fillet weld break test samples shall be prepared and tested in accordance with AWS B4.0. Each specimen tested shall contain a minimum of one weld start and stop.

5.7.6.1 Fillet weld break test acceptance criteria. Fillet weld break test results shall be acceptable if all of the following conditions are met:

a. No evidence of lack of interlayer fusion.

b. No incomplete root fusion.

c. No porosity or inclusions exceeding 0.094 inches (2.4 mm) in any direction.

d. No porosity or inclusions whose largest dimensioned summed does not exceed 1/8 inches (3 mm) in any 2 inch (51 mm) length of weld provided pores and inclusions less than 1/32 inches (0.81 mm) are excluded from the tabulation.

5.7.7 Proofing projectile impact testing. All PPIT for Class I procedure qualifications shall be performed in accordance with Appendix A.

5.7.8 Charpy V Notch (CVN) testing. CVN test data shall be included in the PQRs as required for the class of weld (see table VI), noting fillet weld qualifications shall not require CVN testing. CVN test samples shall be extracted from the test plate as shown in figure 4. Two types of CVN samples shall be extracted from the test plate and are labeled ‘Weld’ and ‘HAZ’. Five samples of each type shall be taken for testing. When qualifying two materials for one PQR, five samples of each HAZ shall be taken for each base material.

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5.7.8.1 CVN test methods. Charpy impact testing shall be conducted in accordance with ASTM E23 with the exception of temperature at the time of test. Specimens shall be removed from the weld metal within 1/16 inch of the surface of the material. The notch shall be normal to the surface of the test coupon. Specimens representing the HAZ shall be removed approximately at a depth midway between the surface and center thickness. The coupons for HAZ impact specimens shall be taken transverse to the axis of the weld and etched to define the HAZ. The notch of the CVN specimen shall be cut normal to the material surface in such a manner as to include as much HAZ as possible in the resulting fracture. If the qualification plate is dissimilar base materials, the HAZ of each side shall be treated separately.

A test shall consist of five specimens, which shall be tested at or below the temperature determined below. The highest and lowest values of these specimens shall be disregarded, and the values of the three remaining specimens shall equal or exceed the impact value as determined below. Alternatively, three specimens may be prepared and tested, and the values of these three specimens shall equal or exceed the impact value determined below.

Test temperature will be determined separately for weld and HAZ samples, and shall be the lowest of the following applicable to that material:

a. That specified in the military filler metal specification.

b. That specified in commercial filler metal specification when there is no applicable military specification.

c. That specified in the base metal specification.

d. -40 °F (-40 °C) when none is defined in any specification.

5.7.8.2 CVN acceptance criteria. For each set of five specimens, three values shall be reported along with the average of those values, dropping the highest and lowest values of the five. The average value from the weld metal shall equal or exceed that specified in the military filler metal specification or, when there is no applicable military specification, the commercial filler metal specification. For the HAZ, the average value shall equal or exceed that of the base metal specification(s). If no minimum applies, the values shall be documented on the PQR for future use to include new procedure development (see 5.3), supplemental PQRs (see 5.3.4) and base material repair qualification (see 5.3.7). If the welded plate uses dissimilar base materials, each heat affected zone shall be treated separately.

5.7.9 Destructive test failures. If a single destructive test specimen, other than CVN, from a qualification plate fails to meet the acceptance criteria, a retest of that plate may be performed. The retest shall include two additional specimens from the same qualification plate for the failed destructive test. The retest shall only be accepted if both additional specimens meet the acceptance criteria. If the original qualification plate lacks material for additional required test specimen, then a supplemental test plate shall be created to using the original parameters, to support additional specimen generation. If any additional test specimen fails to meet the

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acceptance criteria, that qualification attempt shall be deemed unacceptable, no retesting shall be allowed, and a new qualification test plate shall be fabricated.

5.7.10 Destructive testing record retention. All destructive testing records shall be retained for the period of performance of the contract, at a minimum. Destructive testing records shall be delivered to the procuring contract officer upon request.

5.8 Nondestructive Testing (NDT) personnel qualification.

5.8.1 Requirements. Individuals performing any method of NDT shall meet the qualification requirements for the applicable NDT level of certification in accordance with ANSI/ASNT CP-189 or AIA/NAS NAS410, except as listed below. The required NDT level shall be determined by the contractor's NDT written practice (see 5.9.1.1). Alternately, any certification through ASNT Central Certification Program (ACCP) shall be recognized as certified at the NDT level of examination. The certified inspector shall determine compliance of the weld to the requirements of this standard (see 5.9.1). When the results of an inspection process or indications are inconclusive, the weld shall be inspected by another NDT method.

5.8.1.1 Entity responsibility. It is the responsibility of the entity to ensure that all subcontractors are qualified and certified to conduct all nondestructive testing in accordance with this standard.

5.8.2 Visual inspector qualification. Certified visual inspectors (CVI) and authorized visual inspectors (AVI) shall be used for the visual verification of weld quality. All visual inspectors shall undergo a near vision eye exam which shall be administered every three years, unless otherwise specified by the certifying authority for certified visual inspectors. The CVI eye exam shall be administered by a qualified, licensed medical professional, while AVI eye exams may be performed by the entity. Color perception tests are not required nor is failure cause for inspector qualification rejection.

NOTE: A visual inspector is not qualified to perform any other type of NDT unless the requirements of 5.8.3 are met.

5.8.2.1 Certified visual inspectors. Certified visual inspectors (CVI) shall have the authority to sign as certified inspectors in accordance with this document and shall be certified in accordance with at least one of the following:

a. Current certification in accordance with the AWS QC1 for Certified Welding Inspector (CWI) or Senior Certified Welding Inspector (SCWI).

b. Current certified welding inspectors qualified by the Canadian Welding Bureau (CWB) to Level II or the Level III requirements in accordance with CSA W178.2.

c. Current Level II Visual Inspection (VT) certification or greater in accordance with the ANSI/ASNT CP-189.

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5.8.2.2 Authorized visual inspectors (AVI). Welding inspectors qualified in accordance with this method herein may visually examine welds while working under active supervision (see 3.3) with the certified visual inspector, but shall not have the authority to sign as a certified visual inspector on welds. Weld examination approval for visual examination shall be the responsibility of a Certified Visual Inspector, as listed in 5.8.2.1. Visual inspectors qualified to documents other than what is listed in 5.8.2.1 must be trained and qualified by a visual inspector training program that shall be in accordance with AWS B5.1, AWS B5.2, ASNT SNT-TC-1A, or an equivalent document and authorized by the Government procuring contract officer Certified Visual Inspector, see 5.8.2.1.

5.8.2.2.1 Previous CVI certifications. Previous certification in accordance with the 5.8.2.1 precludes the need for the additional training specified in 5.8.2.2 and allows for immediate qualification as an AVI. Those qualified in this manner shall be restricted to within two years following the expiration of their certificates. In cases where the external qualification has lapsed, then the vision test must be passed before being able to perform as an AVI.

5.8.3 Other NDT methods. All NDT personnel shall be qualified and certified in accordance with ANSI/ASNT CP-189 or AIA/NAS NAS410. For NDT methods not covered by ANSI/ASNT CP-189 or AIA/NAS NAS410, personnel shall qualify to comparable levels of competency by the administration of examinations for the particular method involved. A certified NDT Level III inspector shall be responsible for the validity of the examinations and overall personnel qualification and certification (PQ&C) program.

a. Liquid Penetrant Testing (PT)

b. Magnetic Particle Testing (MT)

c. Electromagnetic Testing (ET)

d. Ultrasonic Testing (UT)

e. Radiography Testing (RT)

f. Digital Radiography Testing (DR)

g. Radiograph Interpreter

5.8.4 NDT re-examinations. If the Government procuring contract officer NDT or welding engineering representative has reason to believe that an individual is unable to competently perform at the NDT level that the individual is certified or because of interrupted service the entity shall request a practical or written examination be administered in accordance with ANSI/ASNT CP-189 or AIA/NAS NAS410. If an individual transfers from one entity to another, NDT reexamination shall be required in accordance with the new entity's NDT written practices for personnel qualification and certification. All personnel certified through nationally recognized certification program such as, NAS, AWS, CWB, or ASNT, shall be exempt from reexamination when transferring from one entity to another.

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5.8.5 Alternative NDT specifications and standards. If an alternative inspection program which meets similar requirements of this document is to be used, the alternative program shall be submitted to the procuring contract officer for approval by an applicable Government NDT Level III prior to being used (see 6.2).

5.8.6 Personnel qualification documentation. All personnel qualification documentation shall be maintained for the entire length of employment of the individual or the life of the contract, whichever is greater. The procuring contract officer may conduct reviews of the personnel documentation to assess conformance to this standard at any time.

5.9 Nondestructive testing (NDT).

5.9.1 Nondestructive Testing (NDT) requirements. This section covers the requirements for conducting NDT used in determining the presence of discontinuities. Unless otherwise specified, all NDT shall be performed in accordance with the requirements of this section. This document shall serve as the fabrication document, when a fabrication document is referenced in any other document, unless otherwise required by the technical authority. All classes of welds shall require some form of NDT: see table XXI for inspection method type as required per class of weld. The method of NDT shall be specified in the contract or drawing. When NDT has an indication of a flaw, the process of evaluation shall be in accordance with figure 14.

NOTE: The terms inspection, testing, and examination are commonly used interchangeably in NDT.

5.9.1.1 NDT written instructions. When NDT is performed, written instructions for performing the examination shall be prepared and uniquely identified. The written instructions shall have written certification from an applicable NDT Level III inspector (see 5.8.3). The procuring contract officer may conduct reviews of the NDT written instructions to assess conformance to this standard. All NDT procedures shall be performed in accordance with the NDT written instructions. Typical NDT written instruction should include (but not limited to):

a. Scope, purpose, application.

b. Equipment involved, specifications for equipment, and consumables.

c. Test setup qualification and certification of calibration shall be referenced.

d. Required qualifications of personnel required to perform to said instruction.

e. Reference to external test method procedures, for example ASTM.

f. Details of to which, what, and how the instruction is applied and performed.

g. Acceptance criteria.

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h. Recording and reporting results

5.9.1.2 Hold time prior to NDT. For all non-austenitic high strength steels (see 3.22.2.1), cast homogeneous (see 3.22.1.1), RHA (see 3.22.1.2), and HH (see 3.22.1.3) armor grade steels greater than 0.3" (8mm) in thickness, the final NDT examination for acceptance shall be conducted no less than the following, after the weld has cooled to ambient temperature. When using group A-8 filler metal, there is no hold time required. For all VHA or UHA, the final NDT examination for acceptance shall be conducted no less than the following, after the weld has cooled to ambient temperature, regardless of thickness or filler metal.

a. 24 hours when performing UT,

b. 36 hours when performing RT, or

c. 48 hours when performing VT, PT, or MT

The required visual inspection can be completed immediately prior the completion of UT or RT. Preliminary NDT examinations are advisable to avoid production related delays, by identifying defects not related to delayed hydrogen cracking. Any defects found during the preliminary NDT may be addressed before the hold period is complete, restarting the hold period. Any defects identified and addressed following the first hold period shall be subject to an additional hold period. Alternative strategies to mitigate delayed hydrogen cracking must be supported by data and shall be submitted to the procuring contract officer for review for acceptability prior to amending the length of this hold time.

5.9.1.3 Surface preparation. Weld contours shall not interfere with the NDT process or interpretation of the NDT results. As determined by the responsible Level III NDT inspector, NDT written instructions, or applicable standard, excessive surface irregularities and tightly adhering spatter shall be removed by a suitable mechanical process, such that they will not interfere with the interpretation of the nondestructive test results. All non-permanent backing material shall be removed prior to the performance of any NDT (see 5.1.4).

5.9.1.4 Frequency of NDT examination. All welds shall be visually inspected at a 100% frequency without deviation. Welds to be covered shall be inspected prior to being made inaccessible (see 5.9.2.2.2). A sampling plan shall be developed for all NDT activities in accordance with MIL-STD-1916. The verification level (VL) for Class I and Class II welds shall be VL-VI, Class III welds shall be VL-V and Class IV shall be VL-II. If an alternate acceptance method is requested, the requirements and procedure shall be in accordance with MIL-STD-1916.

5.9.1.5 Inadvertent examination. Any indication that is discovered through inadvertent examination shall follow the NDT evaluation process, see figure 14. Any indication that is interpreted as relevant shall be evaluated for acceptability. If the indication is evaluated as a reject, the defect shall be repaired in accordance with 5.12, and shall be examined after final repair with the same NDT method of the original inadvertent inspection. If the initial examination was performed with RT or UT and the repair does not exceed two weld layers, then

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MT and VT are acceptable methods for examination after the repair. For all high strength steels and armor grade steel, NDT hold time shall apply, see 5.9.1.2. Any repairs shall be recorded in the NDT examination report.

5.9.1.5.1 Inadvertent defects. When an inadvertent examination discovers a rejectable indication in a particular joint, the remainder of that joint shall be examined. Any additional indications shall be interpreted in accordance with 5.9.1.5

5.9.1.6 NDT examination reports. When NDT is performed, an NDT examination report shall be prepared. The NDT examination report shall be made available to the procuring contract officer upon request. The NDT examination reports shall include, at a minimum, the following:

a. Method of examination.

b. NDT written instruction unique identification (see 5.9.1.1).

c. Date and time of inspection.

d. Acceptance criteria used.

e. Weld joint design, classification and unique identification.

f. Welding procedure specification (WPS) identification.

g. Stage of manufacturing (initial inspection, in-process, final, or repair).

h. Test results (accept or reject).

i. Signatures of responsible NDT level II or III inspector.

j. Name of welder or welding operator.

5.9.1.7 NDT symbols. When NDT symbols are used on drawings, the symbols shall be in accordance with AWS A2.4.

5.9.2 Nondestructive Testing (NDT) methods.

5.9.2.1 Visual testing examination (VT). All completed welds shall be visually examined by a visual inspector (see 5.8.2). No unspecified welds shall be added without approval of the procuring contract officer. The visual testing procedure shall contain, as a minimum, the requirements of 5.9.1.6 and the following information:

a. Type of welds or surfaces to be inspected.

b. Specific measuring devices to be used.

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c. Visual aids, reference standards, workmanship standards.

d. List of inspection attributes (visual characteristics).

e. Lighting requirements.

f. Acceptance criteria and classification of defects.

g. Record requirements.

5.9.2.1.1 Visual testing surface illuminance. The surface illuminance shall be 540 lux (lx), or 50 foot-candle (fc), at a minimum. The illuminance source shall be held at a minimum of 6 inches from the piece to be inspected and a maximum distance of 18 inches.

5.9.2.1.2 Visual testing examination access. For direct examination, access to the weld shall be such that the weld inspector's eye shall be within 10-24 inches of the examined surface. The angle of examination shall not be less than 30 degrees from the surface of examination (see figure 15). The usage of cameras, fiber optic cables, mirrors, and borescopes shall be acceptable for visual testing when access in accordance with figure 15 is not possible or when required by the applicable standard or drawing.

5.9.2.1.3 Visual testing acceptance criteria. The acceptance criteria for all welds shall be in accordance with table XX and shall meet the requirements of the applicable fabrication document or material specification.

5.9.2.1.4 Visual testing records. A visual testing record documenting the test results shall be developed for all VT and shall be made available to the procuring contract officer upon request. If a discontinuity is found that requires repair, full traceability and detailed records, reports, photos, as-built drawing mapping of all VT activities shall be undertaken and delivered to the PCO upon request. The entity's CVI or AVI shall make certain only applicable WPSs that are qualified and approved per this document are used and have been documented accordingly. Records of visual inspection shall contain the requirements of 5.9.1.6 as a minimum.

5.9.2.2 Liquid penetrant testing (PT). When required, liquid penetrant testing (PT) shall be used for detecting the presence of surface discontinuities in all classes of weld. PT shall be performed in accordance with ASTM E165/E165M and ASTM E1417/E1417M for all class welds.

5.9.2.2.1 Liquid penetrant testing surface preparation. All surfaces to be examined shall be prepared in accordance with ASTM E1417/E1417M.

5.9.2.2.2 Liquid penetrant testing acceptance criteria. The acceptance criteria for the welds shall be in accordance with table XX.

5.9.2.2.3 Liquid penetrant classification. The penetrant examination process and materials shall be classified in accordance with ASTM E1417/E1417M.

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5.9.2.2.4 Liquid penetrant testing reports. PT examination records shall contain, at a minimum, the requirements of 5.9.1.6 and the penetrant material classification in accordance with 5.9.2.2.3.

5.9.2.3 Magnetic particle testing (MT). When required, magnetic particle testing (MT) shall be used for the detection of surface or near surface discontinuities in ferromagnetic metals. The MT inspection process shall be performed in accordance with ASTM E1444/E1444M or ASTM E3024/E3024M for all class welds.

5.9.2.3.1 Magnetic particle testing surface preparation. All surfaces to be examined shall be prepared in accordance with ASTM E1444/E1444M or ASTM E3024/E3024M.

5.9.2.3.2 Magnetic particle testing acceptance criteria. The acceptance criteria for the welds shall be in accordance with table XX.

5.9.2.3.3 Magnetic particle testing reports. MT examination records shall contain, at a minimum, the requirements of 5.9.1.6 and the instrument manufacturer and model number, or unique equipment identification (yokes excluded).

5.9.2.4 Ultrasonic testing (UT). When required, ultrasonic testing (UT) shall be used for the detection of discontinuities throughout the volume of material, measurement of wall thickness, and evaluation of bond characteristics in basic geometric configurations. When UT is used, the procedure and technique shall be in conformance with AWS D1.1/D1.1M or NAVSEA T9074-AS-GIB-010/271. Procedures developed to NAVSEA T9074-AS-GIB-010/271 shall use an instrument calibration for Class II welds.

5.9.2.4.1 Ultrasonic testing surface preparation. All surfaces to be examined shall be clean and free of dirt, loose scale, loose paint, or other loose foreign matter. Surfaces to be inspected shall have a finish of not greater than 250 Roughness Average (Ra) micro-inches and be free from waviness that may interfere with the test. Welds may be inspected in the as-welded condition, provided the required test sensitivity and inspection coverage can be maintained. The weld reinforcement shall be ground flush to provide a flat surface when ultrasonic inspection is to be accomplished by scanning on the weld surface.

5.9.2.4.2 Ultrasonic testing acceptance criteria. For procedures developed in accordance with AWS D1.1/D1.1M, the acceptance criteria shall be in accordance with the Acceptance criteria for statically loaded nontubular connections section. For procedures developed in accordance with NAVSEA T9074-AS-GIB-010/271 with ultrasonic inspection section of MIL-STD-2035 class 3.

5.9.2.4.3 Ultrasonic testing reports. UT inspection reports shall contain, at a minimum, the requirements of 5.9.1.6 and shall be in accordance with the applicable standard or technical bulletin.

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5.9.2.5 Electromagnetic testing (ET). Electromagnetic testing (ET) shall only be used when required by the drawing, military specification or with approval from the procuring contract officer (see 6.2). ET shall be used for the detection of surface indications. Electromagnetic testing shall be performed in accordance with ASTM E3052.

5.9.2.5.1 Electromagnetic testing surface preparation. All surfaces to be examined shall be prepared in accordance with ASTM E3052.

5.9.2.5.2 Electromagnetic testing acceptance criteria. All discontinuities detected by ET shall be reexamined by MT examination for ferromagnetic materials and by PT examination for nonferromagnetic materials and evaluated in accordance with criteria for that method.

5.9.2.5.3 Electromagnetic testing reports. Reports of ET shall be in accordance with the requirements of 5.9.1.6, ASTM E566 and shall be include with the MT or PT report.

5.9.2.6 Radiographic testing (RT). When required, radiography by X-ray or gamma ray sources shall be used for determining the presence of discontinuities. RT shall be performed in accordance with ASTM E94/E94M and ASTM E1032 or ASTM E1742/E1742M for all class welds.

5.9.2.6.1 Digital radiography. The only digital radiography methods used for examination shall be computed radiography (CR) (Photostimulable Luminescence (PSL) Method), radiography using a digital detector array (DDA) or computed tomography (CT). All CR procedures and examinations shall be performed in accordance with ASTM E2033 and ASTM E2445/E2445M. All procedures and examinations using a DDA shall be performed in accordance with ASTM E2698 and ASTM E2737. All CT procedures and examinations shall be in accordance with ASTM E1441 and ASTM E1570.

5.9.2.6.2 Radiation imaging systems. Ionizing radiation methods other than RT may be used to perform examination of welds. When a radiation imaging system is used to perform examination, the procedures and techniques shall be in accordance with AWS D1.1/D1.1M, Radiation Imaging Systems section.

5.9.2.6.3 Radiographic testing acceptance criteria. Acceptance criteria for RT shall be in accordance with the Acceptance Criteria for Statically Loaded Nontubular Connections section of AWS D1.1/D1.1M.

5.9.2.6.4 Digital radiography acceptance criteria. The acceptance criteria for DR shall be in accordance with the Acceptance Criteria for Statically Loaded Nontubular Connections section of AWS D1.1/D1.1M

5.9.2.6.5 Radiography and Digital imaging reports. RT and DR inspection reports shall contain, at a minimum, the requirements of 5.9.1.6 and shall be in accordance with the applicable standard or technical bulletin.

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5.9.2.6.6 Radiographic film and digital imaging storage. All radiographic film shall be stored in accordance with ASTM E1254. All digital radiography data files shall be in accordance with ASTM E2699.

5.9.2.7 Other NDT methods. If NDT methods, not covered in ANSI/ASNT CP-189, or AIA/NAS NAS410, which meet all the requirements of this document are to be used, the alternative procedure and written practices shall be submitted to the procuring contract officer upon request and reviewed by the applicable Government NDT Level III for approval prior to being used (see 6.2).

5.9.2.7.1 NDT technical manuals. The Tri-services technical manuals (TMs) for NDT, TM 1-1500-366-23 and TM 1-1500-335-23, may be used as a replacement for the requirements in 5.9.2. If the TMs are used, the acceptance criteria shall be agreed upon in the contract documents prior to performing NDT. If the acceptance criteria is not provided in the contract then the acceptance criteria of this standard shall be used for each NDT method. If a new TM is to be developed as NDT written instructions (see 5.9.1.1), then MIL-HDBK-1222 should be used as a guide.

5.9.3 Nondestructive testing record retention. All NDT examination records shall be retained for the period of performance of the contract, at a minimum. NDT records shall be delivered to the procuring contract officer upon request.

5.10 Welding requirements.

5.10.1 Welding environment. Shielding for protection from wind and inclement weather shall be established prior to any welding and shall be maintained until the completed weld has cooled to within 50 °F (10 °C) of the ambient air temperature. Environment temperature differentials of greater than 100 °F (20 °C) along the weld joint during welding and cyclic heating should be avoided to minimize stress differentials within the completed weld.

5.10.1.1 Wind. The maximum wind velocity in the near vicinity of the welding process shall be no greater than 5 miles per hour (8 kilometers per hour).

5.10.1.2 Air temperature. The air temperature in the near vicinity shall not be lower than 60 °F (16 °C).

5.10.1.3 Precipitation. Welding shall not be performed when material surfaces are exposed to any form of precipitation or if the material surfaces are wet.

5.10.1.4 Base material temperature. Welding shall not be performed when material temperature is below 60 °F (16 °C). If the material temperature is below 60 °F (16 °C), welding shall not be performed until one of the two listed practices shall be followed:

a. The material shall be preheated to a minimum of 125 °F or the requirements of table XIII, whichever is greater, and held at that temperature for a minimum of 0.5 hours per inch of thickness, prior to the start of welding. Temperature shall be held below the maximum

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interpass temperature permitted in table XIII. For material thicknesses greater than 1.5 inches, preheat shall be applied and measured on both sides of the material.

b. The material shall be held at a temperature of a minimum of 60 °F (16 °C) for a period of at least 6 hours per inch of thickness, as defined by the total thickness of the stack of plates or the thickest member of the weldment.

5.10.2 Welding Procedure Specification (WPS) accessibility. It shall be the entity's responsibility to ensure that during fabrication, all welders and welding operators shall have ready access to the applicable WPS and understand how to apply the specified parameters of the WPS to fabrication. The entity shall have an established system to ensure that the supervision of any welder or welding operator properly applies the parameters of the applicable WPS to any fabrication done in accordance with this document. This system shall be verified by the entity's welding engineer or quality assurance personnel. Typically the frequency and overall methodology would be defined in an entities auditable Fabrication Operations or Quality procedures. Examples may include a welding variables check log which would correlate verification of measured welding parameters against welding operations for the applicable WPS at hand. Frequency of percentage of workforce checks, days per week, or times per day shall be pragmatic and prudent as fits that entity's ability to have and prove reassurance that WPSs are being adhered to. These weld verification checks as defined in an entity's QMS procedure shall be reviewed the technical authority and approved by the procuring contract officer.

5.10.3 Position. Welding positions shall be in accordance with the qualified position of the applicable WPS. See AWS A3.0M/A3.0 for position descriptions.

5.10.4 Preheat and interpass temperatures. Preheat and interpass temperature monitoring is required and shall be in accordance with table XIII and the WPS. If these temperatures are exceeded, the base material shall be repaired or replaced, see 5.12 for repair limitations. The temperature limits shall be based on the thickness and type of base materials being welded, or the welding filler material being used. Weld joints using group A-8 filler materials are governed by the filler material. All other temperature limits are governed by whichever is more restrictive, the base material or filler material. When temperatures are governed by the filler material, the limit shall apply to all members involved in the joint and shall be based on the joint member with the most restrictive preheat and interpass requirements. When temperatures are governed by the base material, the limits shall be based on the actual thickness and type of the specific material. Welding procedures using preheat and interpass temperatures other than those listed in this section shall require technical authority approval, unless otherwise specified herein.

5.10.4.1 Preheat and interpass calculations.

Example 1:

Governed by filler material; 1.25-inch thick class 1 MIL-DTL-12560 (Member 1) to 1-inch thick class 1 MIL-DTL-46100 (Member 2) using MIL-100S-1 filler material.

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From table XIII, preheat and interpass temperature limits for:
Member 1 – min PH/IP 125 °F, max PH/IP 300 °F
Member 2 – min PH/IP 125 °F, max PH/IP 300 °F
MIL-100S-1 – min PH/IP 150 °F, max PH/IP 300 °F (most restrictive)

The thickness of member 1 requires the filler material preheat and interpass temp to be 150 to 300 °F and therefore the preheat and interpass temperature for the joint shall be 150 to 300 °F.

Example 2:

Governed by base material; 1-inch ASTM A36 (Member 1) to 1½-inch ASTM A514 (Member 2) using MIL-7018-M filler material.
Member 1 - min PH/IP 60 °F
Member 2 – min PH/IP 150 °F, max PH/IP 300 °F (most restrictive preheat and interpass maximum)
MIL-7018-M - not listed

Therefore, preheat and interpass temperature for Member 1 shall be 60 °F minimum and for Member 2, 150 to 300 °F.

5.10.4.2 Preheating and interpass temperature measurement. Preheat temperature shall be checked prior to welding as described in 5.10.4.2.1 and figure 16. Interpass temperature shall be checked prior to initiating another weld pass as described in 5.10.4.2.2 and figure 16. Surveillance of preheat and interpass temperatures shall be in accordance with table XIII. Temperatures of 125 °F and above shall be checked using temperature indicating crayons or other temperature measuring devices. When the ambient temperature is below the required 60 °F, see section 5.10.1.4. If there is evidence of moisture on the material surface, preheat shall be applied until the area is dry and reaches a temperature of 125 °F or greater, depending on the requirements of table XIII. All temperature measuring equipment shall be calibrated or verified to a calibrated meter (see 4.9.1). Temperature indicating crayons shall be applied as to not interfere or contaminate the weld, or contain contaminating elements such as: lead, sulfur, zinc, cadmium, or mercury.

5.10.4.2.1 Preheat temperature measurement. Preheat temperature shall be measured on the surface of the base material on the side from which welding will be performed and 3 inches in all directions from the area to be welded along the joint (see figure 16). For material thicknesses greater than 1.5 inches, preheat shall be applied and measured on both sides of the material.

5.10.4.2.2 Interpass temperature measurement. Interpass temperature shall be measured on the surface of the base material on the side from which welding was performed, within 1 inch of the weld joint edge and along the joint within 3 inches of the start of the next weld pass, or continuation of a weld bead if the weld is not momentarily reinitiated (see figure 16).

5.10.4.3 Heating. The heating shall be of the uniform soaking type, applied preferably by means of electric heaters (resistance or induction), and uniformly distributed on and around

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the area being welded. Spot heating shall be prohibited. The spacing and wattage of heaters shall be such as to ensure that the entire welding area is maintained within the required temperature range. If an oxyfuel torch is used for heating, only an appropriately sized rosebud tip shall be used in conformance with the torch or tip manufacturer's recommendations. Heat should be applied by any of the following suggested methods used either individually or in combinations:

a. Resistance heaters.

b. Radiant and infrared heaters.

c. Electrical induction.

d. Air gas torch.

e. Oxyfuel torch with an appropriately sized rosebud heating tip.

5.10.4.4 Control of minimum temperature. The minimum required temperature shall be established prior to welding. Loss of the minimum required temperature shall require reheating, as necessary, prior to resumption of welding. If the minimum specified preheat temperature for the weld is 125 °F or higher, but then during fabrication the minimum interpass temperature drops below 100 °F and the weld thickness is less than 50% complete, then the partially completed weld(s) shall be inspected in-process with MT (see 5.9.2.3). Reheating to the specified minimum interpass temperature range prior to the resumption of welding shall be done before or after this inspection.

5.10.4.5 Control of maximum temperature. The maximum required temperature shall be checked as specified in 5.10.4.2.2. If at any time during welding (except during deposition of a weld pass) the base metal temperature is found to be above the maximum preheat and interpass temperatures noted in table XIII, the weld area shall be allowed to cool to within the required temperature prior to any subsequent welding.

5.10.5 Post Weld Heat Treatment (PWHT). PWHT shall be prohibited for all quenched and tempered alloy steels and age hardened alloy steels, including all armor materials. When allowed, post heating shall be limited to the maximum temperatures of table XIII.

5.10.6 Heat input calculations. Heat input shall be determined by a. or b. for non-waveform controlled welding, or by b. or c. for waveform controlled welding. All related sections in AWS D1.1/D1.1M and ASME BPVC Section IX should be consulted for further information and guidance.

a. Heat input [joules per inch] = $\frac{\text{Voltage x Amperage x 60}}{\text{Travel Speed [inches per minute]}}$

b. Heat input determined using instantaneous energy:

Heat input [joules per inch] = $\frac{\text{Energy [joules]}}{}$

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Weld Bead Length [inches]

c. For instantaneous power measurements in joules per second or watts:

$$\text{Heat input [joules per inch]} = \frac{\text{Power [joules/second or watts]} \times \text{Arc Time [seconds]}}{\text{Weld Bead Length [inches]}}$$

d. For SMAW the deposit length per unit length of electrode may also be used for measuring and controlling heat input.

5.10.6.1 Alternative heat input limitations. Alternative heat input limitations shall be qualified with adequate supporting data. The proposed test plan to establish alternative heat input limitations shall be submitted to the technical authority for approval. This heat input limit shall be documented on the PQR.

5.10.7 Tack welding. If tack welding is used as an aid in fabrication, the welding shall be performed by a qualified welder or welding operator. All tack welds shall be in conformance with the appropriate WPS, preheat and interpass requirements, attention to heat input, and exhibit the same quality requirements that apply to final welds. Tack welds should be of sufficient length or size to overcome strain issues and be inspected for cracks. Cracked tack welds and those of poor quality shall be removed. For unincorporated tack welds see 5.10.8. In all instances tack welds shall be located in manner that allows for proper joint accessibility and ensuing welding operations. Avoidance of tack welds in corners and areas of stress concentration, as well as use of boxing techniques should all be employed. This applies to both incorporated and unincorporated tack welds.

5.10.7.1 Incorporated tack welds. Incorporated tack welds shall be considered initial increments of the final weld and meet the applicable VT acceptance criteria prior to incorporation. Welds shall be properly prepared for subsequent weld tie-in or additional passes. Incorporated tack welds shall be dressed and blended with minimal geometry and favorable profile before being incorporated with subsequent welding. Alternative conditions for remelting of tack welds with SAW or other deep penetrating high energy welding processes shall be directed to the TA for review.

5.10.8 Temporary welds for fabrication purposes. Temporary attachments, temporary welds, and unincorporated tack welds used as construction aids to facilitate handling, positioning, and distortion mitigation are allowed, but should be used minimally. The same WPS, quality, welding, and workmanship requirements of this document shall apply to temporary attachment welding and removal. Construction aid material shall be compatible to the base metal that is being joined. Sufficient preheat, attention to heat input, and avoidance of damage to the base material during temporary welding and removal is required in accordance with this document. Attachments should be removed a minimum 1/8 inch away from the weld, followed by careful grinding, and blended smooth and flush into the adjacent base material surface. Attachment removal locations shall be documented on an inspection report. The removal area is required to pass a successful VT and MT inspection after attachment removal

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and any subsequent allowable base material repairs to be considered acceptable (see 5.9 and 5.12).

5.10.9 Weld profiles. All weld profiles shall meet the requirements of table XX.

5.10.10 Peening. Welds shall be free of surface matter such as slag, scale, rust and defects prior to peening to prevent entrapment of foreign materials or unfused areas in the weld. If weld bead overlap, excessive roughness or re-entrant bead contour is present, it should be corrected by grinding prior to peening. Peening of the last layer is permitted provided the finished contour is ground prior to subsequent NDT to remove any evidence of peened surface. Peening shall not be performed on the first layer of a multiple layer weld. Peening may be used to correct distortion or minimize residual stresses and may be used for contour grinding purposes.

5.10.11 Arc strikes. Arc strike inspection and limitations shall be in accordance with table XX. For removal and repair of arc strikes see 5.12.

5.10.12 Weave and stringer passes. Weaving and stringer passes may be used interchangeably, so long as the heat input (see 5.4.6.4) and bead width qualification limitations (see 5.3.5 and 5.4.6.4.1) of this document are met, and the essential variable ranges of the qualified WPS (see table V and 5.4) are not violated. Note, the use and need of both stringer beads and weaves beads in a single PQR or WQR qualification test assembly is required, just the same as is practical to facilitate and promote soundness in different joint designs and joint geometry encountered in production (see table XVIII).

5.10.13 Backgouged roots. The joint root of full penetration joints, welded from both sides, shall be backgouged after sufficient welding has been done on one side and before any welding is started on the opposite side. The weld root area shall be backgouged to sound material.

5.11 Workmanship requirements.

5.11.1 Weld joint surface preparation. All paint that would contaminate welding shall be removed for a minimum distance of 3 in. (76 mm) in all directions from the joint to be welded. The addition of rust inhibiting coatings shall be cause for requalification. Surfaces to be welded upon and adjacent surfaces at a minimum distance of 1/2 inch from the expected weld area shall be clean, dry and free of surface matter and defects such as:

a. Paint, except in accordance with the 3 inch minimum requirement stated above.

b. Oil, grease.

c. Moisture.

d. Mill scale.

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e. Oxide or rust.

f. Objectionable nicks, gouges and irregularities. See base material specification or both 5.12 and 5.3.7.1 for depth of allowable repairs.

g. Zinc, galvanizing, or thermal spray aluminum.

h. Excess slag.

5.11.2 Weld cleaning.

5.11.2.1 In-process cleaning. Any welds requiring multiple passes shall be mechanically cleaned of all surface scale, weld spatter, silicon deposits, and weld slag before the next pass is welded. Chemical products that leave a residue shall not be used for cleaning. Other cleaning methods that may exceed the temperatures provided in table XIII shall require the approval of a cleaning procedure by the procuring contract officer (see 6.2).

5.11.2.2 Completed welds. All weld slag, flux, tightly adhering weld spatter greater than 1/8 inch diameter, silicon deposits, and residues shall be removed. Weld spatter that cannot be removed by standard cleaning methods may remain unless there are functional or cosmetic concerns or the spatter would interfere with NDT (see 5.9.1.3). The method or combination of methods used shall be selected to suit the properties of the metal, the type of contaminates and the degree of contamination present. Surface cleaning methods that use de-rusting chemicals or phosphoric acid shall not be used unless test data can substantiate that the combined chemical cleaning or phosphoric acid cleaning and subsequent coating pretreatments are not detrimental to the steels being cleaned.

5.11.3 Welder and welding operator workmanship check. Prior to and during welding, welders and welding operators shall ensure base metal is free of foreign materials and discontinuities that would inhibit successful welding. All weld joints shall receive a baseline visual check by the welder or welding operator prior to, during, and after welding, and prior to involving a NDT inspector. Any concern observed by the welder or welding operator shall be reported to a CVI or AVI for VT inspection. The welder or welding operator does not need to be a qualified weld inspector to perform this quality check. Regardless of qualifications, the welder or welding operator shall not have the ability to approve the final inspection on their own work.

5.11.4 In-process hardness checks. When hardness testing of the base material is desired due to suspected heat input limitations being exceeded, measurements shall be taken at 5/8 inches (16 mm) from any weld toe line. The hardness shall be in accordance with the minimum hardness value for the base material (see 5.1.1). Conversions from one hardness scale to another scale shall be in accordance with ASTM E140. Before hardness tests are performed, the surface to be inspected shall be prepared in accordance with 5.7.5.2.

5.11.5 Joint fit-up inspection. Joint preparation shall be in accordance with approved configurations including those in approved welding procedures. When fit-up tolerances exceed the limitation of approved joint designs, correctional buildup by welding may be employed

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within the limitations of 5.12.6.2. For fillet weld joints, when the fit-up gap exceeds 1/16 inch as a nominal condition along the joint, fillet weld size shall be increased by an amount equal to the excess of opening above 1/16 inch. Where the gap between members exceeds 3/16 inch as a nominal condition along the joint, the condition shall be corrected by weld buildup in accordance with 5.12.6.2 of this document or the joint shall be prepared as a full penetration weld. Weld buildup shall not be performed on armor material.

5.11.6 Distortion control. All efforts to mitigate distortion shall be made on all weldments. Some general practices to mitigate distortion are as follows:

a. Alternating weld passes about the neutral axis to keep a balance of weld deposit on each side of the weldment.

b. The use of a backstep sequence and an overall well planned weld pass sequence.

c. Depositing weld from the middle of the weld joint and working toward the edges of the work piece. Weld starts should be removed and blended to avoid discontinues.

d. Use of optimized welding parameters, larger filler metal diameters, and faster travel speeds.

e. Using the preheat temperatures listed in table XIII, and potentially increasing the preheat temperature within the limitations to slow cooling rates. Conversely, heat sinks may need to be employed as well.

f. Using properly sized weld beads for the application; in effort to avoid both excessively large size beads, too few passes, excessively small beads, and too many weld passes.

g. The use of restraint via strongbacks, fixtures, construction aids, wedges, clips, and clamps to brace the weldment. Allow the weldment to cool to ambient temperature prior to removal of these apparatuses.

h. Pre-bend or preset the weld joint by its anticipated distortion.

i. Limit overwelding by way of best practice weld joint design, welding supervision, and welder training.

j. Use of sub-assemblies, back-to-back methods, and straightening practices before further assembly.

k. Account for both adding material stock to component details and for root openings to serve overall dimensional tolerances and shrinkage concerns.

5.12 Repairs.

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5.12.1 Requirement. This standard is not intended for specific repairs of individual systems. Before the repairing of any component can be accomplished, the base material shall be prepared in accordance with 5.11.1. Repairs 0.625 inches (16 mm) or greater from the weld toe line shall be considered outside the weld zone and considered base material repairs. All materials fabricated out of tolerance shall be repaired until all tolerances and/or requirements are met. All repairs shall be inspected with the applicable NDT process specified for the weld class (see table XXI). Inspection of repairs shall be held to the same requirements for holds prior to examination as all other welds (see 5.9.1.2).

5.12.2 Repair procedures. Welding repair procedures (WRP) shall be developed and performed in accordance with this document. All repair procedures shall utilize a WPS developed and qualified in accordance with this document. Entities that wish to use proprietary repair procedures shall deliver these procedures for approval by the procuring contract officer upon request (see 6.2). System technical bulletins, technical manuals, applicable military standard, or military specification may not be used unless specifically approved by the procuring contract officer. All repair procedures shall be delivered to the PCO for approval upon request. After repair procedures are completed, the fabricated material shall meet all requirements listed in this document, applicable contracts, standards, or drawings.

5.12.3 Repair cycles. Repair cycles shall be referred to and noted on inspection or repair reports as R1 (one repair performed) or R2 (two repairs performed) or more if needed. Minor repairs such as undercut, overlap, and visual blends, shall not be considered in the previous context. Only major defects such as incomplete fusion and cracks or other defects that have failed volumetric NDT and require significant excavation and re-welding shall need documentation. The Government reserves the right to review any or all MRB processes, procedures, and documentation.

MRB disposition shall be required when any one of the following conditions exists:

a. When more than two weld repair attempts have been performed at the same location on armor or high strength steels.

b. When more than three weld repair attempts have been performed at the same location on materials not listed in a).

c. When the wrong filler metal has been used.

d. When a repair weld is required after the weldment has been postweld heat treated.

e. When a repair weld is required after final machining has been completed.

f. When the repair extends outside the original weld zone (fusion zone and HAZ).

g. When a weldment has been direct aged, i.e., aged only without intermediate solution heat treatment, unless authorized by the PCO.

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h. When a weldment has been made with parameters outside the qualified WPS range.

i. All repairs of defective plug welds.

j. Weld repairs following proof or leak test.

k. Any holes repairs on armor material or holes outside of allowances 5.12.6.3.

l. Weld buildup on armor material or outside the allowances of 5.12.6.2.

5.12.4 Mill and foundry defects. Defects in excess of that permitted by the applicable base material specification or this document shall be repaired by grinding (to the maximum extent possible), provided the thickness is not reduced below that allowed by the base material specification or acquisition document. If a discontinuity or defect is the result of poor mill process, repairs of the armor base material shall be made in accordance with MIL-A-11356, MIL-DTL-12560, MIL-DTL-46100, MIL-DTL-46186, or MIL-DTL-32332.

5.12.5 Repair of armor base material. Authorized repairs of armor base materials require a qualified WPS in accordance with 5.3.7.1 and a detailed repair procedure proposal. This repair procedure proposal shall state at a minimum: the extent of the repair, its location, depth, and size, the impact it will have on the mechanical and ballistic properties, and any required qualification documentation to achieve a sound repair, including a volumetric inspection plan.

If a procedure has not been developed in accordance with 5.3.7.1, armor base material repairs shall be permitted for MIL-DTL-12560 and MIL-DTL-46100 for arc strikes, and fabrication scars, etc. less than 1/16 in. (1.5mm) deep, with a repair area limited to 0.1% of the surface area, only for base material thicknesses greater than 1 in. (25 mm). Armor base material repairs for MIL-DTL-46186 and MIL-DTL-32332 shall require prior specific review by the technical authority and approval by the procuring contracting officer for each repair. All weld repairs for MIL-A-11356 materials shall be qualified and performed in accordance with the requirements of 5.3.7.1.

5.12.6 Repairs of non-armor base material. Authorized repairs of non-armor base materials require a qualified WPS in accordance with 5.3.7.2. If the base material standard includes an impact toughness requirement, the filler metal shall meet the impact toughness requirement of the base material. Under matching filler material is prohibited. Repair thickness shall be limited by the qualified WPS.

5.12.6.1 Arc strikes, nicks, gouges, and other fabrication scars. Arc strikes, nicks, gouges, and other fabrication scars up to and including 1/32-inch depth are acceptable without repair. Arc strikes, nicks, gouges and other fabrication scars in excess of 1/32 inch, but within the limitations listed below shall blend smoothly into the base material or shall be repaired by mechanical means. Arc strikes, nicks, gouges and other fabrication scars in excess of the below listed limits shall be repaired by welding. The final repaired surface shall blend smoothly into the base material and shall not interfere with clearance.

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Base material thickness	Maximum allowable depth (inches)
Less than 1/2 inch	1/32
1/2 inch and greater	1/16

5.12.6.2 Weld buildup. Buildup around the weld joint shall not exceed a total of ½ T of the thinner member or 0.5 inches whichever is less. Filler material must meet or exceed the base material specified minimum tensile strength. The buildup shall be considered part of the involved weld and shall be inspected in accordance with its specified classification.

5.12.6.3 Weld repair of holes. Holes cut into or through a member may be welded provided the original hole diameter does not exceed 2½ inches. Holes less than 1/2 inch in diameter shall be opened to 1/2-inch minimum diameter, except that holes in members 1/4 inch thick or less need only be opened to a diameter (d) equal to the thickness (t) of the member (e.g., d=t). The hole shall be beveled to 20 degrees minimum angle, and shall ensure removal of all internal threads when present. Holes greater than 2½ inches in original diameter shall be reported to the procuring contract officer for disposition.

5.12.6.3.1 Partial penetration holes. Partial penetration holes shall have 3/16 minimum material remaining prior to welding, or shall be drilled through and prepared as a full penetration repair as shown in figure 17.

5.12.6.3.2 Full penetration holes. Full penetration holes shall be repaired using a 3/16-inch minimum backing plate of compatible material or approved non-metallic material. Permanent backing may be used to repair holes when approved by the procuring contract officer prior to use.

5.12.7 Weld metal removal. When a defect in the weld metal is discovered during inspection or testing, the defect shall be removed. If the material is removed by a thermal process, the dross, cutting residue, and hardened layer at the cut surface shall be removed by grinding. The cut surface shall be ground an additional 1/16 inch (1.6 mm) to remove the hardened layer. The temperature during excavation shall not exceed the maximum interpass temperature in accordance with table XIII. Weld metal containing the defect shall be completely removed and then prepared with a groove angle in accordance with applicable WPS or 40° minimum, and the end of the weld metal junction shall be tapered to an approximate 45° slope (see figure 18) to allow for ease of reentry. The prepared repair excavation length shall be the original defect length plus 4t, where t is the prepared repair depth thickness. The weld metal shall be removed in a manner that leaves a minimum 5/16 inch (8mm) rounded root opening (joint width). The depth of the removed weld metal shall be the depth of the defect plus an additional 0.125 inch (3.2 mm). Once weld metal has been removed a visual inspection and the appropriate NDT shall be performed on the repaired area to ensure the entire defect has been removed, and no additional discontinuities have been discovered during weld metal removal. The excavated area shall be welded in accordance with the requirements of this document. Excessive reinforcement or convexity may be removed by grinding to conform to profile requirements (see 5.10.9) without re-welding.

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5.12.7.1 Weld repair excavation preparation. Weld repair areas shall be contoured to produce an excavation which is fully visible to the welder and allows accessibility of the electrode to all surfaces to be welded (see 5.10.13). The gouged area configuration shall have side walls sloping without sharp breaks. Prior to resuming welding, inspection of the root or excavated repair area shall be performed with the appropriate NDT method.

5.12.8 Crack repair.

5.12.8.1 Crack identification. Certified NDT personnel shall evaluate a crack indication as relevant or non-relevant by determining appropriate NDT method and identify the beginning and ending locations of relevant crack. See 5.12.1 for cracks that extend beyond the weld zone, which is greater than 0.625 inches (16 mm) beyond the weld toe line.

5.12.8.2 Crack arresting holes. Crack arresting shall be used for partial penetrating cracks as determined by certified NDT personnel or welding engineer. An arresting hole shall be drilled within 0.125 to 0.25 inch past each end of the crack, and should be approximately 0.16 to 0.375 inches (4 to 10 mm) in diameter. These holes shall be prepared and repaired in accordance with 5.12.6.3.

5.12.8.3 Crack removal. Grinding or other mechanical methods shall be used to remove the crack and performed in accordance with 5.12.7. Material temperature shall be monitored during crack removal and shall not exceed the maximum interpass temperature in accordance with table XIII. Cracks starting within 5/8 inches (16 mm) of the weld toe line are weld zone crack repairs and shall be excavated from the end of the crack towards the weld metal in accordance with 5.12.7. Cracks starting outside 5/8 inches (16 mm) shall be considered base material repairs see 5.12.5 or 5.12.6.

5.12.8.3.1 Crack types. There are three types of cracks: partial penetration, full penetration accessible from both sides, and full penetration accessible from one side.

5.12.8.3.2 Partial penetration crack removal. Crack arresting should be used for partial penetrating cracks as determined by certified NDT personnel or welding engineer. The following is a recommended practice:

a. Excavate the crack to the same angle and root opening tolerances as allowed in the WPS, with a 45° tapered end (see figure 18). Do not exceed 66% of the material thickness during excavation, if this thickness is exceeded see 5.12.8.3.3. Excavation should include any drilled holes.

b. There shall be a 5/16 in. (8 mm) minimum root opening at the base of the excavated area.

c. Examination of the excavation site per the original approved NDT method shall be used to ensure the crack is completely removed. If no NDT method was originally used for the weldment, the authorized Level III NDT inspector or welding engineer shall make the decision

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on the method to apply. If the excavation reaches a threshold of 33% of material thickness and shall be repaired as a PJP crack repair joint, see 5.3.1.6.

d. Weld the excavated area in accordance with the requirements of this document. The use of run-off/run-on tabs and removable backing material is encouraged.

e. The repaired crack shall be inspected in accordance with the original approved NDT method and hold time (see 5.9.1.2). If no NDT method was originally used for the weldment, the authorized Level III NDT inspector or welding engineer shall make the decision on the method to apply.

5.12.8.3.3 Full penetration crack removal accessible from both sides. Use crack arresting process as described in 5.12.8.2 to stop propagation of the crack. The following is a recommended practice:

a. Starting on side one of the crack repair, follow the process for partial penetration crack removal in 5.12.8.3.2 steps a, b, and d. The depth of excavation shall be > 0.5 T of the plate to be repaired.

b. Once welding of side one is completed, inspect the excavation site per the original approved NDT method. The original NDT method shall be used to ensure the crack is completely removed. If no NDT method was originally used for the weldment, the authorized welding engineer shall make the decision on the method to apply. The PT inspection process shall not be used at this time. The depth of excavation into sound weld and base material shall be approximately > 0.5 T of the plate being repaired and verified through the NDT examination.

c. Once welding has been completed on side two, final inspection shall be performed. The repaired crack shall be inspected in accordance with the original approved NDT method and hold time (see 5.9.1.2). If no NDT method was originally used for the weldment, the authorized welding engineer or authorized Level III NDT inspector shall make the decision on the method to apply.

5.12.8.3.4 Full penetration crack removal accessible from one side. For materials of any thickness that are only accessible from one side, all repairs shall be in accordance with applicable technical bulletin, technical manual, contract, scope of work, or a repair procedure approved by the procuring contract officer. This is a recommended practice.

5.12.9 Field repairs. Reprocessing of nonconforming material in accordance with approved written repair procedures and operations to address the nonconformity in theater. The purpose of a field repair is to restore functionality to an item being repaired on an as-needed basis. Field repairs are distinguished from other repairs, in that the item, after field repair, is temporarily excused from completely conforming to all of the applicable drawings, specifications or contract requirements. The system will be brought back into full conformity once the repaired item reaches a facility that is capable of performing a complete repair. This type of repair shall not be used in manufacturing of new vehicles or repairs at a facility capable of performing a complete repair.

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5.12.10 Repair record retention. A repair shall be documented in a repair report and shall be updated with pertinent repair locations, repair process used, and inspection results after the repair has occurred. The entity shall deliver this documentation to the procuring contract officer upon request and it shall be retained for the life of the system.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Armor welding covered in this document is intended for military use. Non-government use of this document is authorized.

6.2 Acquisitions requirements. The list of information provided below should be used as a minimum for contract acquisitions and procurement. The information should be evaluated to determine if the requirements should be in the base contract, performance specification or delivered to the government through a contract data requirements list (CDRL) or equivalent.

a. Title, number and date of this standard.

b. Qualification records (see 4.1).

c. Previous qualification records to MIL-STD-3040 revisions (see 4.4).

d. Previous qualification to other standards (see 4.4.1).

e. Additional welding processes (see 4.7.3).

f. Armor base material not listed in a qualified specification (see 5.1.2).

g. Welding of MIL-DTL-32332 materials (see 5.1.2.1).

h. Procedure qualification records (see 5.3.1 and A.4.1).

i. Welding procedure specification (WPS) (see 5.5.1 and A.4.2).

j. WPS revisions (see 5.5.6).

k. Previous welder or welding operator qualifications (see 5.6.6).

l. Requirements for destructive testing of welds (see 5.7.1).

m. Use of an NDT qualification not listed (see 5.8.5).

n. Use of Electromagnetic testing (ET) (see 5.9.2.5).

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o. Use of an NDT method not listed (see 5.9.2.7).

p. Use of an in-process cleaning method (see 5.11.2.1).

q. Use of proprietary repair procedures (see 5.12.2).

6.3 Standard units of measure. This standard makes use of both U.S. Customary Units and the International System of Units (SI). Throughout the standard, the user will find dimensions in U.S. customary units followed by SI in parentheses ( ), or in appropriate columns in tables and figures. SI units are an estimated value based on U.S. Customary Units.

6.4 Subject term (keyword) listing.

Electrodes
Filler metal
High strength steels
Nondestructive testing
Procedure qualification record
Radiography
Ultrasonic
Weld
Welder
Welding operator
Welding procedure
Welding procedure specification
Welder qualification

6.5 Change notations. The margins of this standard are marked with vertical lines to indicate modifications generated by this change. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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TABLE I. Base material groupings.

S-Number	Group	Spec. No.	Class or Type
S-1 – Carbon Steel	--	See AWS B2.1/B2.1M-Annex C- Table List of Base Metal Specifications – Ferrous Alloys Examples include: ASTM A36/A36M ASTM A516 Grade 70 ASTM A572/A572M	
S-2 – Quench and Temper Carbon Steel	--	ASTM A537/A537M	Class 2
S-3 – Carbon Molybdenum Steels	1	ASTM A204/A204M	A, T1, T1A, T1B
		ASTM A213/A213M	T2
		ASTM A217/A217M	WC1
		ASTM A234/A234M	WPI
		ASTM A242/A242M	T-2
		ASTM A335/A335M	P1, P2, P15
		ASTM A352/A352M	LC1
		ASTM A369/A369M	FP1, FP2
		ASTM A387/A387M	Grade 2, Cl1
		ASTM A426/A426M	CP1, CP2, CP15
		ASTM A672/A672M	L65
		ASTM A691/A691M	CM-65; 1/2 Cr, Cl1
	2	ASTM A182/A182M	F1, F2
		ASTM A204/A204M	B, C
		ASTM A336/A336M	F1
		ASTM A387/A387M	2 Cl2
		ASTM A672/A672M	1/2 Cr, Cl.2; CM-70; CM-75; L70; L75
		ASTM A691/A691M	CM-70; CM-75
	3	ASTM A322	4130, 8620, 8720
		ASTM A108	4130, 8620CW
		ASTM A302/A302M	A, B; MnMo, Grade B (plate), C, D
		ASTM A505	4130
		ASTM A507	4130, 4140, 4142, 4145, 4340, 8620, 8630, 8640, 8650
		ASTM A508/A508M	2, 2a, 3, 3a, 4b
		ASTM A519/A519M	4130, 4140, 4145H, 4150H, 4340H, 8620, 8630, 8650H
		ASTM A533/A533M	Type A, Class 1 & 2; Type B, Cl & 2; Type C, Class 1&2, Type D Class 2
		ASTM A541/A541M	2, 2A, 3, 3A
		ASTM A829/A829M	4130
	--	DOD-F-24669/1	CrNi type 8620 (bar and forgings)
		DOD-F-24669/2	CMo class A (forgings); CMo class B (forgings)
		MIL-P-24707/2	ASTM A217 grade WC1 (castings)
		MIL-S-16974	CrNi type 8620 (bar and forgings)
		MIL-S-872	CMo class A (forgings), CMo class B (forgings)
		MIL-T-16286	CMo class D (tube)

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TABLE I. Base material groupings – Continued.

S-Number	Group	Spec. No.	Class or Type
S-4 – Alloy Steels (Cr content 0.75%-2%, Total Alloy Content 2-2.75% max.)	1	ASTM A182/A182M	F11, Cl.2, F11, Cl.3, F12, Cl.1, F12, Cl.2, F11, Cl.1
		ASTM A213/A213M	T3b, T11
		ASTM A336/A336M	F11, F11A, F11B, F12, FP11, FP12
		ASTM A369/A369M	FP3b, FP11, FP12
		ASTM A387/A387M	11, Cl. 1; 11, Cl. 2; 12, Cl. 1; 12, Cl. 2
		ASTM A389/A389M	C23, C24
		ASTM A426/A426M	CP11, CP12
		ASTM A541/A541M	11C
		ASTM A691/A691M	1Cr; 1.25Cr; 1CF Cl.1; 1CF Cl.2
		ASTM A739	B11
	2	ASTM A333/A333M	4
		ASTM A423/A423M	1, 2
	3	ASTM A148/A148M	90-60, 105-85, 115-95, 120-95, 150-135
	--	ASTM A671/A671M	1CR; 1 1/4CR
		SAE AMS6360M	CrNiMo (tube), type 8630
		ASTM A213/A213M	Grade T11 (tube)
		DOD-F-24669/1	CrMo type 4130 (bar)
		DOD-F-24669/2	CrMo, composition F-11 (forgings); class A (forgings)
		ASTM A217/A217M	Grade WC6 (casting)
		MIL-P-24691/2	CrMo, grade P11 (tube and pipe)
		ASTM A217/A217M	Grade 1 CrMo class 1 (casting); WC6; Grade 2 WC9
		ASTM A389/389M	Grade 3 C23
		MIL-S-16974	CrMo, Type 4130 (bar)
		MIL-S-18410	CrMo class A (forgings)
		SAE AMS6350N	CrNiMo type 8630 (plate), CrMo type 4130 (plate)
		SAE AMS6411J	CrMoV, type 4330 (bar, forging)
		SAE AMS6427N	4330 Mod
		MIL-T-18165	CrMo class 1 (tube and pipe)
		SAE AMS6361F	CrMo, Type 4130 (tube)

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TABLE I. Base material groupings – Continued.

S-Number	Group	Spec. No.	Class or Type
S-5 – Alloy Steels (Total alloy content <10% max., Cr-Mo)	1	ASTM A671/A671M	2-1/4CR
		ASTM A182/A182M	F21, F22, F22A
		ASTM A213/A213M	T4, T21, T22
		ASTM A217/A217M	WC9
		ASTM A234/A234M	WP22
		ASTM A335/A335M	P21, P22
		ASTM A336/A336M	F21, F21a, F22, F22a
		ASTM A369/A369M	FP21, FP22
		ASTM A387/A387M	21, Cl. 1; 21, Cl.2; 22, Cl.1; 22, Cl.1
		ASTM A426/A426M	All grades
		ASTM A487/A487M	Cl. A/B/C
		ASTM A508/A508M	Cl. F3V
		ASTM A691/A691M	2-1/4 Cr., Cl. 1; 2-1/4 Cr., Cl. 2; 3 Cr., Cl. 1; 3 Cr., Cl. 2
		ASTM A739	B22
	2	ASTM A182/A182M	F5, F5A, F7, F9, F429
		ASTM A213/A213M	T5, T5B, T5C, T7, T9
		ASTM A217/A217M	C5, C12
		ASTM A234/A234M	WP5, WP7, WP9
		ASTM A335/A335M	P5, P5b, P5c, P9, P91, P7
		ASTM A336/A336M	F5, F5a, F9, CrMo class F5 (bar and forgings)
		ASTM A369/A369M	FP5, FP9
		ASTM A387/A387M	5, Cl.1; 5, Cl. 2
		ASTM A426/A426M	CP5, CP5b, CP7, CP9, CP22
		ASTM A487/A487M	8N
		ASTM A542/A542M	Cl. 3, Cl.4
		ASTM A691/A691M	5 Cr., Cl.1; 5 Cr., Cl.2; F5
	--	ASTM A213/A213M	T22
		DOD-F-24669/2	CrMo, composition F-22 (forgings); class B (forging)
		MIL-F-24669/8	CrMoV, grade F (forgings)
		MIL-P-24691/2	CrMo, grade P-22 (tube and pipe)
		ASTM A217/A217M	Grade 1 CrMo class 2 (casting); WC6; Grade 2 WC9
		ASTM A389/389M	Grade 3 C23
		MIL-S-18410	CrMo class B (forging)
		MIL-F-24669/8	CrMoV, grade F (forgings)
		MIL-T-16286	CrMo class E (tube)
		MIL-T-18165	CrMo class 2 (tube and pipe)

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TABLE I. Base material groupings – Continued.

S-Number	Group	Spec. No.	Class or Type
S-11 – Quenched and tempered alloy steels	A0	T9074-BD-GIB-010/0300	HY-80, HY-100
		MIL-S-23284	Class 1 and Class 2
	A1	ASTM A334/A334M	8
		ASTM A353/A353M	All grades
		ASTM A420/A420M	WPL8
		ASTM A522/A522M	Tp. I, Tp. II
	A2	ASTM A645/A645M	All grades
	A3	ASTM A487/A487M	Cl. 4Q, Cl. 4QA, Gr. 4, Cl. B, Gr. 4, Cl. E
	A4	ASTM A672/A672M	J100
		ASTM A533/A533M	Cl.3 A, Cl.3 B, Cl.3 C, Cl.3 D
	A5	ASTM A352/A352M	LC2-1
		ASTM A508/A508M	Cl. 4, Cl. 4a, Cl. 5, Cl. 5a
		ASTM A543/A543M	Cl. 3 B; Cl. 1 C; Cl. 3 C
	A6	ASTM A542/A542M	Cl. 1, Cl. 2
		ASTM A543/A543M	Tp. B Cl. 1; Tp. B Cl.2; Tp. B Cl.3; Tp. C Cl.1; Tp. C Cl.2; Tp. C Cl.3
	A7	ASTM A732/A732M	All Q grades
	B0	T9074-BD-GIB-010/0300	HY-130
	B1	ASTM A514/A514M	All Grades
		ASTM A709/A709M	100, 100W
	B2	ASTM A513/A513M	E
		ASTM A517/A517M	All Grades
		ASTM A592/A592M	E
	B9	ASTM A592/A592M	9630
		ASTM A709/A709M	G100/G100Q
S-11 – Age hardening alloy steels	C	ASTM A710/A710M	A, Cl. 1 and 3
		T9074-BD-GIB-010/0300	HSLA-80 (plate, sheet, or coil)
	D	T9074-BD-GIB-010/0300	HSLA-100 (plate, sheet, or coil)
S-15 – Armor Grade Steels[1]	A	MIL-DTL-12560	Class 1, Class 2
	B	MIL-DTL-12560	Class 4A
	C	MIL-DTL-12560	Class 4B
	D	MIL-DTL-46100	Class 1
	E	MIL-DTL-46100	Class 2
	F	MIL-A-11356	Class 1, Class 2
	G	MIL-DTL-46186	Class 1
	H	MIL-DTL-46186	Class 2

NOTES:

1: Base material grouping is allowed for Class III and Class IV welds only. A combination of armor steels qualifies the pairing and each material to themselves.

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TABLE II. Welding specification and weld classification cross reference.

Previous standard		New standard[1,2]	
MIL-STD-1261	Class 1	AWS D1.1/D1.1M	Statically loaded
	Class 2	AWS D1.1/D1.1M	Cyclically loaded
	Class 3	MIL-STD-3040	Class II
AWS D1.1/D1.1M	Statically Loaded	AWS D1.1/D1.1M	Statically loaded
	Cyclically Loaded	AWS D1.1/D1.1M	Cyclically loaded
MIL-STD-1941	Armor to Armor	MIL-STD-3040	Class I
	Attachments to Armor	MIL-STD-3040	Class III
MIL-STD-1185	Armor to Armor	MIL-STD-3040	Class I
	Attachments to Armor	MIL-STD-3040	Class III
MIL-STD-2219	Class A	AWS D17.1/D17.1M	Class A
	Class B	AWS D17.1/D17.1M	Class B
	Class C	AWS D17.1/D17.1M	Class C
MIL-STD-248		NAVSEA S9074-AQ-GIB-010/248	
MIL-W-46086	Armor to Armor	MIL-STD-3040	Class I
	Attachments to Armor	MIL-STD-3040	Class III
Ground Combat Vehicle Welding Code-Steel (12479550)	Non-critical Section 6.6.1	MIL-STD-3040	Class IV
	Critical Section 6.6.2	MIL-STD-3040	Class II[3] or III
	Ballistic Section 6.6.3	MIL-STD-3040	Class I
MIL-STD-3040 Base Document	Class III	MIL-STD-3040	Class IV
	Class II	MIL-STD-3040	Class III
	Class I	MIL-STD-3040	Class I

NOTES:
1: The decision between classifications shall be made based on the susceptibility and severity of failure of each weld (see 4.6.1).
2: MIL-STD-3040 listed in the “New standards” column shall be the current revision of MIL-STD-3040.
3: Class II welds are only manufactured from high strength steel welded to high strength steel (see 3.22.1.6). No armor grade steel shall be used for Class II welds.

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TABLE III. Weld classification decision matrix.

Severity / Susceptibility	Catastrophic	Critical	Marginal
Frequent	High (Class I,II)	High (Class I,II)	Medium (Class III)
Probable	High (Class I,II)	High (Class I,II)	Medium (Class III)
Occasional	High (Class I,II)	Medium (Class III)	Low (Class IV)
Improbable	Medium (Class III)	Low (Class IV)	Low (Class IV)

NOTES:
1: Severity refers to either the result or consequence of the failure of the weld.
2: Susceptibility refers to the likelihood of damage resulting in the weld failing during operation or high-strain-rate events.

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TABLE IV. Filler material groupings.

Group	Applicable Document	Filler Material Type	AWS electrode specification acceptable substitution	AWS electrode classification acceptable substitution
A-2A Carbon and low alloy steel (covered electrode)	MIL-DTL-22200/1	MIL-7018-M	A5.5/A5.5M	E7010-A1 E7011-A1 E7018-A1 E7020-A1
A-2B Carbon and low alloy steel (bare electrode and rods)	MIL-E-23765/1	MIL-70S-1 MIL-70S-2 MIL-70S-3 MIL-70S-4 MIL-70S-5 MIL-70S-6	A5.18/A5.18M	ER70S-X
A-2C Carbon and low alloy steel (bare electrode and flux)	MIL-E-23765/1	MIL-70S-1 (wire) MIL-70S-2 (wire) MIL-70S-3 (wire) MIL-70S-4 (wire) MIL-70S-5 (wire) MIL-70S-6 (wire) MIL-70S-7 (wire) MIL-70S-8 (wire) MIL-70S-9 (wire) MIL-70S-XF (flux)	A5.18/A5.18M	ER70S-X
	MIL-E-23765/4	MIL-F6A2-EL12 (flux-electrode) MIL-F6A2-EM12K (flux-electrode) MIL-F7AZ-EM12K (flux/wire) MIL-F7A2-EM12K (flux/wire)	A5.23/A5.23M	F7XA-EXX-XXH8 (flux-electrode) F7XA-EXX-XXH4 (flux-electrode) F7XA-EXX-XXH2 (flux-electrode)

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TABLE IV. Filler material groupings – Continued.

Group	Applicable Document	Filler Material Type	AWS electrode specification acceptable substitution	AWS electrode classification acceptable substitution
A-2D Carbon steel (flux cored electrodes)	MIL-DTL-24403/1	MIL-70T-1 MIL-71T-1 MIL-71T-1A MIL-70T-5 MIL-70T-8 MIL-71T-8 MIL-71T-8A MIL-71T8-K6 MIL-71T8-Ni1	A5.20/A5.20M	E71T1 E71T5 E71T12
A-3A Carbon and low alloy steel (low hydrogen covered electrodes)	MIL-DTL-22200/1	MIL-8018-C3 MIL-9018-M	A5.5/A5.5M	ER8018-C3 ER9018
A-3B Carbon and low alloy steel (bare electrode)	MIL-E-23765/2	MIL-80S-3 MIL-90S-3	A5.28/A5.28M	ER80S-D2 ER90S-D2
A-3C Carbon and low alloy steel (bare electrode and flux)	MIL-E-23765/2	MIL-80S-1 (wire) MIL-80S-2 (wire)	A5.28/A5.28M	ER80S-D2 ER90S-D2
		MIL-80S-1F (flux) MIL-80S-2F (flux)	A5.23/A5.23M	F8XA-EXX-XXH8 (flux-electrode) F9XA-EXX-XXH8 (flux-electrode)
A-3D Low alloy steel (flux cored electrode)	MIL-DTL-24403/1	MIL-80T1-Ni1 MIL-80T1-Ni2 MIL-81T1-Ni1 MIL-81T1-Ni2	A5.29/A5.29M	E81T1 E81T5 E81T12

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TABLE IV. Filler material groupings – Continued.

Group	Applicable Document	Filler Material Type	AWS electrode specification acceptable substitution	AWS electrode classification acceptable substitution
A-5A Low alloy, high strength steel (covered electrode)	MIL-DTL-22200/1	MIL-11018-M	A5.5/A5.5M	E10018 E10718 E11018
	NAVSEA T9074-BC-GIB-010/0200	MIL-10018-M1 MIL-10718-M MIL-12018-M2		
		MIL-14018-M1		
A-5B Low alloy, high strength steel (bare electrode)	NAVSEA T9074-BC-GIB-010/0200	MIL-100S-1 MIL-100S-2 MIL-120S-1	A5.28/A5.28M	ER100S-1 ER100S-G[1] ER110S-1 ER110S-G[1]
		MIL-140S-1		
A-5C Low alloy, high strength steel (bare electrode and flux)	NAVSEA T9074-BC-GIB-010/0200	MIL-100S-1 (wire) MIL-100S-1F (flux) MIL-100S-2 (wire) MIL-100S-2F (flux) MIL-120S-1 (wire) MIL-120S-1F (flux)	A5.28/A5.28M	ER100S-1 (wire) ER100S-G[1] (wire) ER110S-1 (wire) ER110S-G[1] (wire)
			A5.23/A5.23M	F10XA-EXX-XXH8 (flux-electrode) F10XA-EXX-XXH4 (flux-electrode) F11XA-EXX-XXH8 (flux-electrode)
A-5D Low alloy, high yield steel (flux cored electrode)	MIL-E-24403/2	MIL-100TM MIL-100TS MIL-101TS	A5.29/A5.29M	E101T1 E101T5 E101T12
	NAVSEA T9074-BC-GIB-010/0200	MIL-101TM		
A-6A CrMo steel (1.0% to 2.5% Cr, 0.4% to 1.2% Mo) (covered electrode)	MIL-E-22200/8	MIL-8018-B2L or B2 (1.25Cr-0.5Mo) MIL-9018-B3L or B3 (2.25Cr-1Mo)	A5.5/A5.5M	E8018-B2, B2L, B3, B3L, B4L, B5 E9018-B23, B24

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TABLE IV. Filler material groupings – Continued.

Group	Applicable Document	Filler Material Type	AWS electrode specification acceptable substitution	AWS electrode classification acceptable substitution
A-6B CrMo steel (1.0 to 3.0% Cr, 0.4 to 1.2% Mo) (bare electrode and rod)	AWS A5.28/ A5.28M	ER80S-B2L ER80S-B2 ER90S-B3L ER90S-B3	N/A	ER4130[1]
A-7A CrMo steel (4.0-13.5% Cr, 0.4-1.4% Mo) (covered electrode)	MIL-E-22200/8	MIL-410-XXI	A5.4/A5.4M	E410NiMo-15 E410NiMo-16 E410NiMo-17
A-7B CrMo steel (4.0-13.5 percent Cr, 0.4-1.4% Mo) (bare electrode and rod)	MIL-E-19933	MIL-410	A5.9/A5.9M	ER410NiMo-GX ER410NiMo-WX
A-8A High alloy steel (austenitic) (covered electrode)	MIL-E-22200/2	MIL-308-XXI MIL-308L-XXI MIL-309-XXI MIL-309L-XXI MIL-316-XXI MIL-316L-XXI MIL-347-XXI	A5.4/A5.4M	E308 E308L E309 E309L E316 E316L E347
A-8B High alloy steel (austenitic) (bare electrode and rod)	MIL-E-19933	MIL-308L MIL-308HC MIL-309 MIL-316 MIL-316L MIL-347	N/A	ER307Si[1] ER307L[1]
			A5.9/A5.9M	ER307 ER308 ER308L ER309 ER309L ER316 ER316L ER316LSi ER347

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TABLE IV. Filler material groupings – Continued.

Group	Applicable Document	Filler Material Type	AWS electrode specification acceptable substitution	AWS electrode classification acceptable substitution
A-8D High alloy steel (austenitic) (flux cored electrodes)	AWS A5.22/A5.22M	E308LT1-1 E308LT1-4 E309LT1-1 E309LT1-4 E309LNbT1-1 E309LNbT1-4 E316LT1-1 E316LT1-4 E347T1-1 E347T1-4	N/A	N/A
A-9A High alloy steel (duplex stainless) (covered electrode)	AWS A5.4/A5.4M	E2209-15 E2209-16	N/A	N/A
A-9B High alloy steel (duplex stainless) (bare electrode and rod)	AWS A5.9/A5.9M	ER2209	N/A	N/A
A-43A Nickel base alloys (covered electrode)	MIL-E-22200/3	MIL-1N12 MIL-3N12 MIL-4N12 MIL-4N1A MIL-8N12 MIL-8N12H	A5.11/A5.11M	ENiCrFe-1 ENiCrFe-2 ENiCrFe-3 ENiCrMo-3
A-43B Nickel base alloys (bare electrode and rods)	MIL-E-21562	MIL-EN62 MIL-RN62 MIL-EN6A MIL-RN6A MIL-EN82 MIL-EN82H MIL-RN82 MIL-RN82H MIL-EN625 MIL-RN625	A5.14/A5.14M	ERNiCr-3 ERNiCr-6 ERNiCrFe-6 ERNiCrMo-3

NOTES:
1: Filler material used for any classification of weld shall meet the requirements for lot conformance testing as described in 5.2.2.2

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TABLE V. Essential variables.

Paragraph		Description of Variables	Process				
			GMAW	FCAW	GTAW	SMAW	SAW
5.4.2 Base material essential variables	**5.4.2.1**	Base material specification for armor	X	X	X	X	X
	5.4.2.2	Base material specification for non-armor	X	X	X	X	X
	5.4.2.3	Base material thickness (T)	X	X	X	X	X
	5.4.2.4	Deposition thickness (t)	X	X	X	X	X
	5.4.2.5	Backing material	X	X	X	X	X
5.4.3 Filler material essential variables	**5.4.3.1**	Electrode classification	X	X	X	X	X
	5.4.3.2	Low hydrogen electrodes.	X	X	X	X	X
	5.4.3.3	Number of electrodes	X	X	X	X	X
	5.4.3.4	Cold wire and hot wire	X	--	X	--	--
	5.4.3.5	Electrode diameter	X	X	X	X	X
	5.4.3.6	Powder or granular filler material	--	X	--	--	X
5.4.4 Shielding	**5.4.4.1**	Gas	X	X	X	--	--
	5.4.4.1.1	Flow rate	X	X	X	--	--
	5.4.4.1.2	Composition	X	X	X	--	--
	5.4.4.2	Flux	--	--	--	--	X
5.4.5 Joint design	**5.4.5.1**	Joint design limitations for Class I welds	X	X	X	X	X
	5.4.5.2	Joint design limitations for Class II, III and IV	X	X	X	X	X
	5.4.5.5	Backing	X	X	X	X	X
5.4.6 Electrical parameters	**5.4.6.1**	Amperage	X	X	X	X	X
	5.4.6.2	Voltage	X	X	X	X	X
	5.4.6.3	Travel speed	X	X	X	X	X
	5.4.6.4	Heat input	X	X	X	X	X
	5.4.6.4.1	Bead width limitation for weaving	X	X	X	X	X
	5.4.6.5	Current type	X	X	X	X	X
	5.4.6.6	Polarity type	X	X	X	X	X
	5.4.6.7	Output type	X	X	--	--	X
	5.4.6.8	Pulsed current	X	X	X	--	X
	5.4.6.9	Transfer mode	X	X	--	--	--
	5.4.6.10	Tungsten electrode	--	--	X	--	--
5.4.7 General procedure parameters	**5.4.7.1**	Welding process	X	X	X	X	X
	5.4.7.2	Welding method of application	X	X	X	X	X
	5.4.7.2.1	Manual and semiautomatic applications	X	X	X	X	X
	5.4.7.2.2	Mechanized, automatic, and robotic	X	X	X	X	X
	5.4.7.2.2.1	Adaptive control	X	X	X	X	X
	5.4.7.3	Position	X	X	X	X	X
	5.4.7.4	Preheat temperature	X	X	X	X	X
	5.4.7.5	Interpass temperature	X	X	X	X	X
	5.4.7.6	Mulitpass and single pass welds	X	X	X	X	X
	5.4.7.7	Backgouging (for Q&T materials)	X	X	X	X	X
	5.4.7.7.1	Method of backgouging	X	X	X	X	X
	5.4.7.8	Post Weld Heat Treatments (PWHT)	X	X	X	X	X
	5.4.7.9	Cleaning method	X	X	X	X	X

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TABLE VI. Procedure qualification testing requirements.

Class	Item	Type	CJP	PJP & Flare	Fillet
All Classes	1	Visual Testing (5.9.2.1)	Yes	Yes	Yes
	2	Recommended NDT[4] (5.9)	RT or UT	PT or MT	PT or MT
	3	Reduced Section Tensile (5.7.2)	2	2	-
	4	Macroetch Testing[1] (5.7.4)	4	3	3
Class II, III and IV	5	Guided Bends[1,2] (5.7.3)	4	4 (PJP only)	-
		T < 3/8”	2 Root and 2 Face, Trans., or Long.		
		T ≥ 3/8”	4 Trans. Side or 2 Long. Root + 2 Long. Face		
	6	Bend-Break (5.7.6)	-	-	2
Class I, II, and III	7	Charpy V-Notch (5.7.8)	10 or 15[3]	10 or 15[3]	-
	8	Microhardness Traverse (5.7.5)	1		
Class I	9	Proofing Projectile Impact Test (Appendix A)	Yes	-	-
-	A	Blast Test (Prototype TDP)	No relation to destructive testing or PQR/WPS qualification		
	B	Live Fire (Prototype TDP)			

NOTES:
1: Macros may be substituted for the required transverse or longitudinal bends for armor steels only.
2: For T ≥ 3/8", side bend specimens may be substituted for root and face bend specimens. Guided bends are not required for armor MIL-DTL base metals (see 5.7.3), however longitudinal bends may be performed. In either case, when T < 3/8”, two root and two face bends are required.
3: The number of CVNs required is 10 for joints of similar base material and 15 for joints of dissimilar base material (see 5.7.8).
4: Item 2 is a recommended practice but shall be required for greater than or equal to 120ksi tensile strength wires.

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TABLE VII. Procedure qualification positions for plate.

Qualification Testing per Weld Class			Allowable WPS Qualified Positions		
			Groove		Fillet
Weld Class Type		Position Qualified on PQR	Plate	Pipe/Tube	Plate and Pipe/Tube
Plate Groove	Class I	1G	F	F	F
		2G	F, H	F, H	F, H
		3G[1]	V[1]	V[1]	V[1]
		4G	OH	OH	OH
	Class II, III, & IV	1G	F	F	F
		2G	F, H	F, H	F, H
		3G[1]	F, V[1]	F, V[1]	F, V[1]
		4G	F, OH	F, OH	F, OH
Plate Fillet	Class I	1F	-	-	F
		2F	-	-	F, H
		3F[1]	-	-	V[1]
		4F	-	-	F,OH
	Class II, III, & IV	1F	-	-	F
		2F	-	-	F, H
		3F[1]	-	-	F, V[1]
		4F	-	-	F, H, OH

NOTES:
1: Welding prohibitions of 4.10a. shall apply

TABLE VIII. Procedure qualification positions for pipe and tube.

Qualification testing per class			Allowable WPS qualified positions	
			Plate	
Weld class type		Position Qualified on PQR	Groove	Fillet
Pipe/ tube groove	Class I	1G rotated	F	F
		2G	F, H	F, H
		5G[1]	F, V[1], OH	F, V[1], OH
		2G+5G[1]	All[1]	All[1]
		6G	All[1]	All[1]
Pipe/tube fillet	Class II, III, & IV	1F rotated	-	F
		2F	-	H
		2F rotated	-	F, H
		4F	-	F, V[1]
		5F[1]	-	All[1]

NOTES:
1: Welding prohibitions of 4.10a. shall apply.

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TABLE IX. Procedure qualification test criteria for fillet welds.

Test weld size in (mm)	Test plate thickness[1]		Sizes qualified		
	T1 in (mm)	T2 in (mm)	Single pass (W_1) in (mm)	Multiple pass (W_2) in (mm)	Plate thickness
0.125 (3)	0.25 (6)	0.1875 (5)	≤ 0.125 (3)	-	0.125 in. (3mm) to Unlimited
0.1875 (5)	0.5 (13)	0.1875 (5)	≤ 0.1875 (5)	-	
0.25 (6)	0.75 (19)	0.25 (6)	≤ 0.25 (6)	≥ 0.25 (6)	
0.3125 (8)	1 (25)	0.3125 (8)	≤ 0.3125 (8)	≥ 0.3125 (8)	
0.375 (10)	1 (25)	0.375 (10)	≤ 0.375[2] (10)	≥ 0.375 (10)	
0.5 (13)	1 (25)	0.5 (13)	≤ 0.5[2] (13)	≥ 0.5 (13)	
0.625 (16)	1 (25)	0.625 (16)	≤ 0.625[3] (16)	≥ 0.625 (16)	
0.75 (19)	1 (25)	0.75 (19)	≤ 0.75[3] (19)	≥ 0.75 (19)	
> 0.75 (19)	1 (25)	1 (25)	≤ 0.75[3] (19)	> 0.75 (19)	

NOTES:
1: The material thicknesses for the joint design application observed in production and qualification may be used in lieu of tabulated thicknesses above.
2: The maximum single pass using manual SMAW or the other conventional semiautomatic arc welding processes that can repeatedly achieve fusion and penetration soundness under most circumstances.
3: The maximum single pass using unconventional, mechanized, or automated processes with high deposition and high current density characteristics.

TABLE X. Procedure qualification thickness limitations.

PQR Test Assembly Thickness in (mm)	Base Metal Thickness (T) Range Qualified in (mm)[2, 3]		Weld Deposit Thickness (t) Qualified Maximum in (mm)[1]
0.125 ≤ T ≤ 0.375 (3 ≤ T ≤ 10)	½T	2T	2t
0.375 < T < 1 (10 < T < 25)	½T	2T	2t
T ≥ 1 (25)	½T	Unlimited	Unlimited

NOTES:
1: Excludes face and root reinforcement.
2: 1.1T for GMAW-S
3: If any single pass is greater than 3/8 in. thick, then the base metal thickness qualified is 1.1T

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TABLE XI. Procedure cross qualification.

Grouping of Base/Filler Materials For Welding Procedure Cross-Qualification				
Category	**Base Material [3]**		**Filler Material [1,2]**	**Qualified For Categories [4]**
	A	**B**		
Ferritic materials/shielded metal arc process (SMAW)				
1	S-11B	S-11B	A-5A (12018, 11018, or 10718)	1 through 6, 9, and 18 through 20
2	S-11B	S-11A	A-5A (12018, 11018, or 10718)	2 through 6, 9, and 18 through 20
3	S-11B	S-11D	A-5A (12018, 11018, or 10718)	3 through 6, 9, and 18 through 20
4	S-11B	S-11A	A-5A	4 through 6, 9, and 18 through 20
5	S-11B	S-11C	A-5A	5, 6, 9, and 18 through 20
6	S-11B	S-2	A-5A, or the A-3A or A-2A electrodes specifically permitted by table IV	6, 9, and 18 through 20
		S-1		
7	S-11A	S-11A	A-5A (12018, 11018, or 10718)	7 through 9, and 18 through 20
	S-11D	S-11D		
8	S-11A	S-11A	A-5A (10018 only)	8, 9, and 18 through 20
	S-11C	S-11C		
	S-11D	S-11D		
9	S-11A	S-2	A-5A or A-3A	9, and 18 through 20
	S-11C	S-1	A-2A electrodes specifically permitted by table IV	
	S-11D			
10	S-5	S-5	A-7A or A-6A	10 through 15, and 20
11	S-5	S-4	A-7A or A-6A	11 through 15, and 20
13	S-5	S-3	A-6A or A-3A	13, 15, and 20
14	S-4	S-4	A-6A	14, 15, and 20
15	S-4	S-3 S-1	A-6A or A-3A	15 and 20
16	S-3	S-3	A-3A or A-2A	16 through 20
17	S-3	S-1	A-3A or A-2A	17 through 20
18	S-2	S-2	A-3A or A-2A	18 through 20
19	S-2	S-1	A-3A or A-2A	19 and 20
20	S-1	S-1	A-2A	20

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TABLE XI. Procedure cross qualification – Continued.

Grouping of Base/Filler Materials For Welding Procedure Cross-Qualification				
Category	Base Material [3]		Filler Material [1,2]	Qualified For Categories [4]
	A	B		
Ferritic materials/gas tungsten-arc (GTAW), gas metal arc (GMAW)				
1	S-11B	S-11B	A-5B (120S-1 only)	1 through 5, 8, and 16 through 18
2	S-11B	S-11A	A-5B (120S-1 or 110S-1)	2 through 5, 8, and 16 through 18
3	S-11B	S-11D	A-5B (120S-1 or 110S-1)	3 through 5, 8, and 16 through 18
4	S-11B	S-11A S-11C	A-5B	4, 5, 8, , and 16 through 18
5	S-11B	S-2 S-1	A-5B; or A-3B or A-2B types specifically permitted by table IV	5, 8, and 16 through 18
6	S-11A S-11D	S-11A S-11D	A-5B (120S-1 only)	6 through 8, 17, and 18
7	S-11A S-11C S-11D	S-11A S-11C S-11D	A-5B (100S only)	7, 8, 17, and 18
8	S-11A S-11C S-11D	S-2 S-1	A-5B; or A-3B or A-2B types specifically permitted by table IV	8, 17 and 18
9	S-5	S-5	A-7B or A-6B	9 through 13, and 18
10	S-5	S-4	A-7B or A-6B	10 through 13, and 18
11	S-5	S-3 S-1	A-7B or A-6B or A-3B	11, 13, and 18
12	S-4	S-4	A-6B	12, 13, and 18
13	S-4	S-3 S-1	A-6B or A-3B	13 and 18
14	S-3	S-3	A-3B or A-2B	14 through 18
15	S-3	S-1	A-3B or A-2B	15 and 18
16	S-2	S-2	A-3B or A-2B	16 through 18
17	S-2	S-1	A-3B or A-2B	17 and 18
18	S-1	S-1	A-2B	18

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TABLE XI. Procedure cross qualification – Continued.

Grouping of Base/Filler Materials For Welding Procedure Cross-Qualification				
Category	**Base Material [3]**		**Filler Material [1,2]**	**Qualified For Categories [4]**
	A	**B**		
Ferritic materials/submerged-arc process (SAW)				
1	S-11A	S-11A	A-5C (120S-1 wire and 1F flux)	1 through 3, and 6 through 8
	S-11D	S-11D		
2	S-11A	S-11A	A-5C (100S wire and 1F or 2F flux only)	2, 3, and 6 through 8
	S-11D	S-11D		
3	S-11A	S-2	A-5C (100S wire and 1F or 2F flux) or A-3C or A-2C wire and flux specifically permitted by table IV	3, and 6 through 8
	S-11C	S-1		
	S-11D			
4	S-3	S-3	A-3C; or A-2C	4 through 8
5	S-3	S-1	A-3C; or A-2C	5 through 8
6	S-2	S-2	A-2C	6 through 8
7	S-2	S-1	A-2C	7 and 8
8	S-1	S-1	A-2C	8
Category	**Base Material [3]**		**Filler Material [1,2]**	**Qualified For Categories [4]**
	A	**B**		
Ferritic materials/flux cored-arc process (FCAW)				
1	S-11A	S-11A	A-5D (101TC or 101TM only)	1 through 4, and 7 through 9
2	S-11D	S-11D	A-5D (101TC or 101TM only)	2 through 4, and 7 through 9
3	S-11A	S-11A	A-5D (101TC or 101TM only)	3, 4, and 7 through 9
	S-11C			
	S-11D	S-11C		
4	S-11A	S-2	A-5D; or A-2D or 3D electrodes specifically permitted by table IV	4, and 7 through 9
	S-11C	S-1		
	S-11D			
5	S-3	S-3	A-2D or A-3D	5 through 9
6	S-3	S-1	A-2D or A-3D	6 through 9
7	S-2	S-2	A-2D or A-3D	7 through 9
8	S-2	S-1	A-2D or A-3D	8 and 9
9	S-1	S-1	A-2D	9

NOTES:
1: The higher number A-group filler material qualifies the lower number A-group filler material, but not vice versa.
2: In all instances where A-5A, B, or C filler materials are listed, the higher-strength materials are only qualified when the original qualification was performed using the same (or higher) strength filler material.
3: Procedure qualification with any material listed in column A and with any material listed in column B qualifies all the material combinations in that category.
4: Base metal/filler metal combinations not listed in this table require separate procedure qualification

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TABLE XII. Heat input guidance for 120ksi wires.

Welding process	Wire Dia. (in)	Thickness (T) (in) [2]	Position	Heat input (kJ/in) [1]	
				Min	Max
SMAW	1/8	T < 1.0	As Qualified	As Qualified	As Qualified
		T ≥ 1.0	Flat, Horizontal	30	50
			Vertical	40	55
			Overhead	30	55
	5/32	T < 0.75	As Qualified	As Qualified	As Qualified
		0.75 ≤ T <1.0	Flat, Horizontal	30	45
		T ≥ 1.0	Flat, Horizontal	30	55
GMAW Spray	0.045 0.062	T < 0.75	As Qualified	As Qualified	As Qualified
		0.75 ≤ T <0.875	Flat, Horizontal	30	46
		T ≥ 0.875	Flat, Horizontal	30	55
GMAW Pulsed	0.045	T < 0.75	As Qualified	As Qualified	As Qualified
		0.75 ≤ T <0.875	Vertical, Flat, Horizontal	28	45
		0.875 ≤ T ≤ 1.0	Vertical, Flat, Horizontal	28	55
		T > 1.0	Vertical	34	55
			Flat, Overhead	28	55
SAW	0.062 0.093	T < 0.75	As Qualified	As Qualified	As Qualified
		0.75 ≤ T <0.875	Flat	40	50
		T ≥ 0.875	Flat	40	55

NOTES:

1. Minimum and maximum heat input refers to the averages for each weld pass.
2. For welds between members of different thicknesses (T), the welding procedure used shall be acceptable to weld both thicknesses.

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TABLE XIII. Preheat and interpass temperatures.

Material Group Number	Materials Identification	Thickness (in)[2]	Minimum Preheat and Interpass Temperature (°F)		Maximum Preheat and Interpass Temperature (°F)	Supplementary Provisions for Preheat and Interpass Temperature
			Groups 1, 2 (CEN ≤ 0.43)	Groups 3, 4 (CEN > 0.43)		
S-1	Carbon Steel	T ≤ 1.25	60	125	-	For GTAW, GMAW, MCAW, & SAW Process
		1.25 ≤ T ≤ 1.75	60	175		
		T >1.75	150	200		
		T ≤ 1.25	60	150	-	For SMAW and FCAW Process
		1.25 ≤ T ≤ 1.75	150	200		
		T >1.75	200	200		
S-2	Quench and Tempered Carbon Steel	Same as S-1	Same as S-1		300	Same as S-1
S-3	Carbon Molybdenum Steel	All	175		500	Where either the specified min. tensile strength is >70 ksi or thickness is >0.625 in. (5/8)
		All	60		500	For all other S-3 Materials
S-4	Alloy Steels (Cr content 0.75%-2%, Total Alloy Content 2-2.75% max.)	All	250		500	Where either the specified min. tensile strength is >60 ksi or thickness is >0.5 in.
		All	60		500	For all other S-4 Materials
S-5	Alloy Steels (Total alloy content <10% max., Cr-Mo)	All	400		600	Where either the specified tensile strength is >60 ksi or has both specified chromium content >6.0% and thickness >0.5 in.
		All	300		600	For all other S-5 materials
S-11A	Quenched and Tempered alloy steels	T < 1.125	125		300	HY 80 & HY 100 Type Steel
		1.125 ≤ T ≤ 2.75	150			
		T > 2.750	200			

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TABLE XIII. Preheat and interpass temperatures – Continued.

Material Group Number	Materials Identification	Thickness (in)[2]	Minimum Preheat and Interpass Temperature (°F)	Maximum Preheat and Interpass Temperature (°F)	Supplementary Provisions for Preheat and Interpass Temperature
S-11B	Quenched and Tempered Alloy Steels	T ≤ 2.750	150	300	TBD
		T > 2.750	200		
S-11C	Age Hardening Alloy Steel	All	60	300	HSLA 80 Type
S-11D	Age Hardening Alloy Steel	All	60	300	HSLA 100 Type
S-15A	Armor Steel MIL-DTL-12560	T ≤ 0.5	60	500	Class 1-2
		T > 0.5	125		
S-15B		T ≤ 0.5	60	500	Class 4A
		T > 0.5	125		
S-15C		All	125	300	Class 4B
S-15D	Armor Steel MIL-DTL-46100	All	125	300	Class 1
S-15E		All	125	300	Class 2
S-15F	Armor Steel MIL-A-11356	All	125	500	Class 1
A-2	Carbon and low alloy steel filler materials	All	-	-	-
A-3	Carbon and low alloy steel filler materials	All	60	400	-
A-5[1]	Low alloy, high strength steel filler materials	t < 0.5	60	300	-
		0.5 ≤ t < 1.125	125		
		1.125 ≤ t ≤ 2.75	150		
		t > 2.750	200		
A-6[1]	CrMo steel filler materials (1.0% to 2.5% Cr, 0.4% to 1.2% Mo)	All	300	500	-
A-7[1]	CrMo steel filler materials (4.0-13.5% Cr, 0.4-1.4% Mo)	All	300	500	-
A-8	High alloy steel (austenitic) filler materials	All	60	350	-
A-9	High alloy steel filler materials (duplex stainless)	All	-	400	-
A-43	Nickel base alloy filler materials	All	60	300	-

Note 1: Any 120ksi tensile strength or greater filler material requires a 225 ° F minimum preheat and 300 ° F maximum interpass with no exceptions.
Note 2: Thickness as denoted by T indicates the plate thickness of basematerial; thickness denoted by t indicates thickness of weld deposit.

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TABLE XIV. Minimum travel speed example (IPM).

Volts / Amps	20	20.5	21	21.5	22	22.5	23
170	5.8	6.0	6.1	6.3	6.4	6.6	6.7
175	6.0	6.2	6.3	6.5	6.6	6.8	6.9
180	6.2	6.3	6.5	6.6	6.8	6.9	7.1
185	6.3	6.5	6.7	6.8	7.0	7.1	7.3
190	6.5	6.7	6.8	7.0	7.2	7.3	7.5
195	6.7	6.9	7.0	7.2	7.4	7.5	7.7
200	6.9	7.0	7.2	7.4	7.5	7.7	7.9
205	7.0	7.2	7.4	7.6	7.7	7.9	8.1
210	7.2	7.4	7.6	7.7	7.9	8.1	8.3

NOTE: As qualified heat input = 35000 J/in

TABLE XV. Welder and welding operator qualification criteria.

Type of test weld	**T of test plate** in (mm)	**Destructive tests**[1]	**Qualifies welder for T inclusive** in (mm)	
			Min	**Max**
Welder groove weld for plate[2]	0.375 (10) ≤ T ≤ 0.5 (13)	2 bends	0.125 (3)	0.75 (19)
	T ≥ 1” (25)	2 bends	0.125 (3)	Unlimited
Welding operator groove weld for plate	1 (25)	2 bends	0.125 (3)	Unlimited
Plug	0.375 (10) or greater	1 macroetch	0.125 (3)	Unlimited
Fillet	0.5 (13)	1 fillet break test and 2 macroetch	0.125 (3)	Unlimited
Pipe or tube	Any	4 macroetch	0.125 (3)	Unlimited

NOTES:
1: Volumetric examination may be substituted for bend testing (see 5.6.4.1).
2: For a previously qualified WQR with .5” < T < 1”, the range qualified shall be .125 (3mm) to 2T.

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TABLE XVI. Welder and welding operator qualification positions for plate.

Qualification testing per class		Allowable qualified positions		
		Groove		Fillet
Weld class type	**Test position**	**Plate**	**Pipe/tube[1]**	**Plate and pipe/tube[1]**
Plate groove	1G	F	F	F
	2G	F, H	F, H	F, H
	3G	F, V[2]	F, V[2]	F, H, V[2]
	4G	F, OH	F, OH	F, OH
	3G[2] & 4G	F, V[2], OH	F, V[2], OH	All[2]
	2G, 3G[2], & 4G	All[2]	All[2]	All[2]
Plate fillet	1F	-	-	F
	2F	-	-	F, H
	3F	-	-	F, V[2]
	4F	-	-	F, OH
	3F[2] & 4F	-	-	All[2]

NOTES:
1: Only qualifies for pipe or tube ≥ 24-in in diameter.
2: Welding prohibitions of 4.10a. shall apply.

TABLE XVII. Welder and welding operator qualification positions for pipe and tube.

Qualification testing per class		Allowable qualified positions		
		Groove		Fillet
Weld type	**Test position**	**Plate**	**Pipe/tube**	**Plate and pipe/tube**
Pipe/tube groove	1G Rotated	F	F	F
	2G	F, H	F, H	F, H
	5G[1]	F, V[1], OH	F, V[1], OH	F, V[1], OH
	(2G+5G)[1]	All[1]	All[1]	All[1]
	6G	All[1]	All[1]	All[1]
Pipe/tube fillet	1F Rotated	-	-	F
	2F	-	-	F, H
	4F	-	-	F, OH
	5F[1]	-	-	All[1]

NOTES:
1: Welding prohibitions of 4.10a. shall apply.

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TABLE XVIII. Welder and welding operator essential variables.

Essential variable	Requalification of personnel is required when this change occurs	Personnel	
		Welders	Welding operators
Weld classification	A change from Class II, III, or IV to Class I, except as permitted by this document	X	X
Welding process (see 4.7)	Change to an unqualified process	X	X
Filler material group (see table IV)	Change from one filler A-Group to another, except as permitted by table XIX	X	X
Filler material form	A change in filler material form from solid to cored wire, or powdered.	X	X
Position (see table XVI and table XVII)	Change to an unqualified position	X	X
Base material thickness (see table X)	Change to an unqualified thickness	X	X
Backing	Exclusion of the use of backing	X	X
Stringer vs. Weave Passes (see 5.10.12)	A change from stringer to weave, or vice versa; if not combined in the same test coupon	X	X
Number of electrodes	When going from single electrode to multiple, but not from multiple to single	X	X
Change in method of application	Change from manual, semiautomatic, mechanized, or automatic welding to any other method of application	X	X
Arc voltage control	A deletion of automatic arc voltage control (GTAW only)	-	X
Adaptive control	Addition or deletion of adaptive controls	-	X

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TABLE XIX. Welder and welding operator cross qualification categories.

	Qualified with:	Qualified for:	
Category	**Filler metal group**	**Shielded metal arc**	
1	A-5A	1 through 3	
2	A-2A, A-3A, A-6A	2	
3	A-7A	3	
4	A-8A, A-9A	4	
5	A-43A, A-45A	5	
Category	**Filler metal group**	**Gas metal-arc**	**Gas tungsten-arc**
1	A-5B	1 through 3	1 through 3
2	A-2B, A-3B	2	2
3	A-7B	3	3
4	A-8B, A-9B	4	4
5	A-43B, A-45B	5	5
Category	**Filler metal group**	**Submerged-arc**	
1	A-5C	1 through 2	
2	A-2C, A-3C	2	
3	A-8B	3 through 4	
4	A-43B	3 through 4	
Category	**Filler metal group**	**Flux-cored-arc**	
1	A-5D	1 and 2	
2	A-2D	2	
3	A-8D	3	

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TABLE XX. Visual testing acceptance criteria.

Discontinuity Description	Base Material Thickness (T) or Weld Size (S)	Discontinuity Limit[1]	Weld Classification	
			Class I & II	Class III & IV
Arc strikes, armor	All thicknesses	≤ 1/32 in (1 mm) in depth after removal and blend	X	X
Arc strikes, non-armor	T < 0.5 in (12 mm)	≤ 1/32 in (1 mm) in depth after removal and blend	X	X
	T ≥ 0.5 in (12 mm)	≤ 1/16 in (2 mm) in depth after removal and blend		
Burn-through[5]	All thickness	None	X	X
Concavity, fillet welds	All thickness	Throat and size shall be met[2]	X	X
Convexity, fillets welds	See Weld Profile		X	X
Corner-melt	All thickness	≤ 1/32 in (1 mm) or 10%T, whichever is less	X	
		≤ 1/16 in (2 mm) or 10%T, whichever is less		X
Cracks	All thickness	None	X	X
Crater cracks	All thickness	None	X	X
End/edge melt	All thickness	≤ 1/32 in (1 mm) or 10%T, whichever is less	X	
		≤ 1/16 in (2 mm) or 10%T, whichever is less		X
Depth of fusion	All thickness	Shall meet the requirements of the fabrication document or drawing	X	X
Incomplete fusion	All thickness	None	X	X
Incomplete joint penetration	All thickness	None[8,9]	X	X
Inclusions	All thickness	Shall be treated as porosity	X	X
Joint mismatch, butt joints	All thickness	20%T, but not to exceed 1/8 in (3mm)	X	X
Meltback[4,5]	All thickness	None	X	X
Melt-through[5]	All thickness	≤ .2T	X	X
Missing weld(s) or Short Welds	All thickness	None	X	X
Notching[4,5,6]	All thickness	None	X	X

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TABLE XX. Visual testing acceptance criteria – Continued.

Discontinuity Description	Base Material Thickness (T) or Weld Size (S)	Discontinuity Limit[1]	Weld Classification	
			Class I & II	Class III & IV
Off-joint welds[4,5]	T ≤ 3/16 in (5 mm)	25%W (width of weld face)		X
Overlap	All thickness	None	X	X
Porosity, single (max)	All thickness	≤ 3/32 in (2.5 mm)	X	X
Porosity, single (min)		≥ 1/32 in (1 mm)		
Porosity, summation		≤ 3/8 in (9.5 mm) in 1 in (25 mm) of linear weld and ≤ 3/4 in (19 mm) in 12 in (305 mm) of linear weld		
Porosity, cluster		1 in 4 in (10 cm) of linear weld		
Porosity, aligned		none		
Reentrant (toe or bead) angle, fillet	See Overlap and Weld Profile		X	X
Reentrant (toe) angle, groove				
Skip[5,7]	All thickness	None	X	X
Spatter	All thickness	See 5.11.2.2	X	X
Undercut	T < 1/8 in (3 mm)	≤ 10%T	X	X
	T ≥ 1/8 in (3 mm)	≤ 1/32 in (0.8 mm)		
	1/8 in (3 mm) ≤ T < 1.0 in (25 mm)	≤ 1/32 in (0.8 mm), max of 1/16 in (1.6 mm) for 2 in (50 mm) in 12 in (305 mm) of linear weld		X
	T ≥ 1.0 in (25 mm)	≤ 1/16 in (1.6 mm)		
Underfill, groove and flare groove welds	All thickness	None, or shall meet the requirements of the fabrication document or drawing	X	X
Undersized welds, fillet welds	S ≤ 3/16 in (5 mm)	≤ 1/16 in (2 mm)	X	X
	1/4 in (6 mm)	≤ 3/32 in (2.5 mm)		
	S ≥ 5/16 in (8 mm)	≤ 1/8 in (3 mm)		
	Total weld length ≤ 10%[3]			
Unspecified/added welds or weld length	All thickness	None	X	X
Weld profile, general	All thickness	See AWS D1.1/D1.1M	X	X
Weld reinforcement, groove welds	T ≤ 1.0 in (25mm)	≤ 1/8 in (3 mm)	X	X
	1.0 in (25 mm) < T ≤ 2.0 in (50 mm)	≤ 3/16 in (5 mm)		

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TABLE XX. Visual testing acceptance criteria – Continued.

Discontinuity Description	Base Material Thickness (T) or Weld Size (S)	Discontinuity Limit[1]	Weld Classification	
			Class I & II	Class III & IV
Weld size, plug, slot, and edge	All thickness	Shall meet the requirements of the fabrication document or drawing	X	X

NOTES:
1. Weld deposit thickness (t), base metal thickness (T).
2. An oversized leg size may be deposited with a concave throat with no further disposition as long as the required leg size and throat from the drawing are achieved. In all cases, excessive concavity should be considered with respect to weld bead centerline cracking via surface profile induced cracking. In cases where the leg size is achieved but the throat is not, the undersized throat dimension allowance and underrun (weld length) limit shall revert to the same values for undersized fillet weld leg sizes.
3. The total length limitation for undersized welds shall be for the above weld sizes.
4. Most commonly associated with mechanized or automated processes only.
5. Most commonly associated with sheet steel applications.
6. A weld heat related burn back of material at the end of the joint resulting in a notch (see AWS D8.8).
7. An unwelded portion of a designated weld (see AWS D8.8)
8. VT for this applies during welding of a CJP double-sided weld when the weld root and backgouge (or no backgouge, as-applicable) are to be inspected prior to welding on side two.
9. Also applies when VT is accessible to the weld root and root surface of a CJP weld made from one side; with or without temporary backing and without permanent backing.

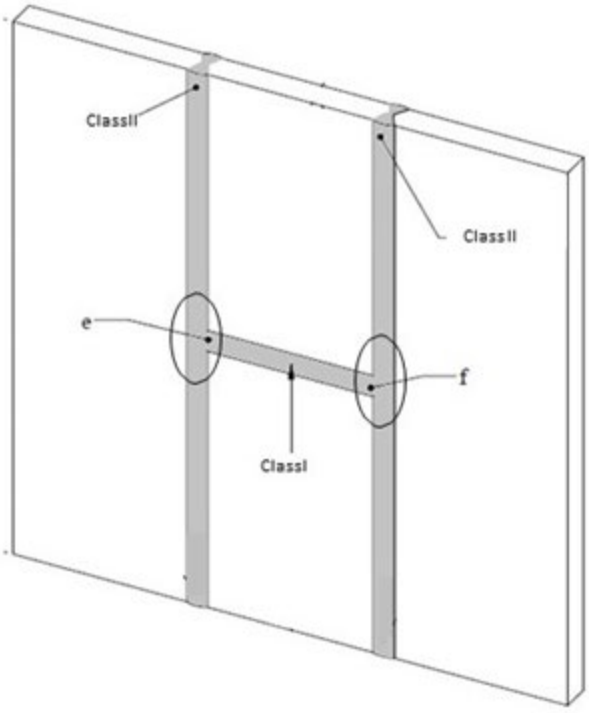
TABLE XXI. NDT per weld classification.

Class I[2] and II[2]	
CJP	***PJP/Fillet***
Volumetric	Surface
Class III[2]	
CJP	***PJP/Fillet***
Surface	Surface
Class IV	
CJP	***PJP/Fillet***
Visual	Visual

Volumetric = RT, UT, CT
Surface = MT, PT[1], ET
Visual = VT[2]

NOTES:
1: PT shall not be used for in-process inspection
2: All classes of welds require visual inspection

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Example:

If Class I weld at center section butt joint is done as a sub-assembly and then later gets married to the outer plate details with welds originally designed as Class II, then the intersection about a 2" radius at points e and f would now be subjected to all the higher classification inspection requirements.

FIGURE 1. Overlap visual.

WELD PER MIL-STD-3040. WELD CLASSES SHALL BE AS INDICATED AT THE WELD SYMBOL

FIGURE 2. Recommended engineering drawing weld note.

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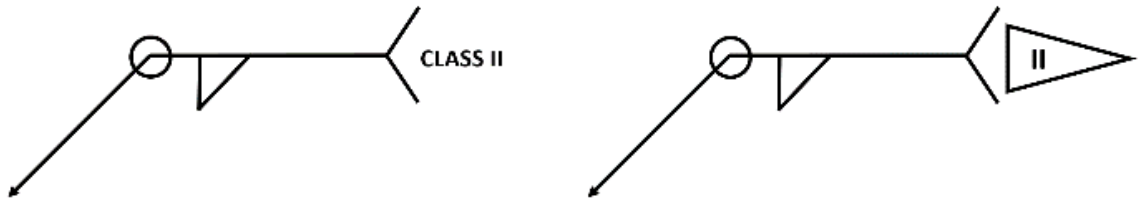


FIGURE 3. Welding symbol classifications.

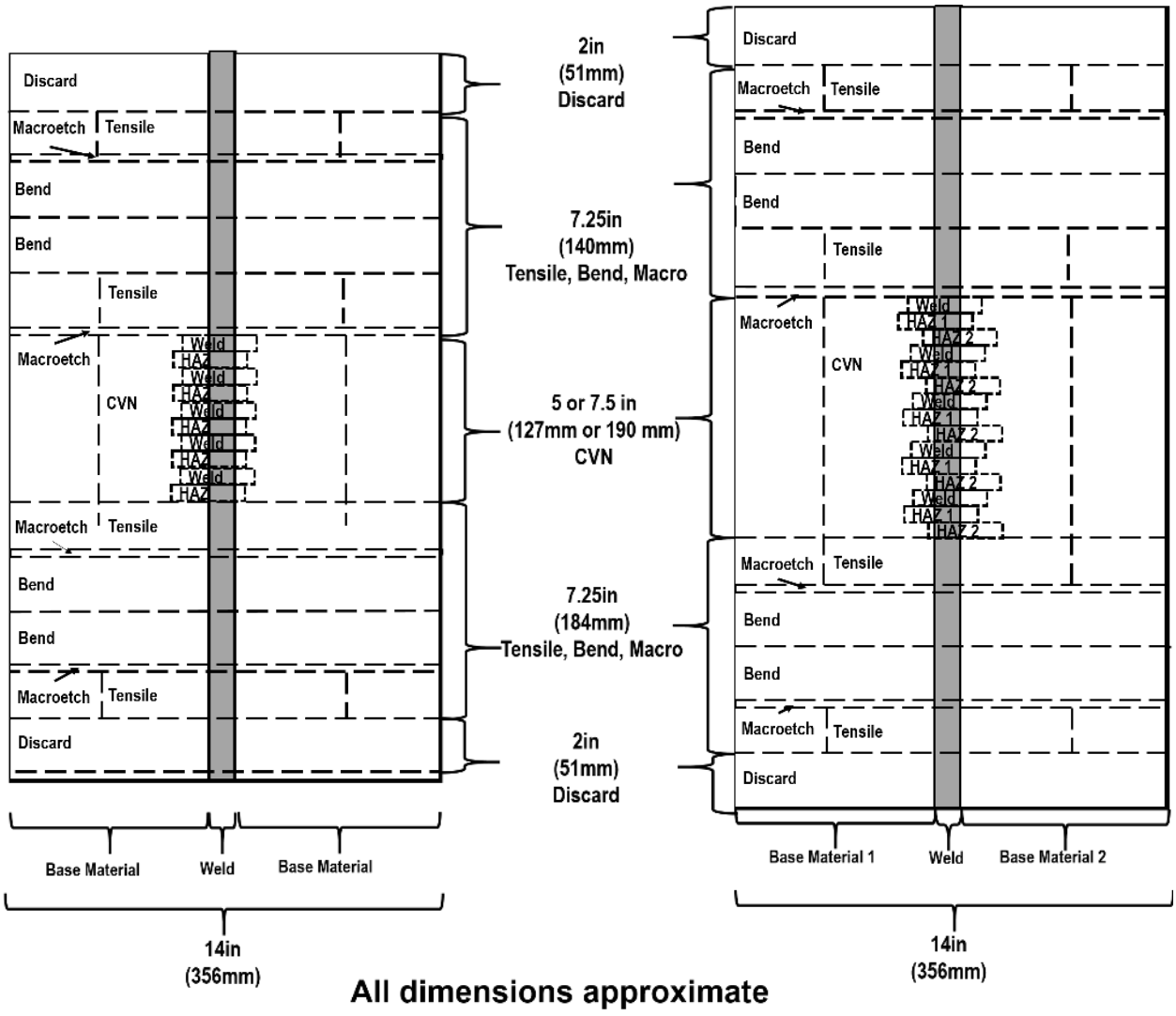
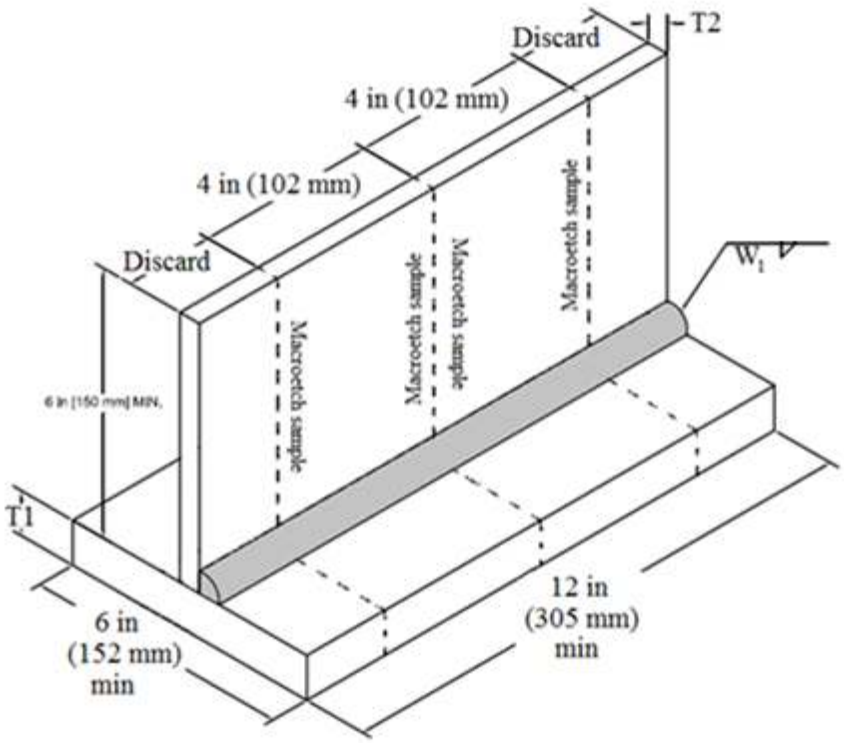


FIGURE 4. Groove procedure qualification mechanical test plate.

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Note:

1. Three total macroetch tests shall be completed: each cut end and either side of the center cut.
2. Both center pieces shall be used for fillet weld break test macroetch testing.

FIGURE 5. Fillet procedure qualification test plate.

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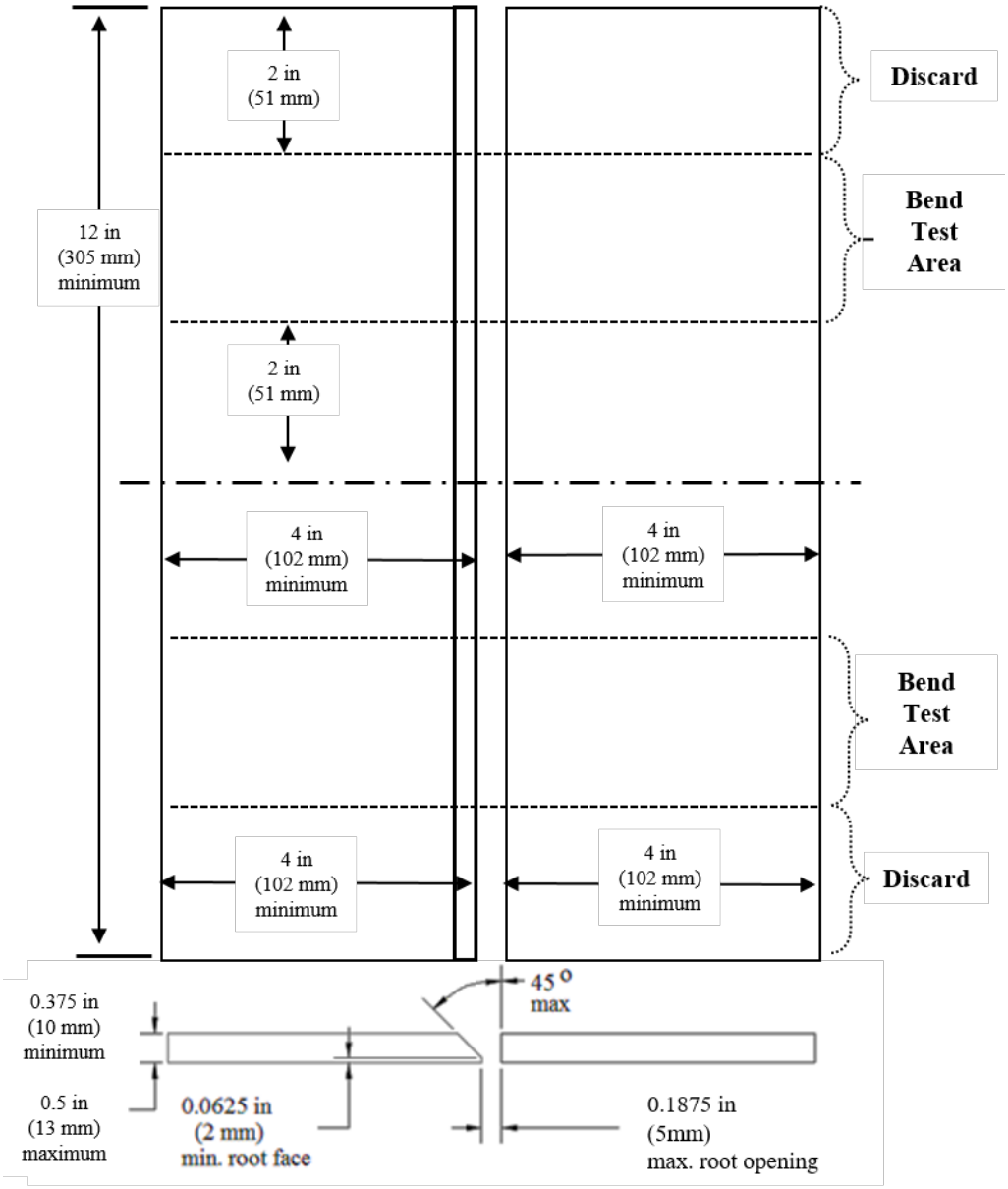


FIGURE 6. Class I welder and welding operator qualification plate thickness less than 0.5 inches inclusive.

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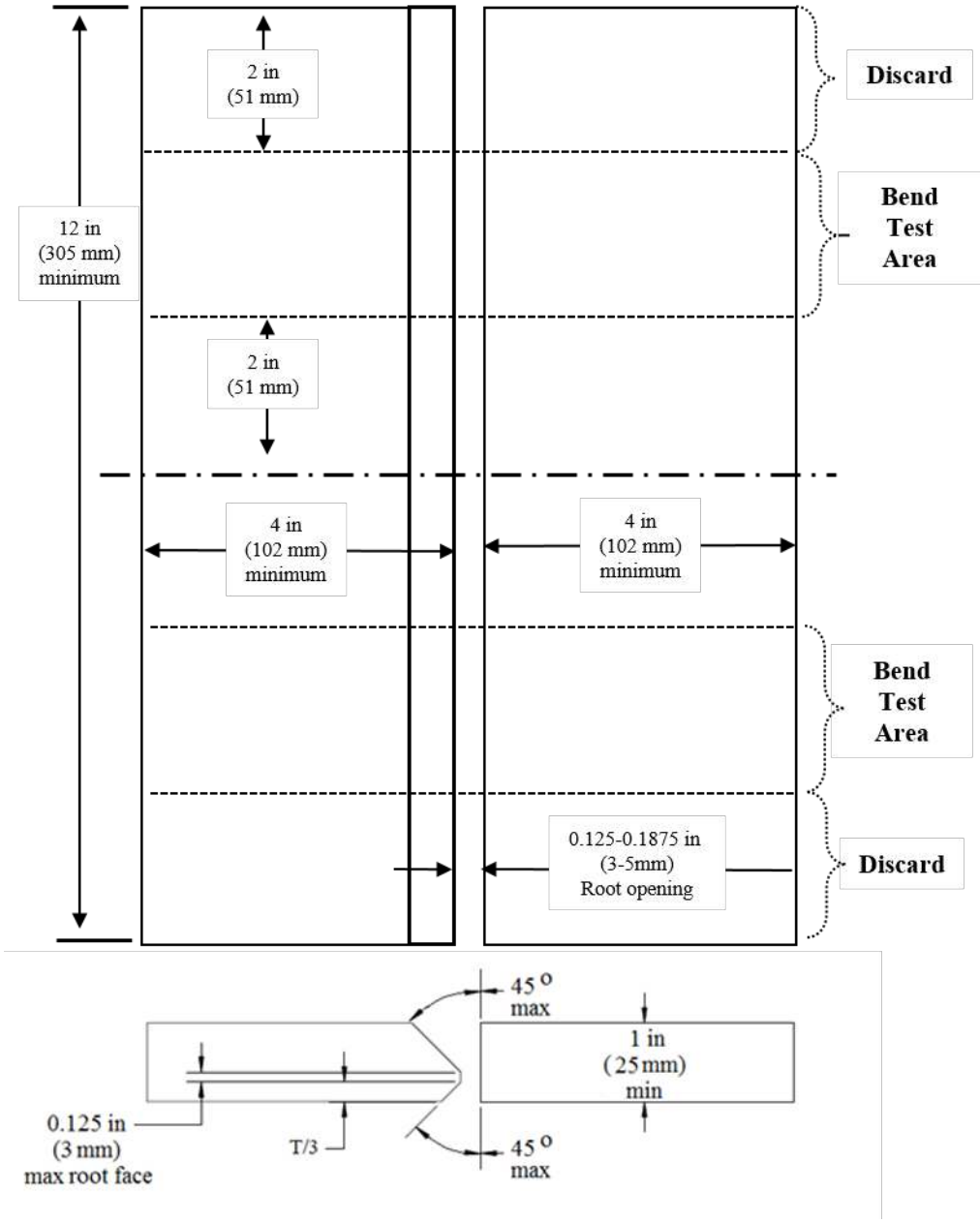
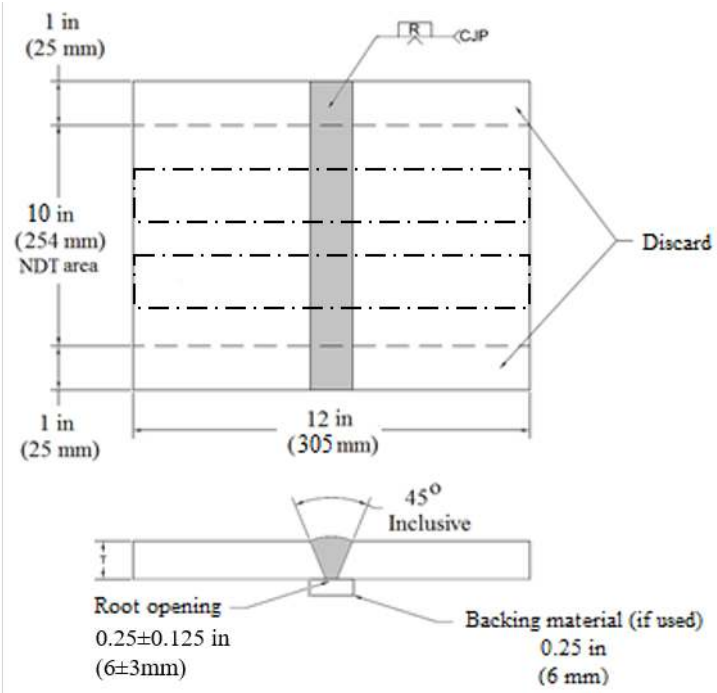


FIGURE 7. Class I welder and welding operator qualification plate thickness greater than 1.0 inches.

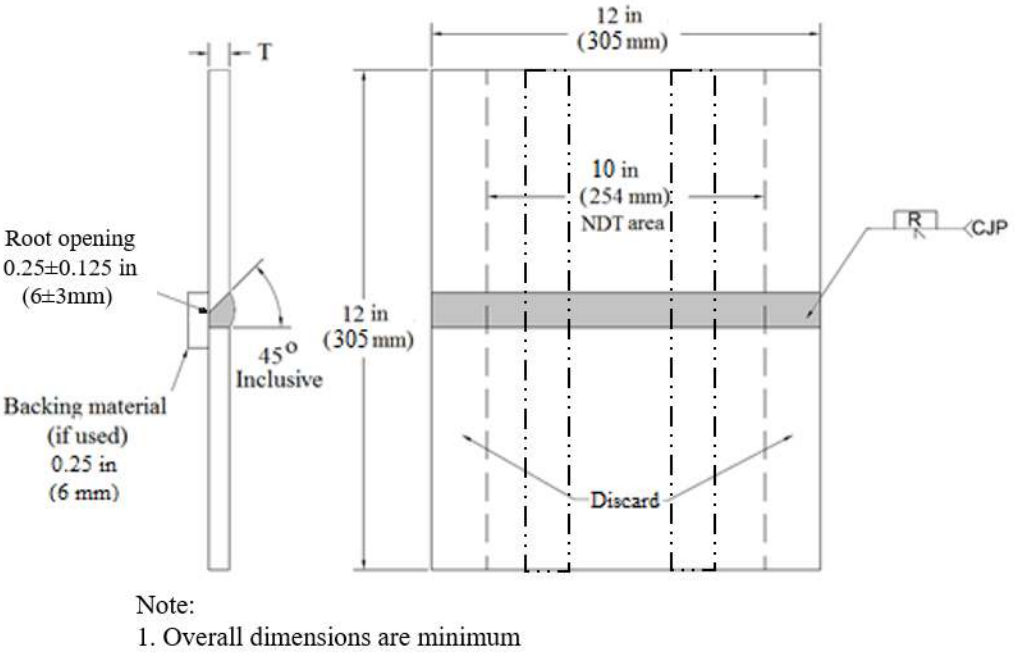
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Note:
1. Overall dimensions are minimum

FIGURE 8. Class II, III, and IV welder qualification groove plate.

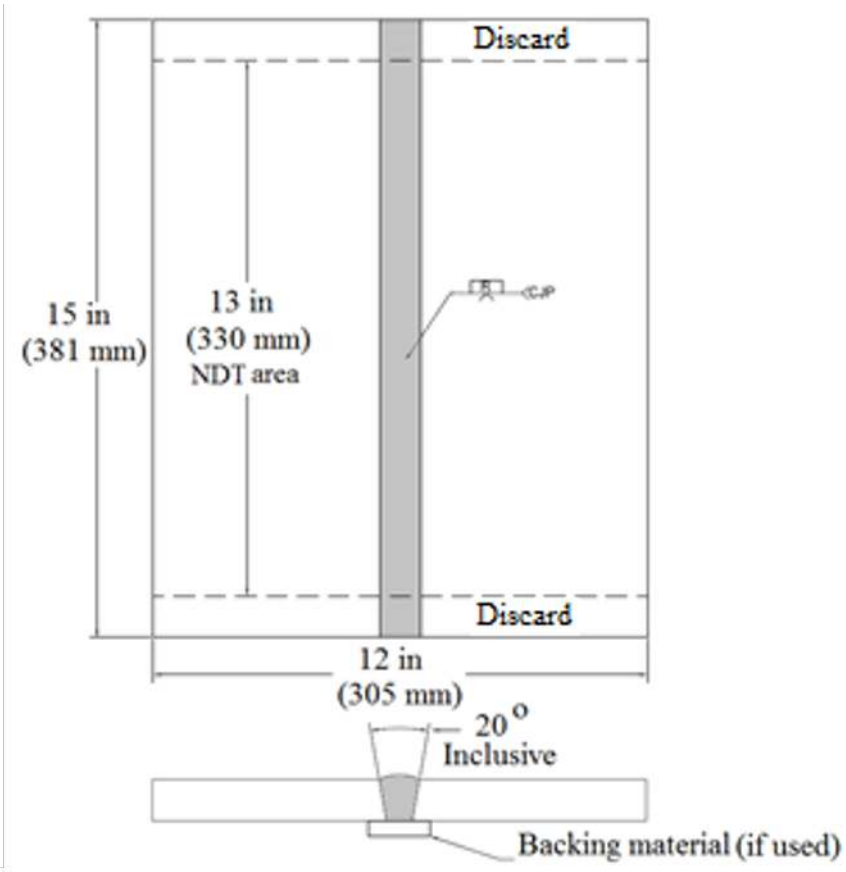
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Note:
1. Overall dimensions are minimum

FIGURE 9. Class II, III, and IV welder qualification groove plate horizontal.

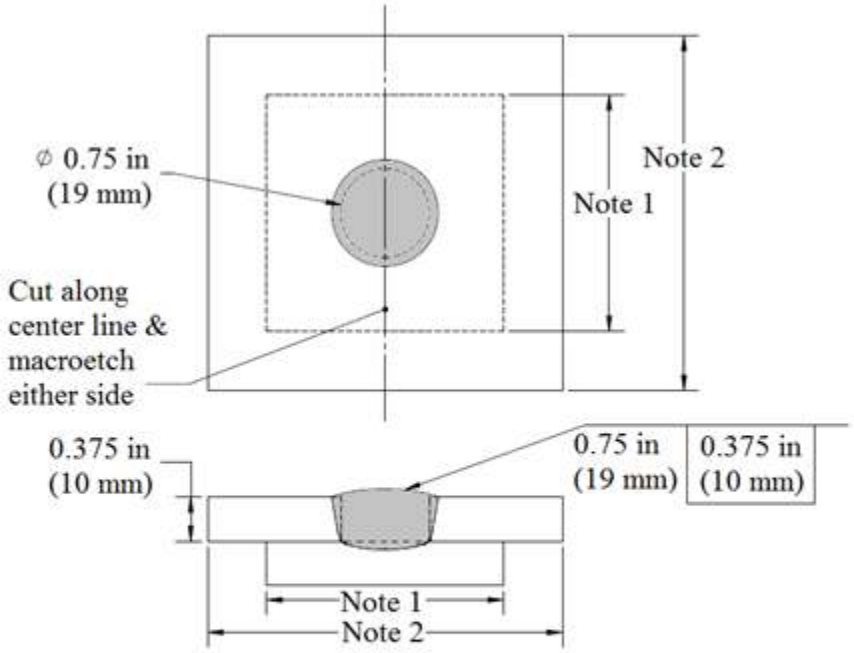
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Note:
1.Overall dimension are minimum.
2. Root opening must be a minimum of 3/16" (5mm) if backing material is used
3. If backing is used, it shall be 3/8" (10mm) to 1/2" (13mm) thick and a minimum width of 1.5" (38mm) when mechanical testing is intended, or 3" (76mm) when RT is to be used.

FIGURE 10. Class II, III, and IV welding operator qualification groove plate.

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NOTE:
1. 2 in. (51 mm) minimum for welder, 3 in. (76 mm) minimum for welding operator.
2. 3 in. (76 mm) minimum for welder, 5 in. (127 mm) minimum for welding operator.

FIGURE 11. Class II, III, and IV welder and welding operator qualification plug weld assembly.

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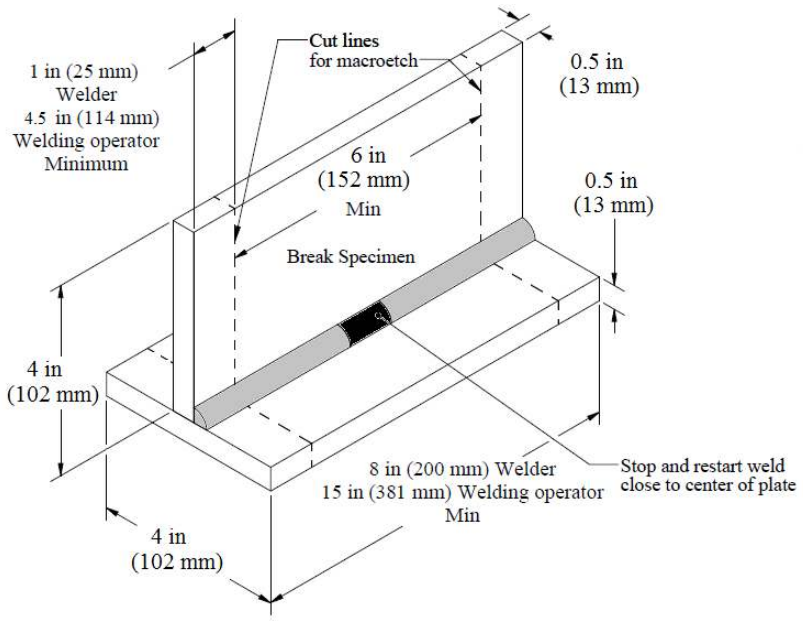
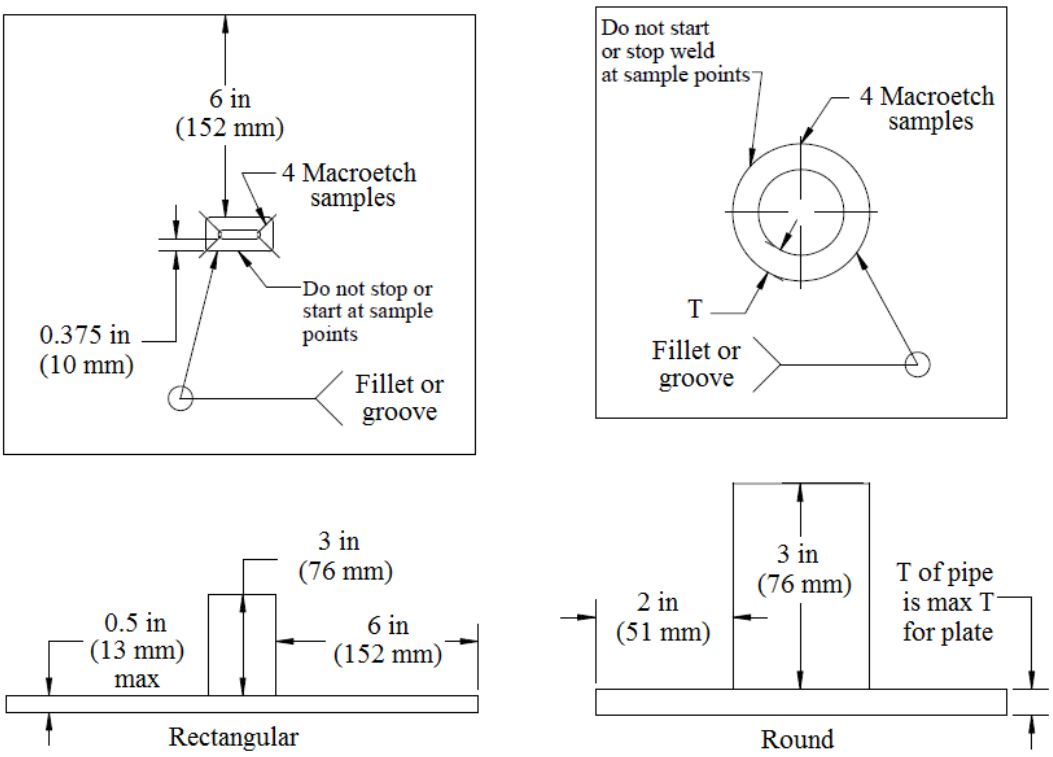


FIGURE 12. Class II, III, and IV welder qualification fillet weld assembly.



Note:
1. Unless specified, all dimensions are minimum.

FIGURE 13. Class II, III, and IV PQR, welder, and welding operator qualification pipe or tube assembly.

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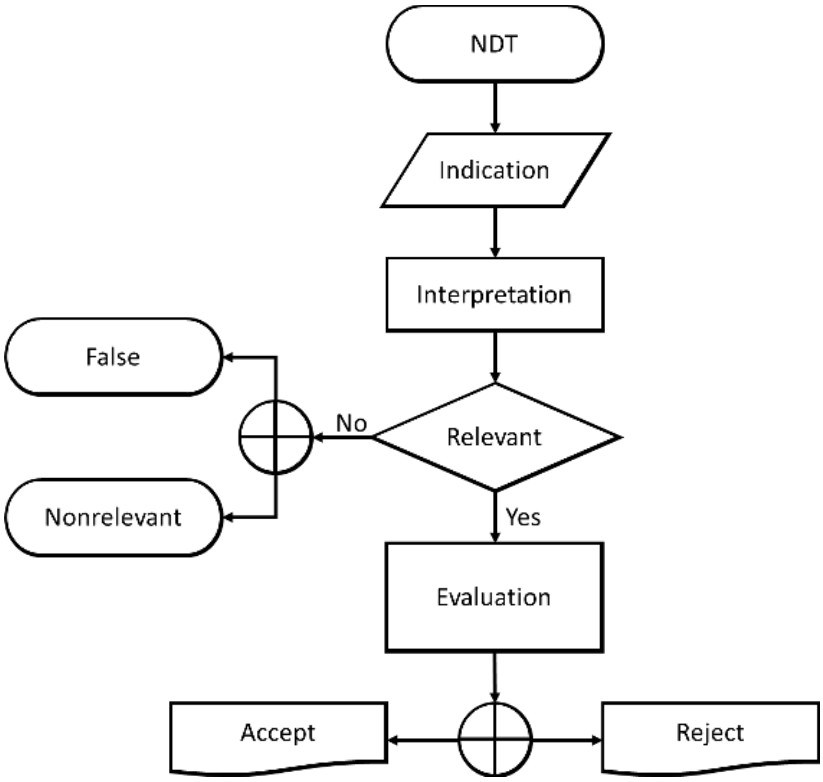


FIGURE 14. Excavated weld metal flow chart.

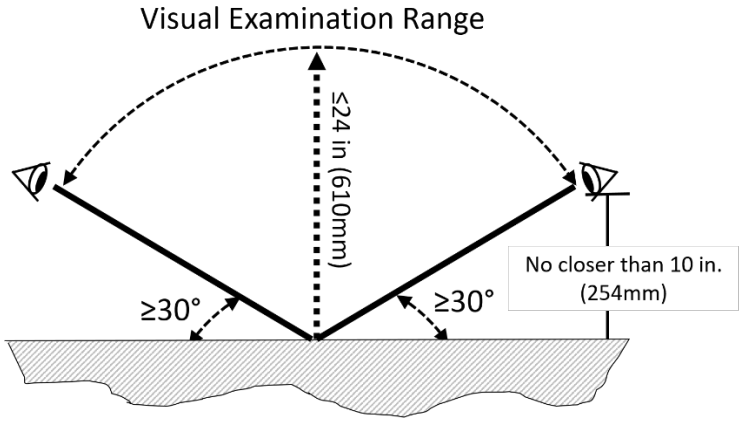


FIGURE 15. Visual inspection.

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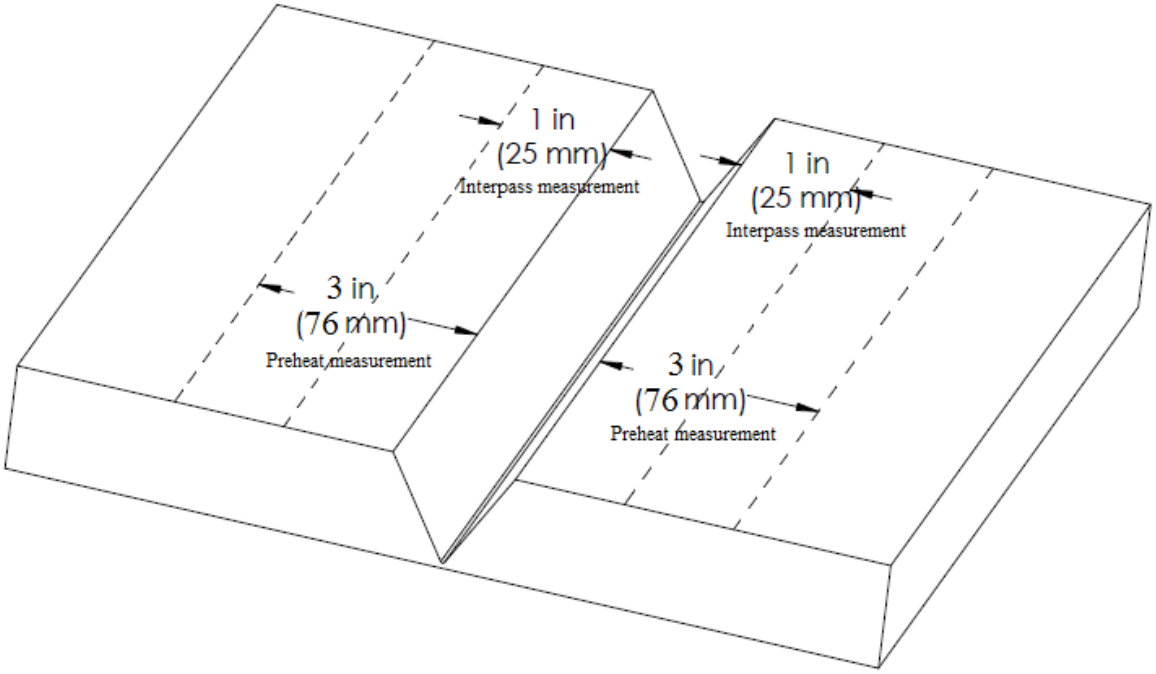


FIGURE 16. Temperature measurement technique.

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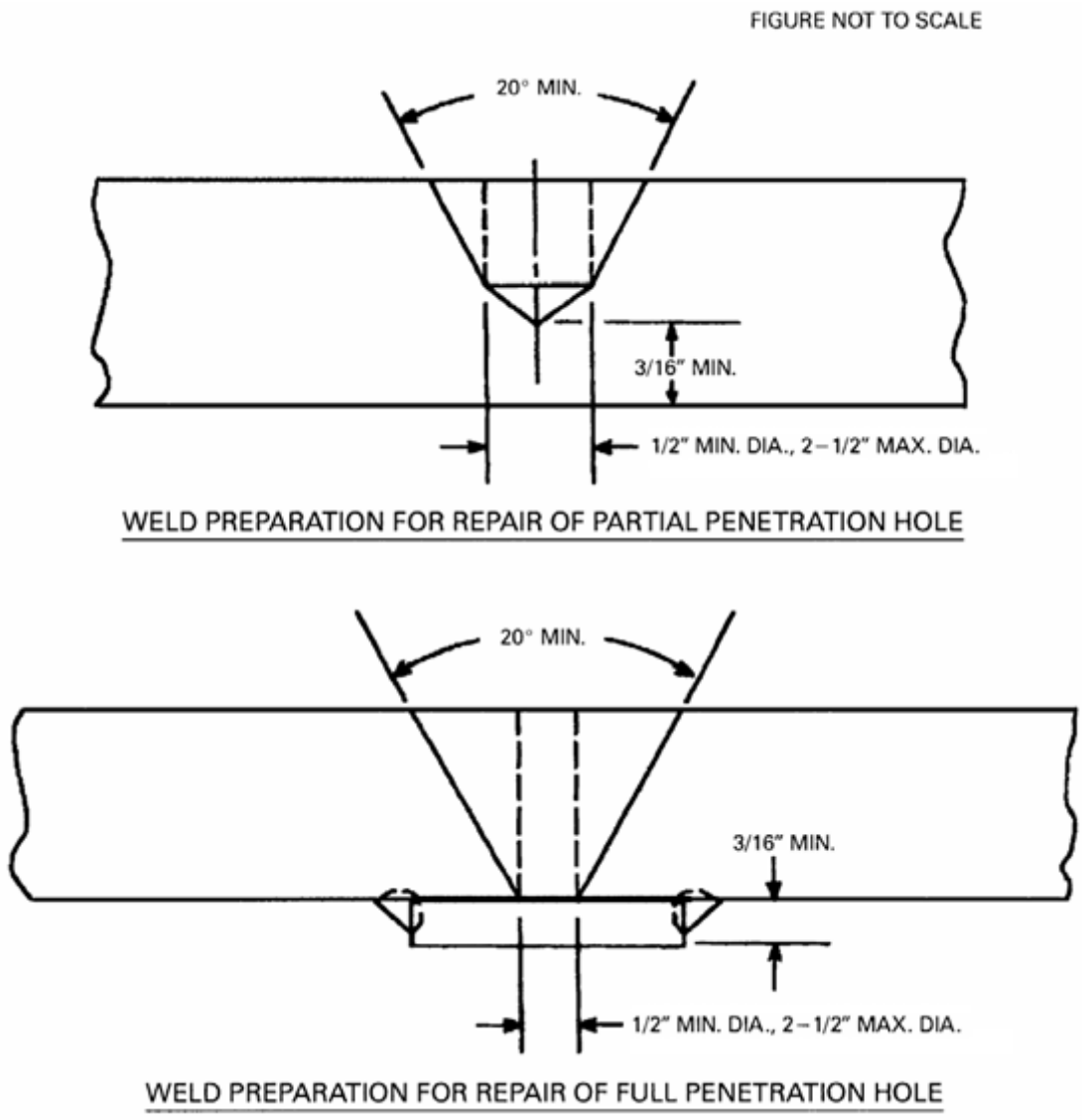


FIGURE 17. Hole repair.

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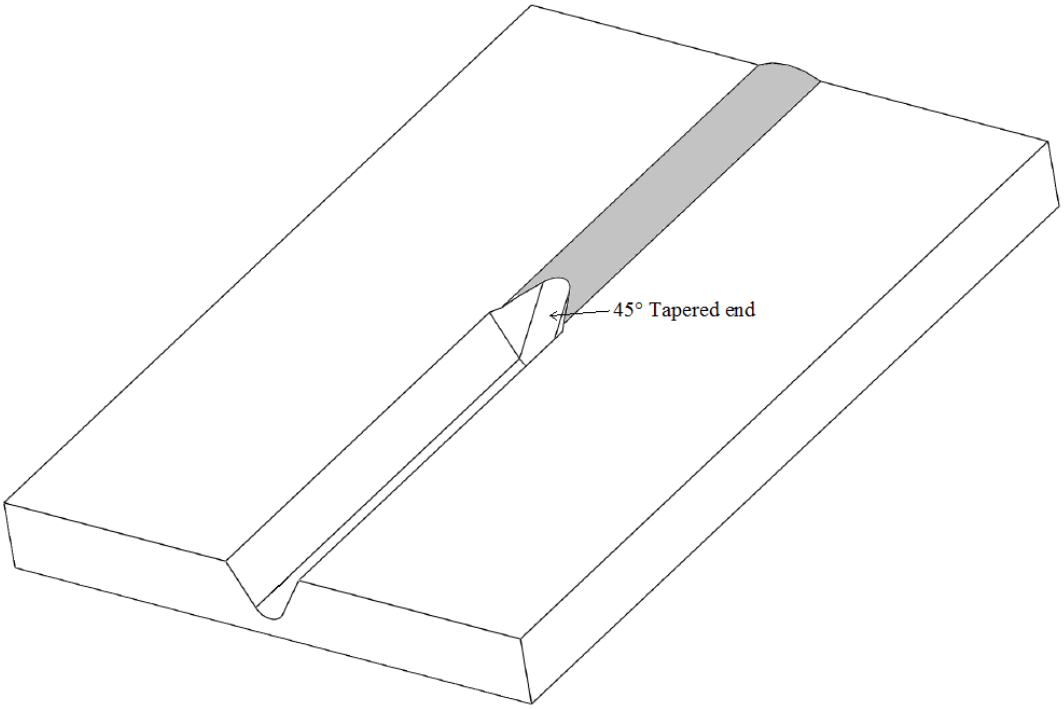


FIGURE 18. Excavated weld metal.

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APPENDIX A
WELDING PROCEDURE QUALIFICATION OF CLASS I WELDS

A.1 SCOPE

A.1.1 Scope. The information contained herein is required for welding procedure, welder and welding operator qualification of Class I welds.

A.1.2 Application. The PPIT shall provide verification that a proposed combination of weld process, filler metal, and base material can withstand PPIT. The CJP joints specified in this appendix are intended for use with the PPIT coupons only and do not represent the production representative samples.

A.2 APPLICABLE DOCUMENTS

A.2.1 General. The documents listed in section 2 of this standard shall also be required for Class I welds.

A.3 REQUIRMENTS

A.3.1 Requirements. All relevant general requirements (see 4) and detailed requirements (see 5) shall apply to Class I welding procedures and Class I welder and welding operator qualification.

A.4 PROCEDURE RECORDS

A.4.1 Procedure Qualification Records (PQRs). Prior to any production, a welding Procedure Qualification Record (PQR) shall be prepared for each Class I weld in accordance with 5.3, and shall be submitted to the procuring contract officer (see 6.2) for approval prior to usage in production. In addition to destructive (see 5.7) and nondestructive (see 5.9) testing required, each Class I PQR shall also require PPIT in accordance with this appendix.

A.4.1.1 Previous qualifications. If a previously qualified Class I PQR is to be used, all requirements of 4.4 and 5.3.5 shall apply. If all requirements are not met, a supplemental PQR may be developed in accordance with 5.3.4.

A.4.1.2 PQR base material. The base materials used for a Class I procedure qualification shall meet the requirements of 5.1.2.

A.4.1.3 Welding materials. All welding materials used for a Class I procedure qualification shall meet the requirements of 5.2.

A.4.1.4 Process qualification essential variables. All required applicable essential variables shall be fulfilled (see 5.4 and table V) for a Class I procedure qualification.

A.4.1.5 Nondestructive Testing (NDT). Prior to delivery of the qualification test plate to a Government approved testing agency (see A.6.2), the test plate shall be volumetrically

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inspected in accordance with 5.9. PPIT shall not be performed until all NDT requirements have been met and all inspection results shall be documented within the PQR.

A.4.1.6 Destructive testing. Prior to PPIT, all destructive testing requirements of 5.7 shall be met. The test plate may be a separate test plate or may be cut from the H-plate so long as the final dimensions of the H-plate meet the requirements of A.5.1. All testing results shall be documented within the PQR.

A.4.2 Welding Procedure Specification (WPS). Prior to any production, a welding procedure specification (WPS) shall be prepared for each weld in accordance with 5.5 and shall be submitted to the procuring contract officer for approval (see 6.2).

A.5 TEST PLATE

A.5.1 Dimensions of test plate. The size and shape of all plate samples shall be fabricated in accordance with figure A-1.

A.5.2 Plate flatness. All welded H-Plates shall have a maximum out of flatness tolerance of 3/8 in. (9.5mm) in any direction. Mechanical straightening methods are allowed to meet flatness tolerances.

A.5.3 Test plate thickness. The production thickness ranges from table A-I shall be used to determine the thickness of the test plate required for qualification. Qualification to the minimum thickness and maximum thickness ranges does not qualify any other thickness range unless the ranges of the two qualification procedures overlap.

A.5.3.1 Dissimilar thickness welds. When a production weld is to be fabricated from two dissimilar thicknesses, a minimum of one test plate of each thickness involved shall be prepared. For example, a 1 inch (25 mm) plate welded to a 0.375 inches (10 mm) plate would require two test plates, both a 1 inch (25 mm) and a 0.375 inches (10 mm) test plate.

A.5.4 Test plate armor types. A separate PPIT test plate shall be prepared for each type of armor used for production. Armor is defined as:

a. Armor manufactured by different methods, i.e. such as rolling or casting

b. Armor qualified under different material specifications, i.e. MIL-DTL-12560 or MIL-DTL-46100. The only groupings of armor grade materials shall be MIL-DTL-12560 Class 1 and Class 2 alloys (see 5.1.2 and 5.1.3)

c. Any material not qualified under MIL-A-11356, MIL-DTL-12560, or MIL-DTL-46100 that is approved by the procuring contract officer to be used as armor (see 5.1.2)

A.5.5 Test plate joint type. Regardless of actual application joint type (CJP or PJP), all test plates shall be fabricated utilizing CJP single or double V groove welds. The test samples

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shall be Type I-A, Type I-B or Type II in accordance with thickness limitations of A.5.3 and table A-I.

A.5.6 Preparation of test plate. There shall be a unique PPIT test plate for each PQR developed for Class I welds.

A.5.6.1 Preheat and interpass temperature. Preheat and interpass temperatures shall be in accordance with 5.10.4 and table XIII. Preheat and interpass temperature shall be measured immediately before deposition of each bead in accordance with 5.10.4.2 and figure 16.

A.5.6.2 Order of welding. The PPIT test plate shall be prepared by welding sections “a-b” and “c-d” completely on both sides, prior to welding crossbar “e-f” (see figure A-1). Each section and the crossbar shall be started at the same preheat temperature, within ±5 °F (±3 °C). Welding in the crossbar prior to completion of sections “a-b” or “c-d” shall not extend for more than 2 in. (51 mm) in either direction and shall be feathered to leave at least 8 in. (200 mm) of continuous, open groove before starting to weld the crossbar. When welding crossbar “e-f”, the plate shall be rotated to weld the crossbar in the same position as other sections (see 5.10.3).

A.5.6.3 Backgouging of welds. All test plates without backing shall have a backgouging technique used (see figure A-2 and figure A-4). Each plate shall be prepared by backgouging the root on the second side after sufficient weld metal has been deposited on the first side.

A.5.6.4 Tack welds. If tack welds are used during preparation of the test plate, the tack welds shall be made with the same electrode as the final weld. Tack welds shall be of the same quality and shall be deposited so as to facilitate incorporation into the final weld or removed prior to welding (see 5.10.7).

A.5.7 Identification marking of test plates. Each PPIT test plate shall be clearly and permanently marked in the upper right corner for easy identification on the front surface of the plate. All markings shall be a minimum of 0.25 inches (6 mm) in height and shall include the plate identification, the manufacturer's name, the contractor's name (if different from manufacturer’s name), process and position, and the designation “IMPACT SIDE” to indicate the front of the plate (see figure A-1). All markings shall be fully legible or testing shall not occur. The front of the ballistic plate shall be determined as follows:

a. For Type I joints, the front of the PPIT test plate is considered to be the outside or ballistically exposed surface containing the face of the groove.

b. For Type II joints, the front of the PPIT test plate is considered to be the surface on which welding (excluding tacking) was begun on each leg and on the crossbar.

A.5.8 Marking of retest plates. All retest plates shall be marked with the number of the original rejected plate, as well as the new test plate number with the suffix “T” indicating retest.

A.6 PROOFING PROJECTILE IMPACT TESTING

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A.6.1 Proofing Projectile Impact Testing (PPIT) requirements. PPIT is required for Class I welds in addition to the requirements of section 5 of this document. The completed welding procedure for the process used shall be delivered with each PPIT test plate. The test plate shall be tested and meet the requirements in accordance with TOP 2-2-711.

A.6.2 Proofing projectile impact testing address. Unless otherwise specified, ballistic testing shall to be completed at a certified testing facility. Plates shall be directed to either ATTN: Survivability Armor Ballistic Lab (SABL), 6501 E. 11 Mile Road, RDTA-RS/MS 263, Warren, MI 48397-5000, or Commander, U.S. Army Aberdeen Test Center, 400 Colleran Road. Bldg. 358, ATTN: CSTE-DTC-AT-SL-V Armor Acceptance - B690, Aberdeen Proving Ground, MD 21005-5059.

A.6.3 NDT examination of test plate: Prior to shipping the test plate to the testing facility, all finished welds shall have visual and volumetric examination performed. All testing requirements and acceptance criteria shall be in accordance with this document. Final acceptance testing shall take place in accordance with 5.9.1.2. Should the test plate fail to meet the NDT examination requirements, the PPIT shall not be performed until the defective weld area has been repaired (see 5.12), and able to pass examination. Alternative strategies to mitigate delayed hydrogen cracking must be supported by data and shall be submitted to the technical authority for acceptability prior to amending the length of this hold time as required in 5.9.1.2.

A.6.3.1 Visual examination of test plates. Prior to PPIT, all completed H-plates shall meet the visual inspection requirements of 5.9.2.1 and table XX.

A.6.3.2 Volumetric NDT examination. Prior to PPIT, all completed H-plates shall pass a volumetric inspection (see 5.9).

A.7 TEST RESULTS

A.7.1 Evaluation of test. The test plate shall be tested and meet the requirements in accordance with TOP 2-2-711.

A.7.2 Nondestructive testing after test. All test plates shall be examined with VT (see 5.9.2.1) within 24 hours after testing to indicate a crack and measure the crack length. All test plates shall have PT (see 5.9.2.2) or MT (see 5.9.2.3) performed to assure an accurate evaluation of the crack length if the test director of the testing facility cannot determine the crack length by VT. All PT or MT shall be performed by certified NDT inspector in accordance with 5.8. This second examination shall occur within a 24 hour period of VT, otherwise the original crack length determined shall be used as the result of the test.

A.7.3 PPIT Reports. The testing facility shall document results of the PPIT and provide the results of the evaluation crack length within 10 working days after final NDT after testing is completed.

A.7.3.1 Photographic record. Photographs shall be taken of the entire front and back of the plate as it sits in the test fixture prior to and after any impacts. Additional close-up

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photographs of the front and back of the impact area showing any cracks present shall also be taken after each impact. Front, back and length of cracking shall be clearly identified for each photographic record. Each photograph shall include an embedded reference scale.

A.7.4 Repair of test plate. Weld repair on a test plate shall not exceed a total length of 8 in. (200 mm) and any defective weld zone shall not be repaired more than once. Repairs shall be to correct discontinuities identified during NDT. No repairs shall be allowed after PPIT. An amendment to the PQR (see A.4.1) shall be submitted with the repaired test plate, and shall include the repair record in accordance with 5.12.10.

A.7.5 Rejection of test plate. Failure of any test plate to pass either the PPIT, or the second radiographic inspection after allowable repairs, shall be cause for rejection of the recorded welding procedure.

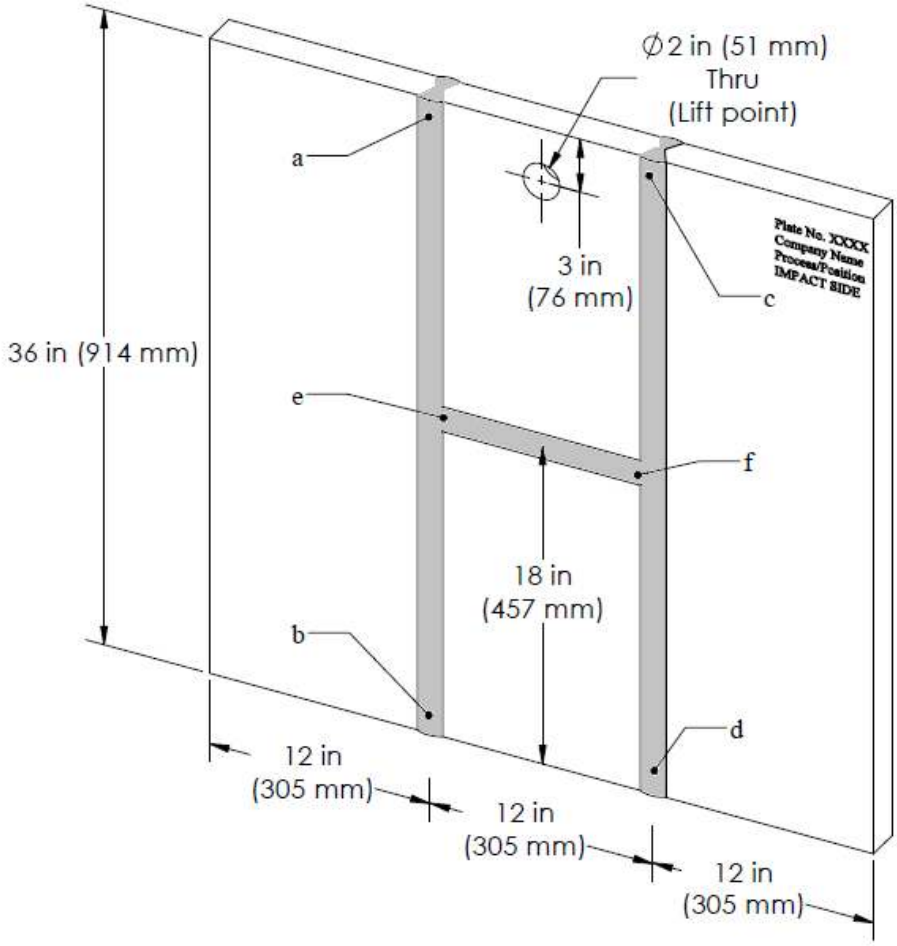
A.7.6 Retest. Retests of a rejected test plate may be made upon the request of the contractor. Two additional test plates shall be fabricated using the same WPS as the previously rejected test plate, marked in accordance with A.5.7 and submitted to the testing facility for retest. Failure of either of these plates shall be cause for rejection of the WPS represented. Only one retest for a welding procedure shall be allowed.

TABLE A-I. H-test plate thickness and qualification range.

Thickness (T) of H-test plate		Joint type[1]	Thickness (T) range qualified
Inches	Millimeters		
0.25	6	I-A or I-B	1/2T - 2T
0.5 [1]	13	I-A, I-B or II	
1	25	II	
1.5	51	II	1/2T - unlimited

NOTE:
0.5 inch thick plate accepts for a single or double V groove weld deposit and sequence.

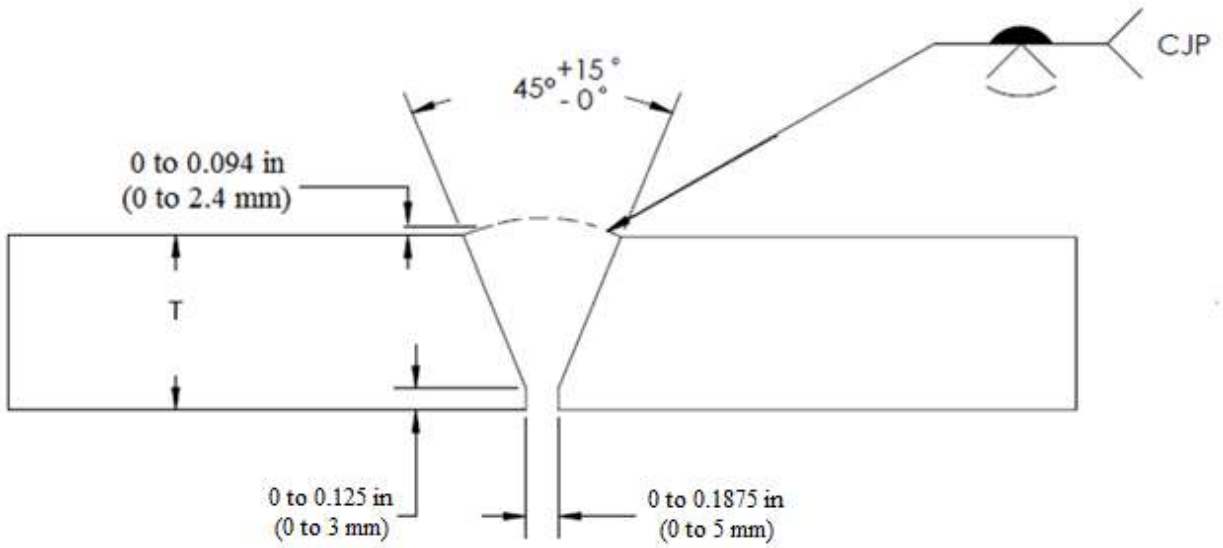
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Note:
1. Overall plate dimensions are ±0.5 in.
2. Lift point dimensions are -0 +0.25 in.

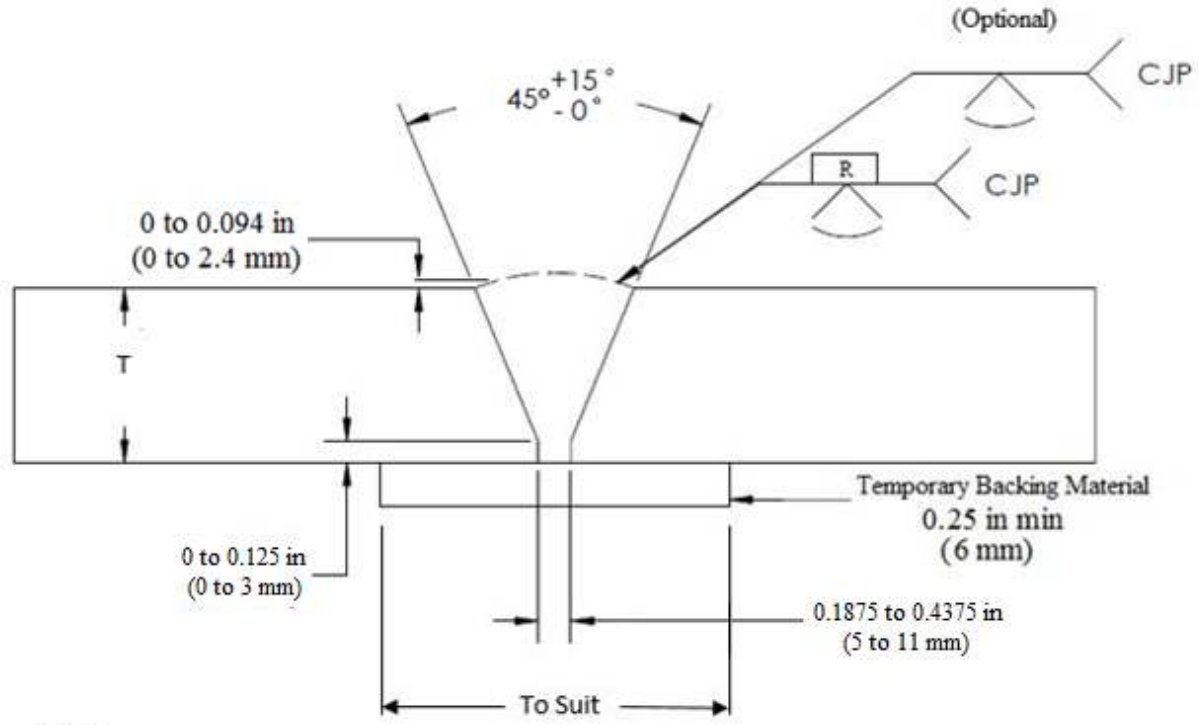
FIGURE A-1. Proofing projectile impact testing test plate (H plate).

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NOTE:
CJP requires backgouging (see A.5.6.3)

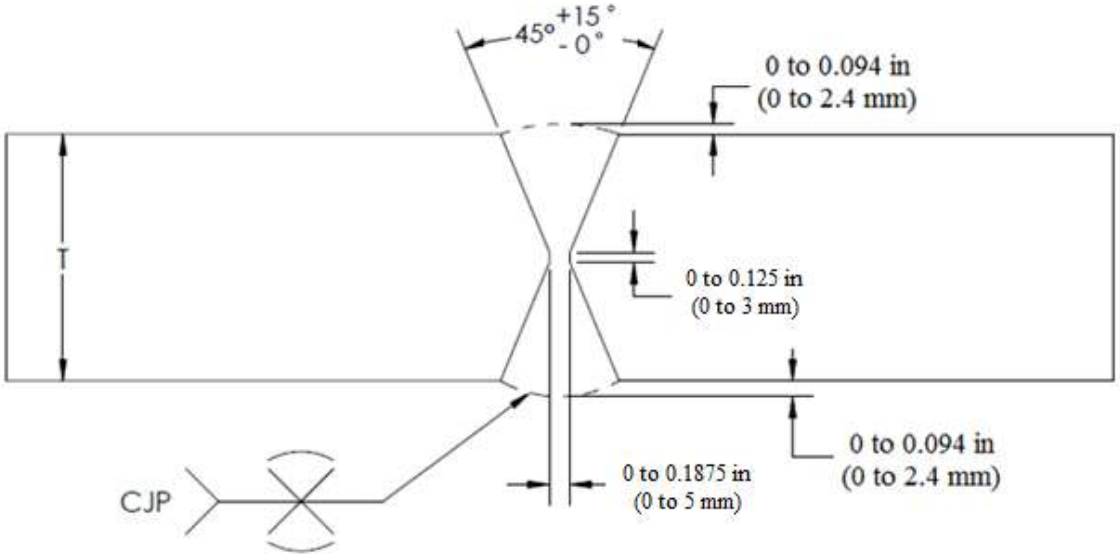
FIGURE A-2. Type I-A, single V-groove butt joint, without backing.



NOTE:
CJP requires backgouging (see A.5.6.3)

FIGURE A-3. Type I-B, single V-groove butt joint with backing material.

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NOTE:
CJP requires backgouging (see A.5.6.3)

FIGURE A-4. Type II, double V-groove butt joint.

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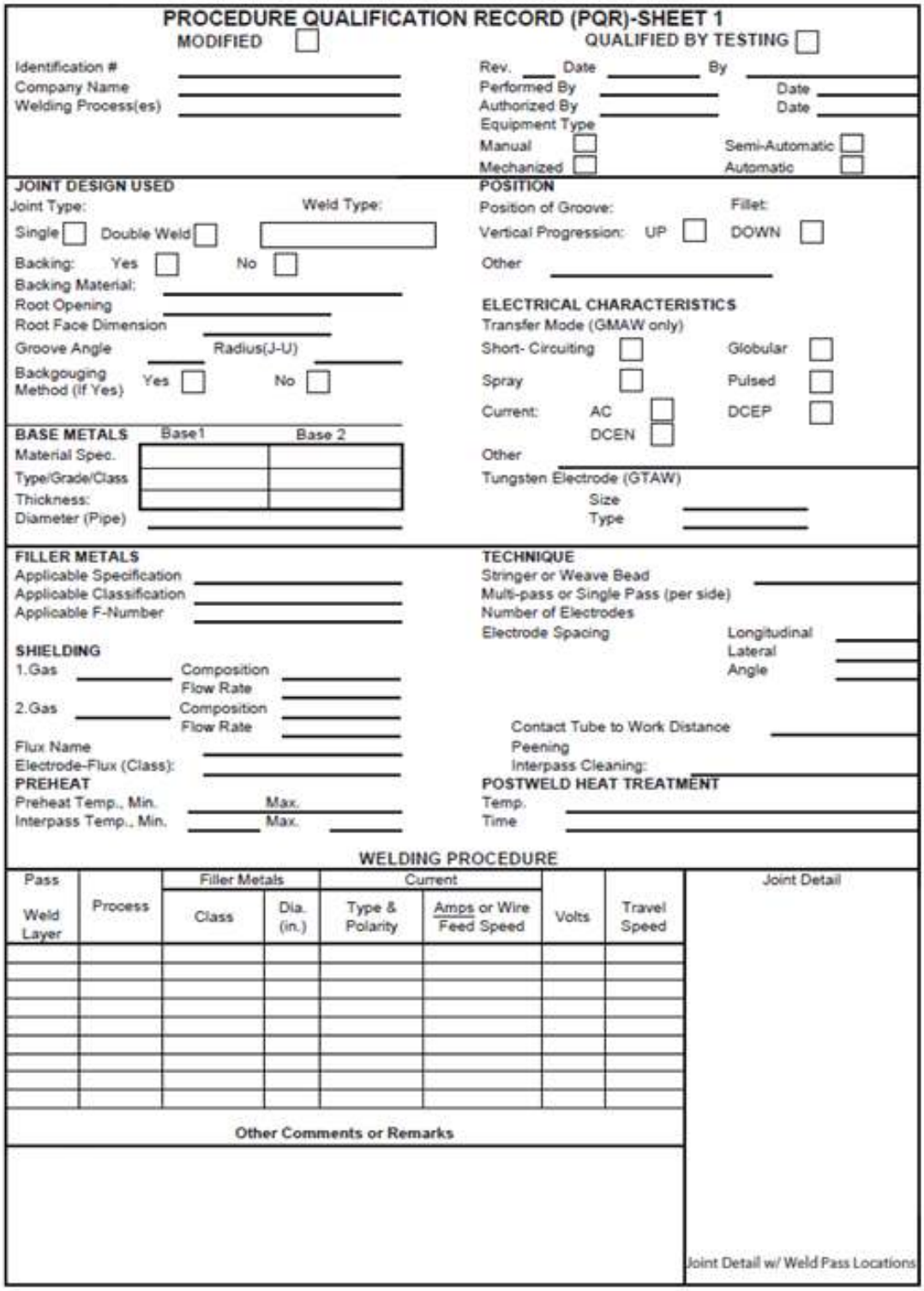
REFERENCE MATERIALS

B.1 SCOPE

B.1.1 Scope. The documents contained herein are examples of various welding records and forms.

B.1.2 Application. The forms in this appendix may be used in whole or partially as needed.

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PROCEDURE QUALIFICATION RECORD (PQR)-SHEET 1

MODIFIED ☐ QUALIFIED BY TESTING ☐

Identification # _____ Rev. ___ Date _____ By _____
Company Name _____ Performed By _____ Date _____
Welding Process(es) _____ Authorized By _____ Date _____
Equipment Type
Manual ☐ Semi-Automatic ☐
Mechanized ☐ Automatic ☐

JOINT DESIGN USED
Joint Type: Weld Type:
Single ☐ Double Weld ☐
Backing: Yes ☐ No ☐
Backing Material: _____
Root Opening _____
Root Face Dimension _____
Groove Angle _____ Radius(J-U) _____
Backgouging Method (If Yes) Yes ☐ No ☐

POSITION
Position of Groove: Fillet:
Vertical Progression: UP ☐ DOWN ☐
Other _____

ELECTRICAL CHARACTERISTICS
Transfer Mode (GMAW only)
Short- Circuiting ☐ Globular ☐
Spray ☐ Pulsed ☐
Current: AC ☐ DCEP ☐
DCEN ☐
Other _____
Tungsten Electrode (GTAW)
Size _____
Type _____

BASE METALS

	Base1	Base 2
Material Spec.		
Type/Grade/Class		
Thickness:		

Diameter (Pipe) _____

FILLER METALS
Applicable Specification _____
Applicable Classification _____
Applicable F-Number _____

SHIELDING
1.Gas _____ Composition _____
Flow Rate _____
2.Gas _____ Composition _____
Flow Rate _____
Flux Name _____
Electrode-Flux (Class): _____

TECHNIQUE
Stringer or Weave Bead _____
Multi-pass or Single Pass (per side) _____
Number of Electrodes _____
Electrode Spacing Longitudinal _____
Lateral _____
Angle _____
Contact Tube to Work Distance _____
Peening _____
Interpass Cleaning: _____

PREHEAT
Preheat Temp., Min. _____ Max. _____
Interpass Temp., Min. _____ Max. _____

POSTWELD HEAT TREATMENT
Temp. _____
Time _____

WELDING PROCEDURE

Pass or Weld Layer	Process	Filler Metals: Class	Filler Metals: Dia. (in.)	Current: Type & Polarity	Current: Amps or Wire Feed Speed	Volts	Travel Speed

Other Comments or Remarks

Joint Detail

Joint Detail w/ Weld Pass Locations

FIGURE B-1. Procedure Qualification Record (PQR) example.

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PROCEDURE QUALIFICATION RECORD (PQR)-SHEET 2

Test Results

Tensile Test

Specimen	Width (in.)	Thickness (in.)	Area (in.)	Unlimited tensile Load lb.	Ultimate Unit stress, psi	Character of failure and location
#1						
#2						
#3						
#4						

GUIDED BEND TEST

Specimen No.	Type of bend	Result	Remarks

VISUAL
Appearance _____
Undercut _____
Piping porosity _____
Convexity _____
Test date _____

RADIOGRAPHIC-ULTRASONIC EXAMINATION
RT report no.: _____ Result _____
UT report no.: _____ Result _____

Test conductor and interpreter _____ Print
_____ Signature
_____ Title
_____ Date

Stamp Location

ALL-WELD-METAL TENSION TEST
Tensile Strength _____
Yield Strength _____
Elongation in 2 in. % _____
Laboratory tests no. _____

FILLET/GROOVE WELD TEST RESULTS
Minimum size multiple pass = N/A
Macro-etch 1.) _____ 3.) _____ 2.) _____
Bend Break 1.) _____ 2.) _____

Welders name _____ Clock No. _____ Stamp No. _____
Test Conducted By _____

Test number Per _____

IMPACT TEST
Size of Specimen _____ 1.) 4.) Average
Test Temp _____ 2.) 5.) High
Ft-lb ☐ Joules ☐ 3.) 6.) Low

Signed by _____
Title _____
Date _____

CWI Stamp Location

Signed By _____
Title _____
Date _____

FIGURE B-1. Procedure Qualification Record (PQR) example – Continued.

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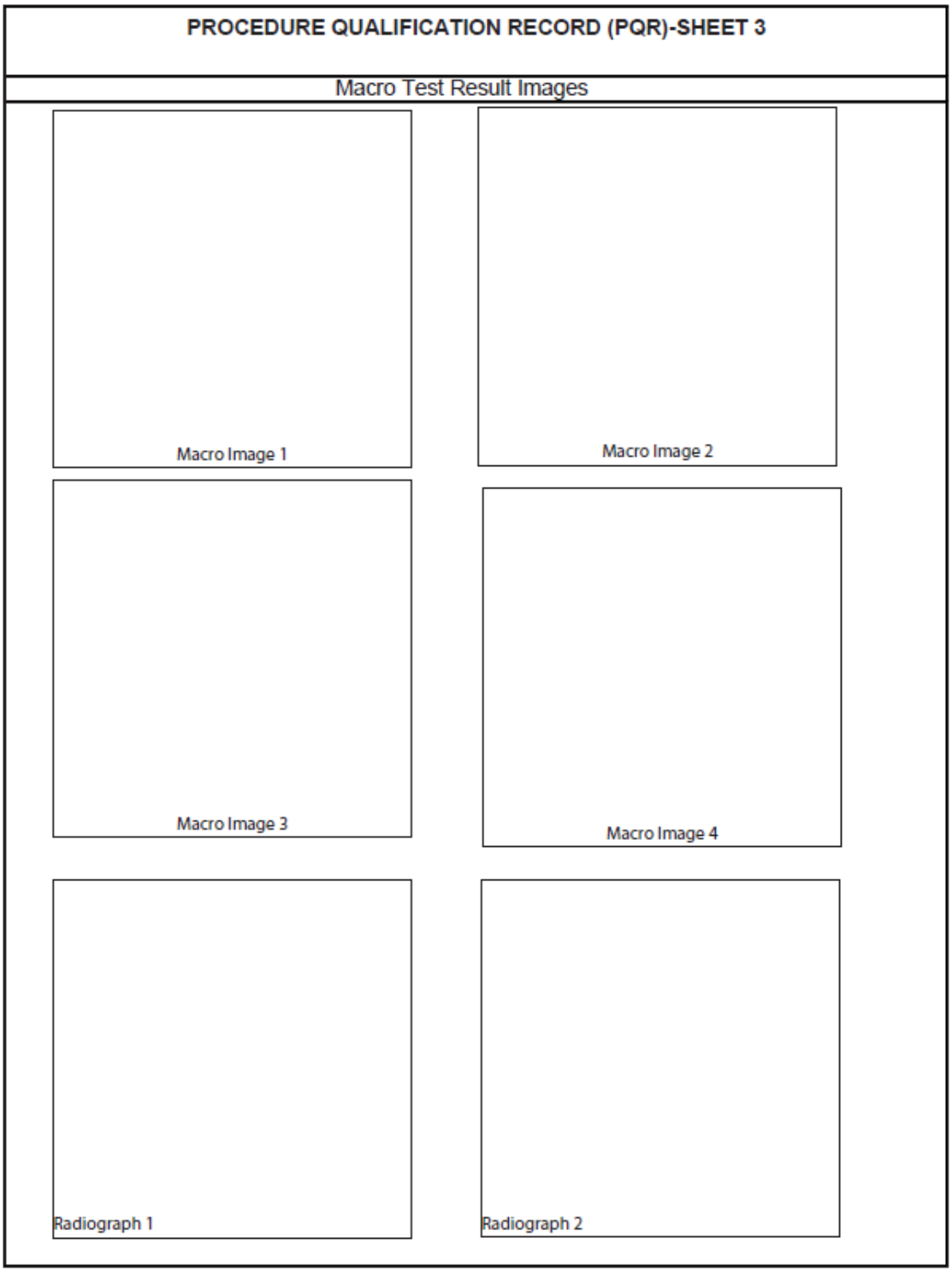


FIGURE B-1. Procedure Qualification Record (PQR) example – Continued.

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WELD PROCEDURE SPECIFICATION (WPS)

MODIFIED ☐ QUALIFIED BY TESTING ☐

Identification # _____ Rev. ____ Date _____ By _____
Company Name _____ Performed By _____ Date _____
Welding Process(es) _____ Authorized By _____ Date _____
Equipment Type
Manual ☐ Semi-Automatic ☐
Mechanized ☐ Automatic ☐

JOINT DESIGN USED
Joint Type: Weld Type:
Single ☐ Double Weld ☐
Backing: Yes ☐ No ☐
Backing Material: _____
Root Opening _____
Root Face Dimension _____
Groove Angle _____ Radius(J-U) _____
Backgouging Method (If Yes) Yes ☐ No ☐

POSITION
Position of Groove: Fillet:
Vertical Progression: UP ☐ DOWN ☐
Other _____

ELECTRICAL CHARACTERISTICS
Transfer Mode (GMAW only)
Short- Circuiting ☐ Globular ☐
Spray ☐ Pulsed ☐
Current: AC ☐ DCEP ☐
DCEN ☐
Other _____
Tungsten Electrode (GTAW)
Size _____
Type _____

BASE METALS

	Base1	Base 2
Material Spec.		
Type/Grade/Class		
Thickness:		

Diameter (Pipe) _____

FILLER METALS
Applicable Specification _____
Applicable Classification _____
Applicable F-Number _____

SHIELDING
1.Gas _____ Composition _____
Flow Rate _____
2.Gas _____ Composition _____
Flow Rate _____
Flux Name _____
Electrode-Flux (Class): _____

PREHEAT
Preheat Temp., Min. _____ Max. _____
Interpass Temp., Min. _____ Max. _____

TECHNIQUE
Stringer or Weave Bead _____
Multi-pass or Single Pass (per side) _____
Number of Electrodes _____
Electrode Spacing Longitudinal _____
Lateral _____
Angle _____
Contact Tube to Work Distance _____
Peening _____
Interpass Cleaning: _____

POSTWELD HEAT TREATMENT
Temp. _____
Time _____

WELDING PROCEDURE

Pass Weld Layer	Process	Filler Metals Class	Filler Metals Dia. (in.)	Current Type & Polarity	Current Amps or Wire Feed Speed	Volts	Travel Speed

Joint Detail

Other Comments or Remarks

Joint Detail w/ Weld Pass Locations

FIGURE B-2. Welding Procedure Specification (WPS) example.

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WELDER/ WELDING OPERATOR QUALIFICATION RECORD (WQR)

Name _____ Identification # _____

Welder Type _____ Date _____

Welding Procedure Specification (WPS) ID# _____

Variables	Actual Recorded Values Used During Qualification	Range Qualified
Process Type		
Automatic/Machine		
Electrode		
Number of Electrodes		
Current/Polarity		
Position		
Weld Progression (Vertical)		
Backing		
Base Material 1 Grade/Class		
Base Material 2 Grade/Class		
Base Material 1 Thickness		
Base Material 2 Thickness		
Base Material 1 Type		
Base Material 2 Type		
Base Material Weld Configuration		
Filler Material Specification # (MIL-E-XXXXX,MIL-DTL-XXXXX)		
Filler Material Class		
Gas/Flux Composition		
Arc Voltage Control		
Joint Tracking Software		

FIGURE B-3. Welder/Welding Operator Qualification Record (WQR) example.

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Welding Procedure Specification (WPS) ID# _____

WQR Test Results

VISUAL INSPECTION

Appearance _____ Pass/Fail _____

Undercut _____

Piping porosity _____ Test date _____

Convexity _____

GUIDED BEND TEST

Specimen No.	Type of bend	Result	Remarks

RADIOGRAPHIC-ULTRASONIC EXAMINATION

UT report no.: _____ Result _____ RT report no.: _____ Result _____

Result _____ Result _____

FILLET WELD TEST RESULTS

Macro-etch Results: 1.) _____ 2.) _____

Break Test Results: 1.) _____

Fillet Size (two locations): 1.) _____ 2.) _____

MACRO/RADIOGRAPH IMAGES

Macro Image 1	Macro Image 2	Radiograph 1	Radiograph 2

Test conductor and interpreter

Welders Name _____ Clock No. _____ Stamp No. _____

Test Conducted By _____

Welder Signature _____

Title _____

Date _____

CWI Stamp Location

Test Conductor Signature _____

Title _____

Date _____

FIGURE B-3. Welder/Welding Operator Qualification Record (WQR) example – Continued.

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CONCLUDING MATERIAL

Custodians:
Army – AT
Navy – SH
Air Force – 20

Preparing activity:
Army – AT

(Project THJM-2024-001)

Review activities:
Army –AV, MI, MR, TE
DLA – DH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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